

ICOHP Descriptors and Thermodynamic Stability Across a 597-Structure Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

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Report generated as part of the `viability` project

August 16, 2026

The project’s central question is viability discrimination based on $\Delta(\text{ICOHP})$ (§6.2), across a 597-structure dataset.

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1 Objective and Problem Statement

Central research question (see §6.2): does $\Delta(\text{ICOHP})$ — the reaction-ICOHP descriptor for a compound’s decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?

2 Methodology

2.1 Dataset Selection

Materials Project query: unary and binary chemical systems, non-magnetic ($|\text{total magnetization}| \leq 0.01 \mu_B/\text{formula unit}$ — MP’s categorical `magnetic_ordering` field is "Unknown" for $\sim 98\%$ of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements. The dataset combines two complementary selection tracks: a first set of 186 compounds (≤ 20 atoms/cell, 60 per stability family, a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system), and a second set of 163 compounds/elements (≤ 40 atoms/cell) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N/O/F/P/S/Cl, chosen to build genuine same-composition polymorph groups and to test descriptors far from equilibrium; and a third set of 45 binary compounds, each pairing an experimentally known, maximally metastable compound (farthest above the convex hull among its chemical system) with 1–3 theoretical polymorphs of the same formula sitting even farther above the hull, a robustness test deliberately targeting stability edge cases.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by default by the `classify()` function (elemental composition + `is_metal`). A second, independent classification is built directly from each compound’s ICOHP/ICOB data (§5): `is_metal` for metallic character, then the mean ICOBI per bond (bond order, covalency) to separate ionic from covalent among non-metallic compounds. The two classifications are compared in §5.

2.3 VASP + LOBSTER Computation

Same template across all structures (static run, `ISYM=-1`, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`): NBANDS is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 ICOHP/ICOB Quantities: Integration, Cutoffs, Units

LOBSTER generation (`lobsterin`, identical across the dataset): `pbeVaspFit2015` basis; COHP integration window from -35 to $+5$ eV around the Fermi level (`COHPstartEnergy/COHPendEnergy`); bond detection by interatomic distance from 0.1 to 6.0 Å (`cohpGenerator`, orbital-resolved); 0.05 eV Gaussian smearing (`gaussianSmearingWidth`).

Integrated ICOHP/ICOB (`ICOHPLIST.lobster/ICOBILIST.lobster`). A bond’s integrated ICOHP is, by LOBSTER’s own definition, the integral of $\text{COHP}(E)$ from the bottom of the window above up to the Fermi level (occupied states only), in eV; negative = bonding, positive = antibonding. ICOBI is the same integral over COBI (bond order index), dimensionless.

Per-compound aggregation: `icohp_sum/icobi_sum` over every symmetrically inequivalent bond label in `ICOHPLIST.lobster/ICOBILIST.lobster` (each periodic bond counted once, not once per direction); `icohp_mean/icobi_mean`, the corresponding per-bond average — this intensive version, comparable across compounds of different cell sizes, is what is reported in Table 10 and the “ICOHP”/“ICOB” columns of Appendix B. These quantities feed §5 and the classic correlations of §6.

Which bonds get summed — three distinct conventions, not one. This project uses three different rules for which atom pairs enter an ICOHP/ICOB sum, and conflating them is a real risk when reading numbers across sections:

1. **Unfiltered (default):** `icohp_sum/icohp_mean` and `icobi_sum/icobi_mean` — used throughout §6.2–6.2, Table 10, Appendix B’s “ICOHP”/“ICOB” columns, the elemental-reference table, and every $\Delta(\text{ICOHP})$ reaction total in the full case-1 reaction list — sum **every** symmetrically inequivalent bond label LOBSTER reports within its 0.1–6.0 Å detection window (`cohGenerator` above); no further coordination-shell filtering. This is the project’s working default because it requires no chemistry-specific tuning and treats every compound identically.
2. **First-coordination-shell only:** an adaptive, relative rule (`reaction_analysis/nearest_neighbor.py`, gap ratio 2.5) sorts every observed bond distance for a species pair and cuts the shell at the first gap exceeding $2.5\times$ the largest gap seen so far within the putative shell — **no fixed Å cutoff is hardcoded anywhere**; the boundary is self-calibrating per compound. Used in exactly two places: the 7 Reitz & Dronskowski validation reactions (§6.1, matching the manuscript’s own “first gap in the bond-distance spectrum” convention) and the bond-type classification’s **dominant species-pair** ICobi/ICOHP (§5, the “ICOB”/“ICOHP” columns of Appendix B are this dominant-pair value, *not* the compound-wide mean over every LOBSTER-reported neighbor).
3. **No per-pair filtering at all:** the antibonding-population descriptor below does not sum discrete bond pairs by distance or shell; it integrates LOBSTER’s own pre-aggregated “average” COHP trace (every bond, LOBSTER’s internal per-bond weighting already applied) and restricts only by an *energy* window near E_F/VBM — coordination never enters this descriptor’s definition.

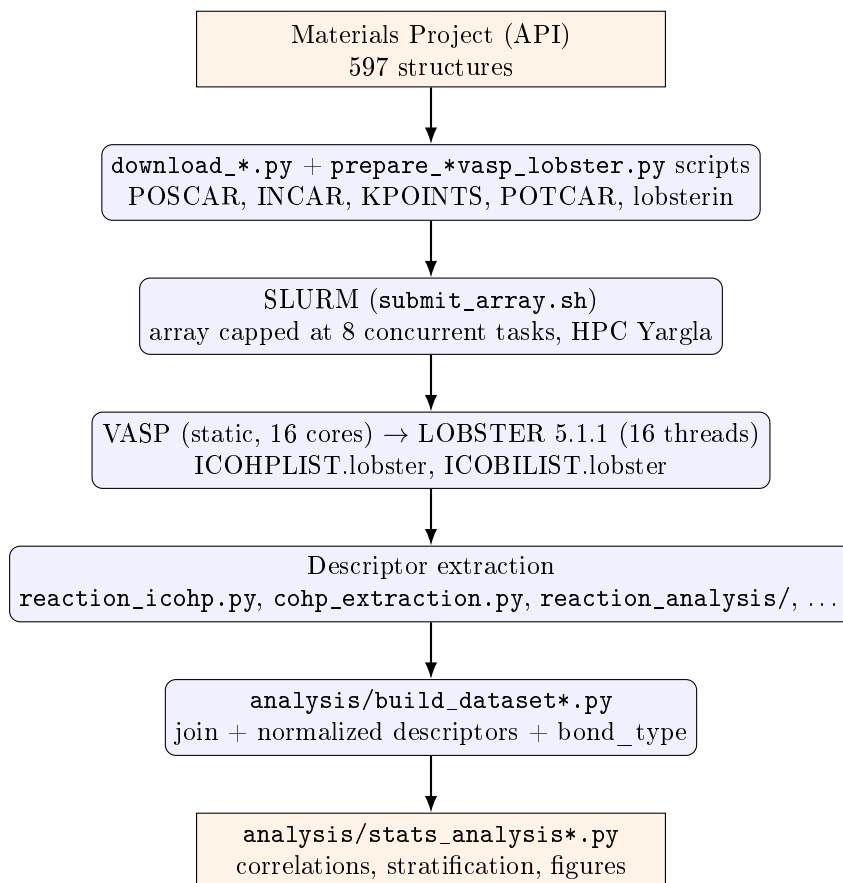
Reaction $\Delta(\text{ICOHP})$ (`reaction_analysis/`). For a balanced reaction (typically compound $\rightarrow$ elements), $\Delta = \text{sum of products' ICOHP} - \text{sum of reactants' ICOHP}$, “products – reactants” convention, in three normalizations: per formula unit (extensive), per transferred atom (the normalization used throughout this report unless noted), and per bond (diagnostic only, non-conservative — bonds are not conserved between differently-bonded structures). Positive sign (“endobondic”): breaking the reactant’s bonds costs more than the products’ bonds recover — a possible kinetic barrier. Negative sign (“exobondic”): no such barrier is visible from bonding alone.

Antibonding population near the Fermi level (`coh_extraction.py`) — a quantity distinct from the $\Delta(\text{ICOHP})$ above. Rather than integrating the whole occupied COHP, this quantity integrates only its antibonding part ($\text{COHP} > 0$) over a one-sided window ($E_{\text{ref}} - \Delta E, E_{\text{ref}}$], where $E_{\text{ref}} = E_F$ (metallic compounds) or the VBM (gapped compounds), and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV ($\Delta E = 1.0$ eV as the primary window) — only occupied states can destabilize the ground state via antibonding, hence the one-sided rather than symmetric window. Reported both raw (same “eV” unit convention as LOBSTER, though it is a Hamilton population integral, not literally an energy) and normalized by the total occupied ICOHP magnitude at the same E_{ref} . This quantity, and only this one, feeds the descriptor in §6.2 — the reaction $\Delta(\text{ICOHP})$ (§6.2–6.2) integrates the whole occupied COHP with no restriction to a window near E_F .

2.5 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOB I descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the descriptor), overall and per subgroup (`bond_type`, `is_metal`); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §9).

3 Workflow Diagram



4 Environment

Identical across all 597 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §8.

5 Dataset

The full set of computed compounds and elements totals **597 structures**.

Bond type. Two independent classifications exist (§2.2) — `classify()` and a classification built directly from ICOHP/ICOB I data — presented and compared in step 3 of the Results section (§6.3), not here: classification is not an end in itself, it exists solely to condition the viability-descriptor correlation tested in that same section.

Table 1: Dataset composition by element count.

Composition	<i>n</i>
Single-element references	69
Multi-element compounds	522
Validation structures without compound metadata*	6
Total	597

* isolated gas dimers (H_2 , N_2 , O_2 , F_2 , Cl_2) and CaO [sphalerite], used only for the Reitz & Dronskowski validation (Appendix M), outside the main dataset.

Table 2: Stability class (by E_{hull} and experimental/theoretical character).

Class	<i>n</i>	E_{hull} range (eV/atom)
stable ($E_{\text{hull}} = 0$)	160	0.0000
experimental metastable	183	0.0000347 – 2.1080
theoretical metastable	242	0.0000843 – 3.1688
unknown*	12	—
Total	597	

* 2 COD references with no E_{hull} (S_4N_2 , S_4N_4), 6 validation structures without metadata (see Table 1), 1 further COD reference (ZnSn/NiAs), and 3 compounds inherited from the original campaign, labeled "metastable" before the `theoretical` flag existed.

6 Results

Logic of this section

Four steps, in this order: (1) validate this project’s methodology by reproducing the 7 reference reactions chosen by Reitz & Dronskowski; (2) test this project’s central correlation — viability (a compound’s thermodynamic/kinetic stability) against descriptors derived from LOBSTER data (ICOHP, antibonding ICOHP, ICOBI, antibonding ICOBI); (3) classify compounds and elements into metallic/ionic/covalent/mixed directly from that same ICOHP/ICOBI data, and compare that classification against `classify()` — a tool in service of the next step, not an end in itself; (4) determine whether step 2’s correlation is strengthened or weakened once conditioned on step 3’s bond type.

6.1 Step 1 — Methodological validation: the 7 Reitz & Dronskowski reactions

Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

Per-bond-type ICOHP values for 7 reactions with independently known ΔICOHP (transcribed independently of any calculation in this project) are run end to end through `balance.py/delta.py/classify.py`:

The 7 cases reproduce the reference ΔICOHP to within a few kJ/mol (source per-bond value rounding) and every `BondingLabel` matches. The Mn_2O_7 case is a useful guardrail on its own: exobondic under this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than through abrupt bond breaking — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than special-casing Mn_2O_7 .

This validation only compares the arithmetic of the manuscript’s own published ICOHP values, independent of any local crystal structure — Appendix M documents, for the 16 species

Table 3: Validating the endobondic/exobondic classification on 7 reactions with independently known ΔICOHP .

Reaction	Reference (kJ/mol)	Computed (kJ/mol)	Label	
			Ref.	Computed
$\text{Pb}(\text{N}_3)_2 \rightarrow \text{Pb} + 3 \text{N}_2$	+1345	+1344.3	endo	endo
$\text{S}_4\text{N}_2 \rightarrow \frac{1}{2}\text{S}_8 + \text{N}_2$	+258	+258.5	endo	endo
$\text{S}_4\text{N}_4 \rightarrow \frac{1}{2}\text{S}_8 + 2\text{N}_2$	+399	+397.3	endo	endo
$\text{ZnSn} \rightarrow \text{Zn} + \text{Sn}$	-337	-336.6	exo	exo
$\text{CaO}[\text{sphalerite}] \rightarrow \text{CaO}[\text{rocksalt}]$	-79	-78.9	exo	exo
$\text{CaN} \rightarrow \frac{1}{3}\text{Ca}_3\text{N}_2 + \frac{1}{6}\text{N}_2$	-205	-204.8	exo	exo
$\text{Mn}_2\text{O}_7 \rightarrow 2\text{MnO}_2 + \frac{3}{2}\text{O}_2$	-186	-187.7	exo	exo

involved, which MP/COD structure was used in this dataset and whether it could be verified to match the manuscript’s description (3 real corrections identified: N_2 , the $\text{Pb}(\text{N}_3)_2$ polymorph, ZnSn ’s competing tin phase). For N_2 and ZnSn , a DFT+LOBSTER calculation of this project’s own (not just the manuscript’s arithmetic) was additionally compared directly against the manuscript’s numbers (Table S21): N_2 reproduces the manuscript’s ICOHP to within 0.7%; ZnSn reproduces the same sign (bonding-favorable decomposition) but with $\sim 2.9\times$ larger magnitude — a gap judged not significant at this stage, since a bulk intermetallic’s ICOHP sum is far more sensitive to basis/pseudopotential/cutoff choices than a simple gas dimer. Reproducing the manuscript’s full DFT methodology (PBEsol+D3, Table S1’s own k-point sets) has since been carried out for all 16 species (2026-08-19): 10/16 are computed and compared directly (Table S21), the remaining 6 (Ca_3N_2 , Mn_2O_7 , $\text{Pb}(\text{N}_3)_2$, S_4N_2 , S_4N_4 , S_8) still computing at the time of writing (§9).

Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even with a large systematic offset or scale factor. **Lin’s concordance correlation coefficient** (CCC) specifically tests agreement with the identity line (computed = reference), the claim actually being tested here:

Table 4: Computed-vs-reference concordance on the 7 validation reactions.

Statistic	Value
n	7
Lin’s CCC	0.999998
Pearson r	0.999999 ($p = 3.5 \times 10^{-15}$)
Sign agreement	7/7
Mean residual	0.76 kJ/mol
Max residual	1.70 kJ/mol
RMSE	0.98 kJ/mol

A CCC this close to 1 means the two methods agree essentially to the reporting precision of the reference values themselves (rounded to 2–3 decimals), not merely that they move together — the implementation is validated, not just spot-checked. It is this implementation, validated here independently of any of this project’s own calculations, that is applied to the full dataset in the next step.

6.2 Step 2 — Viability versus LOBSTER descriptors

Classic ICOHP aggregates (baseline). Before any constructed descriptor, the raw ICOHP aggregates (sum, minimum per compound) are tested against E_{hull} as-is, on the original 186-

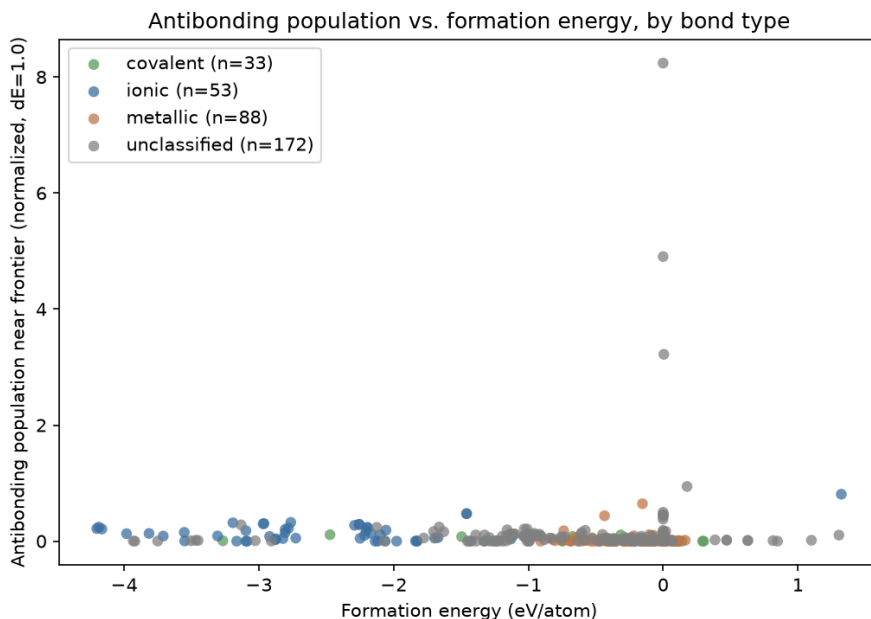
compound campaign’s subset:

Table 5: Spearman correlations, classic ICOHP aggregates (sum, minimum) against E_{hull} , 186-compound subset.

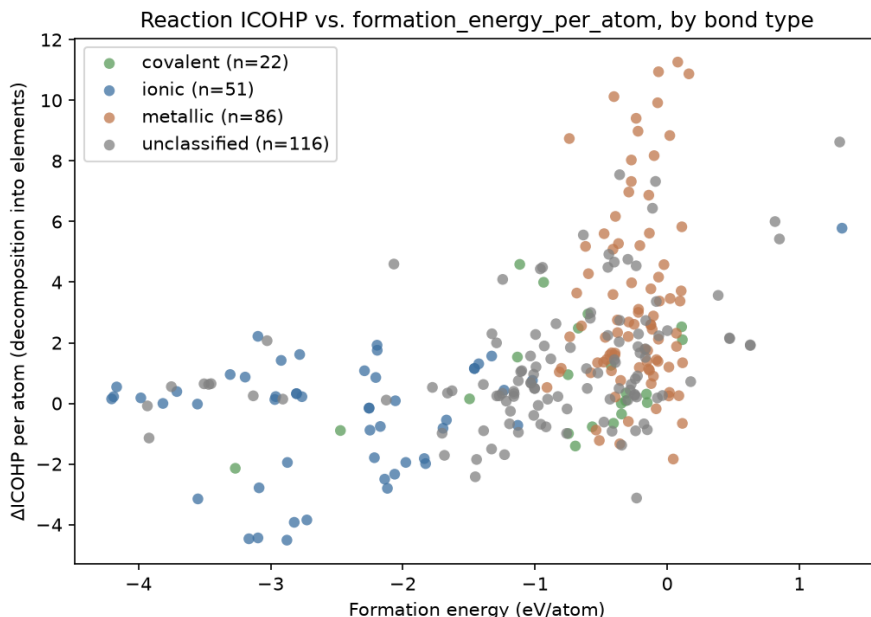
Group	Metric	n	ρ	p
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

These four rows correlate little or not at all with E_{hull} , except in the metallic subgroup — a weak baseline the descriptors below need to beat.

Antibonding population near the Fermi level (ICOHP). A distinct question from the fully-integrated ICOHP (§2.4): not *how much* bonding a compound has, but *how* COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn-Teller-type electronic instabilities. Result, against **formation_energy_per_atom**: $\rho = -0.155$, $p = 1.8 \times 10^{-4}$ ($n = 579$) — weaker than at smaller scale ($\rho = -0.282$, $n = 346$) but still real and significant; the sign remains the *opposite* of the naive Peierls/Jahn-Teller reading (§6.4 for the bond-type-stratified reading, where this descriptor’s headline stratified result no longer survives at the current scale).



Reaction ICOHP. An ICOHP analog of **formation_energy_per_atom**: is a compound’s total ICOHP "cheaper" than that of the same atoms as reference elements? Result: $\rho = 0.295$, $p \approx 0$ ($n = 510$) — a real, highly significant correlation given the sample size, with an intuitive sign (less bonding than in elemental form $\Rightarrow$ less stable formation). On a 288-compound subset shared with every other descriptor in the project, the antibonding population above slightly edges out reaction ICOHP on both targets — the latter remains a strong descriptor, but no longer unambiguously *the* strongest. Case 2 (direct same-formula polymorph comparison, 74 groups): 47.3% agreement between the most-bonded and the most-stable member, against a 42.4% chance baseline (exact Poisson-binomial test, $P = 0.22$) — no signal.



$\Delta(\text{ICOHP})$, applied to the full dataset, pooled. 517 case-1 reactions, every element and compound with a computed decomposition-into-elements reaction, pooled into a single population (grouping by selection provenance would be exactly the kind of between-group confound this project’s methodology is meant to catch):

Table 6: Spearman correlations, $\Delta\text{ICOHP}/\text{atom}$ (case 1, 517 reactions) against two continuous stability targets.

Target	n	ρ	p
formation_energy_per_atom	510	-0.2948	1.1×10^{-11}
energy_above_hull_eV_per_atom	514	-0.0564	0.202

(Sign: `reaction_analysis`’s products–reactants convention, opposite to `reaction_icohp.py`’s own column above — same magnitude, flipped sign.) Both correlations are weaker than at smaller scale: the formation-energy result was $\rho = -0.50$ before the marginal-formation-energy and max-hull-distance batches were folded in, and the hull-distance result moved from borderline ($p = 0.068$) to clearly non-significant ($p = 0.20$) — consistent with deliberately adding compounds with marginal or extreme thermodynamics, which dilute a rank correlation partly carried by the original population’s narrower stability range.

Sign-based discrimination (endobondic/exobondic). Experimental viability indicator (`theoretical=False` means experimentally realized):

Table 7: Endobondic/exobondic label against experimental/theoretical status.

	endobondic	exobondic
experimental ($n = 267$)	79	188
theoretical only ($n = 244$)	46	198

Fisher’s exact test: $p = 0.0054$, odds ratio = 1.81 — **significant** (endobondic is $\sim 1.8\times$ more frequent among experimentally realized compounds, the expected direction). Mann–Whitney on the continuous $\Delta\text{ICOHP}/\text{atom}$, same split: experimental median -0.868 eV, theoretical-only median -1.454 eV (more exobondic), $p = 0.0109$ — same direction, also significant. This is

exactly the result the marginal-formation-energy and max-hull-distance batches were built to test: the pooled 281-reaction population did not have enough near-stability-boundary compounds to reveal this signal ($p = 0.10$ at that scale).

Reading trap, explicitly flagged by `classify.py` itself: Table 7 shows 188 *experimental* (i.e. genuinely existing) compounds labeled *exobondic*. This is **not** a contradiction — exobondic only implies "not viable" once combined with `delta_energy` (`classify_viability()`): a compound is only `UNSTABLE_NONEXISTENT` if decomposition into elements is *both* exobondic *and* thermodynamically favorable ($\Delta E < 0$), which is rare (decomposing a real compound into pure elements is almost never favorable). Verified directly on the same population as Table 7:

Table 8: Computed `ViabilityLabel` (`classify_viability()`) on the same population as Table 7 ($n = 518$).

	experimental ($n = 267$)	theoretical ($n = 244$)
<code>STABLE_ON_HULL</code>	233	131
<code>UNSTABLE_NONEXISTENT</code>	26	95
<code>METASTABLE_VIABLE</code>	8	17

7 additional reactions (COD-sourced or otherwise `mp_id`-less, so no `formation_energy_per_atom`) excluded, insufficient data.

Collapsing `STABLE_ON_HULL+METASTABLE_VIABLE` ("viable") against `UNSTABLE_NONEXISTENT`: Fisher’s exact test $p = 2.9 \times 10^{-15}$, odds ratio **6.0** ($\chi^2 = 59.0$, $p = 1.6 \times 10^{-14}$) — **a much larger effect than the sign-only test** (Table 7, $p = 0.0054$, odds ratio 1.81). An experimentally realized compound is roughly $6\times$ less likely to be predicted `UNSTABLE_NONEXISTENT` than a theoretical-only one. **Of the 188 experimental exobondic compounds, only 26 are predicted `UNSTABLE_NONEXISTENT`** — the other 162 are `STABLE_ON_HULL`: their decomposition into elements, though exobondic, is not thermodynamically favorable. Reading Table 7 in isolation as "exobondic = not viable" is therefore misleading — it is Table 8, combining sign with thermodynamics, that actually answers the viability question the **theoretical** flag is a proxy for, and it is by far the strongest result of this step.

Table 9: Median $\Delta\text{ICOHP}/\text{atom}$ and % endobondic by stability class ($n = 177$).

Class	n	median $\Delta\text{ICOHP}/\text{atom}$ (eV)	% endobondic
experimental metastable	59	−1.038	28.8%
stable	59	−1.391	22.0%
theoretical metastable	59	−1.601	13.6%

Kruskal–Wallis across the three groups: $H = 2.56$, $p = 0.278$ — **not significant** (population unchanged at $n = 177$: the extension/marginal-formation-energy/max-hull batches carry a **theoretical** flag but not yet a **family** label, so this particular test has not benefited from the dataset’s growth), but the ordering remains exactly what physics predicts: experimentally metastable compounds have the highest endobondic fraction; never-synthesized theoretical metastable compounds have the lowest.

Step reading. The 7 reference reactions reproduce almost exactly (step 1) — the implementation is correct. Extended to the current full dataset (517 reactions, up from 281), the coarse sign-based discrimination is significant, and the full `ViabilityLabel` machinery — sign combined with thermodynamics — is far more strongly so. The continuous correlations against `formation_energy_per_atom` and `energy_above_hull` weakened over the same period — the expected price of adding chemically diverse, thermodynamically extreme compounds to what

was a narrower population; a different kind of test from the categorical results above, so the two are not in tension. Whether the `ViabilityLabel` result is itself confounded by bond type (step 4), or by the fact that max-hull’s theoretical polymorphs were deliberately selected far above the hull — a `formation_energy_per_atom`-adjacent selection effect, not necessarily an independent confirmation of the bonding-kinetics mechanism — remains to be checked.

A second, independent viability signal: mission 4b’s reaction-level antibonding descriptor, $\Delta(\text{ICOHP antibonding})$ — never previously tested against `ViabilityLabel`, only against continuous thermodynamic targets — also discriminates viable from non-viable compounds significantly, overall and within every large-enough subgroup (Mann–Whitney $p = 6.7 \times 10^{-12}$, Fisher $p = 2.4 \times 10^{-7}$, Appendix L.5).

Still no SISSO at this stage: detailed reports, `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

6.3 Step 3 — Bond-type classification from ICOHP/ICOB data

A tool in service of the next step, not an end in itself: this classification exists solely to test, in step 4, whether step 2’s viability-descriptor correlation depends on a compound’s bond type. Two independent classifications (§2.2): the `classify()` function (elemental composition + `is_metal`) and a classification built directly from each compound’s ICOHP/ICOB data (`is_metal` for metallic character; otherwise the ICobi of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within a wider cutoff, which strongly dilutes the signal for dense covalent solids such as diamond — compared against a threshold calibrated at the midpoint of the median ICobi of compounds `classify()` already labels ionic/covalent, i.e. 0.5315, to separate ionic from covalent among non-metallic compounds). A fourth label, **mixed** (Zintl-type), identifies compounds where a *homoatomic* pair above the covalent threshold (e.g. N–N in an azide, P–P in a polyphosphide, S–S in a polysulfide) coexists with a markedly weaker *heteroatomic* pair linking a distinct cation — the structural signature of Zintl–Klemm chemistry, where a single ionic/covalent label would be misleading.

Table 10: Bond type: `classify()` vs. the ICOHP/ICOB classification (concordance on the subset `classify()` labels: 276/311 = 88.7%).

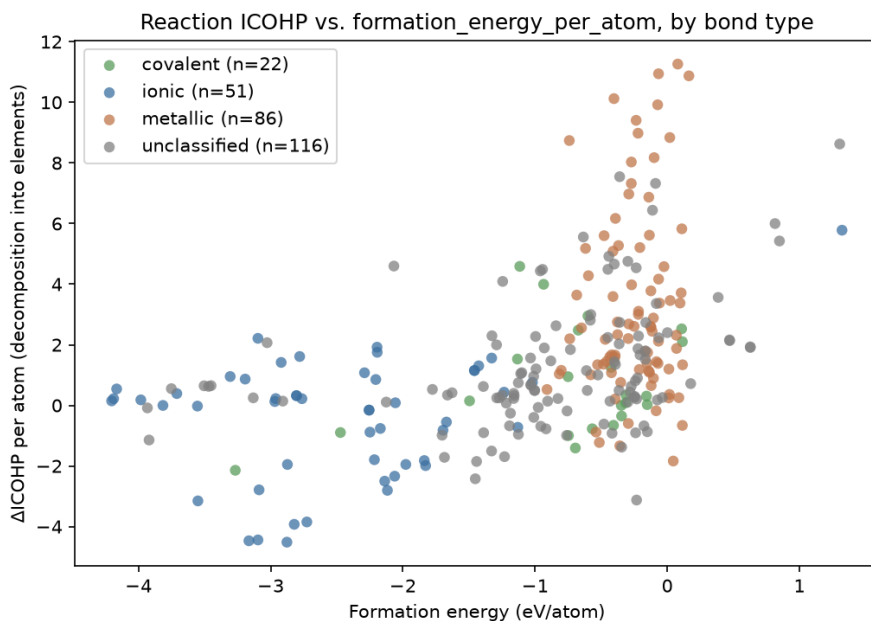
<code>classify()</code> \ ICOHP/ICOB	metallic	ionic	covalent	mixed	unclassified
metallic	193	0	0	0	0
ionic	0	48	2	3	0
covalent	0	14	35	16	0
unclassified	179	36	19	43	0
Total	372	98	56	62	0

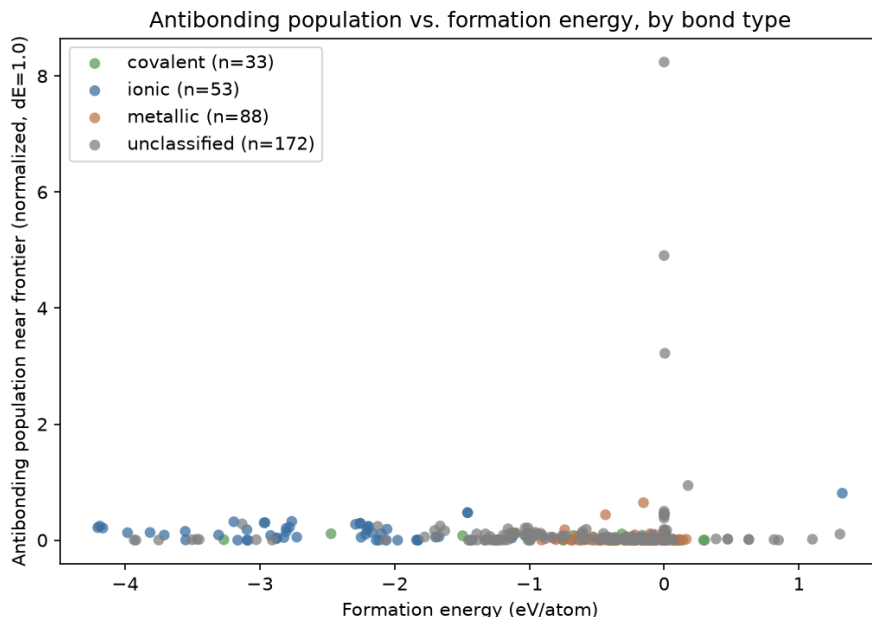
Of the 277 compounds `classify()` leaves "unclassified" (essentially alkali/alkaline-earth $\times \{\text{N, P, S}\}$ pairs outside `IONIC_ANIONS`, and metals bonded to an anion-like element that `classify()` deliberately excludes from its "metallic" bucket), the ICOHP/ICOB classification resolves **55** into ionic (36) or covalent (19), and identifies **43** as mixed (Zintl-type); the remaining 179 are metallic, already covered by `is_metal`. Where both methods render a verdict (metallic/ionic/covalent), agreement is strong (88.7%) but not total: 16 ionic/covalent disagreements, and 19 compounds `classify()` labels ionic or covalent but that the ICobi classification identifies as mixed (their dominant homoatomic bond coexists with a markedly weaker heteroatomic link) — worth checking case by case before using either classification as a stratification variable in step 4. It is the ICOHP/ICOB classification (broader coverage, built on the same data as step 2’s descriptors) that is used below, not `classify()`.

6.4 Step 4 — Does the viability-descriptor correlation depend on bond type?

Antibonding population (ICOHP). The project’s previously most robust result is **retracted**: `bond_type=covalent`, which survived clearly at $n = 33$ ($\rho = -0.551$, $p = 0.0009$), no longer survives once the sample grew to $n = 54$ ($\rho = -0.046$, $p = 0.74$) — a failure mode already seen elsewhere in this project (a subgroup result that does not replicate once the sample grows), including this same descriptor’s own earlier `bond_type=ionic` subgroup. **New at this scale**: `bond_type=mixed` (Zintl) is the best-surviving stratum ($\rho = -0.538$, $p = 6.6 \times 10^{-6}$, $n = 62$) — **strengthened** relative to step 2’s pooled correlation — a category that barely existed when the covalent result was first reported. `is_metal=False` remains the most consistently surviving stratum across this project’s successive scales ($\rho = -0.333$, $p = 6.7 \times 10^{-7}$, $n = 212$) — **strengthened**, not weakened, by the dataset’s growth. `is_metal=True` has also converged onto `bond_type=metallic` (identical $n = 367$, $\rho = 0.056$, not significant), the same phenomenon documented below for reaction ICOHP.

Reaction ICOHP. `bond_type` stratification (four categories, including `mixed`/Zintl) no longer uniformly dilutes step 2’s pooled correlation ($\rho = 0.295$): `ionic` ($\rho = 0.360$, $p = 0.0004$) and `mixed` ($\rho = -0.525$, opposite sign, $p \approx 0$) both survive clearly — **strengthened** in magnitude — `covalent` too but flips sign ($\rho = -0.354$, $p = 0.023$); only `metallic` (the majority class, 315/510 rows) stays non-significant — **weakened** to the point of disappearing in that stratum. `is_metal` stratification — the one that used to survive — has converged onto the same population as `bond_type=metallic` (identical n , ρ , p in every row) and so inherits its non-significant result: it is `bond_type`, not `is_metal`, that carries information at the current dataset composition. Against `energy_above_hull`, `bond_type=mixed` again survives on its own ($\rho = -0.307$, $p = 0.016$).





Synthesis. In both cases (antibonding and reaction ICOHP), this step’s answer is *yes, but not uniformly*: bond type does not merely weaken or strengthen a single global correlation, it changes the *sign* of the relationship depending on the stratum (ionic positive, covalent/mixed negative for reaction ICOHP) — the useful signal is conditional on bond type. The antibonding population’s covalent-result retraction is itself a direct illustration of why this step exists: a stratified number is only trustworthy once re-checked as the dataset grows, not accepted once and for all. A pooled feature search (SISSO) would therefore be actively misleading here; any future search should be run separately per `bond_type`, not on the pooled population.

7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the first 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to the whole dataset):

- **Al₁₂W, BW, WO₂, TcW, TiW** (all tungsten-containing): the local mirror’s `W_sv` POT-CAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **Al₁₂W** failed again after that fix, this time with a VASP SIGSEGV — a known issue (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`’s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 597 structures computed to date: **597/597** for VASP; **596/597** for LOBSTER (1 trivial-cell reference element with no usable LOBSTER step, §8).

Metadata bug, not a compute failure: the 4 compounds of the original 6-compound pilot campaign that are not single elements (NaCl, AlNi, LiBr, BeCu) stored their Materials Project identifier under the legacy key `material_id` instead of `mp_id`, which every later script reads — their `formation_energy_per_atom` lookups silently returned nothing even though the correct value was already present in `formation_energies.json` under the exact same material id encoded in their own directory name. Fixed by adding `mp_id` alongside the existing `material_id`;

raises the n on every table joined against formation energy that includes these 4 compounds (e.g. Table S17, $n = 434 \rightarrow 438$) with no change to any correlation’s sign or significance.

8 Compute Cost

597 structures computed in total: 69 single-element references, 522 multi-element compounds, and 6 further validation structures without compound metadata but with their own VASP+LOBSTER run (§5).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1, srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The original SLURM array (`submit_array.sh`, main campaign) is capped at 8 concurrent tasks; the batches added since (`extension/marginal-formation-energy/ max-hull/widen`) were submitted individually (`sbatch` per directory), without that cap of their own but sharing the cluster with other, unrelated workloads.

Table 11: VASP and LOBSTER compute time (597 structures).

	VASP	LOBSTER
n	597	596*
Mean (s)	188	578
Median (s)	72	157
Max (s)	5715 (1.6 h)	22555 (6.3 h)
Sequential total	31.2 h	95.8 h

* 1 of 597 structures had no usable LOBSTER step (trivial-cell reference element).

Combined sequential total (VASP + LOBSTER, 597 structures): **127.0 h**. The main campaign (capped at 8 concurrent tasks) gave an order-of-magnitude elapsed time of roughly one-eighth of the sequential total; that factor no longer applies uniformly across the whole dataset — the individually-submitted batches added since have no cap of their own but shared the cluster queue with other projects, making a single elapsed-time estimate for the full dataset not meaningful.

9 Current Limits and Next Steps

- **One compound excluded for LOBSTER quality:** `extension_Cu4Au_mp-1225761` was dropped from every analysis (but not from the raw dataset – its VASP+LOBSTER calculation is kept for reference) — `lobsterout` reports 343/343 k-points failed orthonormalization (100%, not a borderline case), and `bandOverlaps.lobster` shows a maximum deviation of 11.4 (this project’s own established red-flag threshold is 0.1). Direct consequence: physically implausible ICOHP values (third- neighbor Cu–Cu contacts at 4.25–5.04 Å with ICOHP values of 7.6–11.6 eV) and a $\Delta\text{ICOHP}/\text{atom}$ of -380.5 eV, $\sim 27\times$ the next-worst case in the rest of the population. A systematic sweep of all 597 structures (`analysis/audit_lobster_quality.py`, criterion: 100% k-point failure *and* deviation > 0.5 *and* robust z-score > 10) confirms this is a unique case, not a wider problem — the second-worst case (`Ir4W`) is already $29\times$ less extreme and plausibly real chemistry (a dense, d-electron-rich intermetallic). Every statistic in this report covering the 518 case-1 reactions has been regenerated on the remaining 517 — the effect on any given number is negligible (3rd–4th decimal place), no conclusion or significance threshold is affected.

- **classify() coverage:** 277/588 compounds (47%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth $\times$ {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§5) — purely informative here, but a real limitation for any analysis that would want to stratify on it.
- **No true dynamical "unstable" class:** `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **Per-subgroup sample size:** remains this project’s structural limit at every new stratification (`bond_type`, `is_metal`) — see §6.2 for a concrete example where an $n < 15$ subgroup did not replicate once the sample grew.
- **Why not SISSO yet:** reaction ICOHP’s signal (§6.2) is conditional on bond type and opposite-signed across strata (ionic positive, covalent/mixed negative) — a pooled feature search would be misleading here; any future search should be run separately per `bond_type`.
- **family not yet extended to the newer batches:** the theoretical-based ViabilityLabel/Fisher discrimination (§6.2) fully benefits from the dataset’s growth, but the finer three-way split (`exp_metastable/exp_stable/theo_metastable`) stays at $n = 177$ for lack of a `family` label on the extension/marginal-formation-energy/max-hull batches — the direct next step to test whether that finer split also turns significant.
- **The ViabilityLabel result (§6.2, $p = 2.9 \times 10^{-15}$) has not yet been checked for confounds:** neither stratified by `bond_type/is_metal`, nor distinguished from a selection effect (max-hull’s theoretical polymorphs were deliberately chosen far above the hull).
- **Manuscript-validation DFT+LOBSTER comparison (Appendix M), 5/16 species still computing:** Mn_2O_7 , $\text{Pb}(\text{N}_3)_2$, S_4N_2 , S_4N_4 , S_8 were still on SLURM as of this writing (2026-08-20). Of the 11 species already compared, three (Pb, MnO_2 , O_2) show open ~ 2 –10% residuals under both the manuscript’s own methodology (PBEsol+D3) and this project’s default one (PBE) alike — investigated and ruled out as instances of Zn’s shell-detection bug, left as unresolved, likely methodology-level questions (spin-orbit coupling for Pb; AFM/Hubbard-U treatment and spin-polarization for MnO_2/O_2) rather than forced to zero. Ca_3N_2 (landed 2026-08-20) showed an initial 24% residual traced to a second, distinct occurrence of Zn’s near-degenerate-shell issue and is now resolved (0.1%, Appendix M, note f).
- **Whether Zn’s shell-detection bug (Appendix M) affects any other compound’s `bond_type` classification remains open:** two quick heuristic sweeps of the full dataset’s first-shell distance spectra (2026-08-19) each flagged 40–47% of pair-groups under thresholds loose enough to be dominated by ordinary within-shell fine structure and floating-point-near-duplicate distances, not by genuine Zn-like near-degenerate double shells — inconclusive by construction, not evidence of a wide problem. Ca_3N_2 ’s independent occurrence of the same pattern (2026-08-20, four sub-shells rather than two) adds one more confirmed data point to that ground-truth set but does not itself touch `nearest_neighbor.py` or the dataset-wide classification, which remain unchanged. A calibrated audit (ground truth needed per compound, which only exists for the 16 manuscript species) is still to be done before this can be closed either way.

10 Conclusion

Reitz & Dronskowski’s endobondic/exobondic methodology (§6.2) reproduces faithfully across the 7 independently known reference reactions (Lin’s CCC 0.999998) and applies to this report’s full dataset (518 decomposition-into-elements reactions). On that population, sign-based

Δ ICOHP discrimination is significant (Fisher’s test, $p = 0.0054$), and the full `ViabilityLabel` machinery — sign combined with thermodynamics — is far more strongly so ($p = 2.9 \times 10^{-15}$, odds ratio 6.0): an experimentally realized compound is roughly $6\times$ less likely to be predicted `UNSTABLE_NONEXISTENT` than a theoretical-only one (§6.2). Reaction ICOHP otherwise remains a strong continuous correlate against formation energy ($\rho = 0.295$, §6.2), though antibonding population slightly edges it out on the subset shared by both descriptors.

Two concrete directions follow. First, checking whether the `ViabilityLabel` result above survives stratification by `bond_type/is_metal` and is not simply carried by max-hull’s deliberate selection effect — the same verification discipline this project applies systematically to every new descriptor. Second, restricting the reaction Δ (ICOHP) to a window near the Fermi level rather than the fully integrated ICOHP, modeled on the antibonding population of §6.2: two variants of this construction are detailed in Appendix L (antibonding part only, and the full signed ICOHP restricted to the window) — the first survives every stratification tested, a result no other descriptor in this report reaches.

Source code: <https://github.com/John-Akapulco/viability>

Data: `analysis/` (this report’s datasets and scripts)

Detailed reports: `analysis/REPORT_*.md`

Appendix (SI)

A Elemental Reference Structures

The 62 elemental reference structures (`analysis/compute_reaction_icohp_case1.py`’s `ELEMENT_REFERENCE`) used as the products side of every case-1 (decomposition-to-elements) reaction in this report — space group, this project’s own VASP total energy per atom (OSZICAR, extrapolated $T \rightarrow 0$ E_0), Materials Project’s own tabulated `energy_per_atom` (fetched live via the MP API), and the ICOHP/ICOHP-antibonding descriptors of the element’s own structure. The Δ column (campaign — MP, meV/atom) is **not** a DFT-accuracy check: for elements whose PAW potential differs in valence-electron convention from a default choice (semi-core electrons folded into valence, e.g. a `_d/_pv/_sv` variant — common for heavier metals), the two absolute total energies differ by up to several tens of eV/atom purely because they count a different number of explicit electrons — absolute VASP total energies are only meaningful within one consistent pseudopotential set, never across two independently chosen ones. This project’s real cross-code/cross-pipeline energetic accuracy check is §6.1 (the 7 Reitz & Dronskowski reactions, Lin’s CCC= 0.999998) and Appendix M (independent lattice-parameter/ICOHP reproduction). N has no Materials Project entry (hand-built isolated-dimer box, see §B below for the other two hand-built references, ZnSn and the CaO sphalerite polymorph).

Table S1: The 62 elemental reference structures used as the products side of every case-1 (decomposition-to-elements) reaction. $\Delta = E_{\text{campaign}} - E_{\text{MP}}$ per atom: the large offsets (up to several tens of eV/atom for heavy metals) reflect a PAW pseudopotential valence-electron convention difference (semi-core electrons explicitly included, `_d/_pv/_sv` variants), NOT a DFT-pipeline error: absolute total energies are only meaningful within one consistent pseudopotential set. This project’s real cross-code energy consistency check is §6.1 (the 7 Reitz & Dronskowski reactions, CCC=0.999998). N has no MP entry (hand-built dimer box, §B).

Element	mp-id	Space group	MP E/atom (eV)	Campaign E/atom (eV)	Δ (meV/at.)	ICOHP (eV)	IC
Ag	mp-124	Fm-3m	-21.3538	-2.7139	18639.9	-1.2496	
Al	mp-134	Fm-3m	-6.7321	-3.7530	2979.1	-1.4193	
As	mp-158	Cmce	-14.7075	-4.6474	10060.2	-3.9683	
Au	mp-81	Fm-3m	-50.5850	-3.2282	47356.7	-1.3381	
B	mp-160	R-3m	-7.3464	-6.7055	640.8	-4.7960	
Ba	mp-122	Im-3m	-1.9190	-1.9004	18.5	-0.0122	
Be	mp-87	P6 ₃ /mmc	-3.7394	-3.7668	-27.4	-1.8975	
Br	mp-23154	Cmce	-1.6369	-1.6249	12.0	-3.5362	
C	mp-48	P6 ₃ /mmc	-9.9441	-9.2252	718.9	-11.5387	
Ca	mp-21	Im-3m	-7.0327	-1.9093	5123.4	-0.0304	
Cd	mp-94	P6 ₃ /mmc	-20.1050	-0.7327	19372.3	-1.6119	
Cl	mp-22848	Cmce	-1.8485	-1.8312	17.4	-5.5728	
Co	mp-102	Fm-3m	-13.2342	-7.0237	6210.5	-1.9150	
Cr	mp-90	Im-3m	-15.0733	-9.3756	5697.8	-0.6245	
Cs	mp-1	Im-3m	-25.0891	-0.8512	24237.9	-0.1230	
Cu	mp-30	Fm-3m	-10.8468	-3.7349	7111.9	-0.2488	
F	mp-561203	C2/c	-1.9115	-1.8021	109.3	-6.9536	
Fe	mp-13	Im-3m	-8.4700	-8.2638	206.2	-0.5370	
Ga	mp-142	Cmce	-11.4428	-2.8954	8547.4	-2.6378	
Ge	mp-32	Fd-3m	-13.8720	-4.5037	9368.3	-4.6450	
H	mp-730101	P2 ₁ 2 ₁ 2 ₁	-3.4551	-3.3884	66.7	-5.1222	
Hf	mp-103	P6 ₃ /mmc	-45.1514	-9.9280	35223.4	-1.1612	
Hg	mp-10861	P6 ₃ /mmm	-49.3648	-0.1779	49186.9	-0.9652	
I	mp-23153	Cmce	-1.5240	-1.5085	15.5	-0.1504	
In	mp-1055994	I4 ₁ /mmm	-22.5741	-2.5393	20034.8	-1.1562	
Ir	mp-101	Fm-3m	-52.3594	-8.8646	43494.8	-2.4130	
K	mp-10157	Fm-3m	-6.0113	-1.0370	4974.3	-0.0808	
Li	mp-1018134	R-3m	-2.3867	-1.9056	481.1	-0.6931	
Mg	mp-153	P6 ₃ /mmc	-4.1693	-1.4360	2733.3	0.0000	
Mn	mp-35	I-43m	-14.6833	-8.8422	5841.2	-0.6082	
Mo	mp-129	Im-3m	-26.0720	-10.8984	15173.5	-1.1059	
N	—	—	—	-8.3294	—	—	
Na	mp-10172	P6 ₃ /mmc	-3.5642	-1.3111	2253.1	-0.0839	
Nb	mp-75	Im-3m	-24.7341	-10.0844	14649.6	-1.1439	
Ni	mp-23	Fm-3m	-11.9817	-5.4842	6497.5	-0.4204	
O	mp-1524462	C2/m	-5.9725	-4.8611	1111.4	-17.7956	
Os	mp-49	P6 ₃ /mmc	-52.9852	-11.2254	41759.9	-1.1579	
P	mp-568348	P2/c	-9.0021	-5.4039	3598.2	-5.0760	
Pb	mp-20483	Fm-3m	-56.2851	-3.5759	52709.3	-0.8519	
Pd	mp-2	Fm-3m	-22.9605	-5.2009	17759.6	-1.2162	
Pt	mp-126	Fm-3m	-51.4877	-6.0992	45388.6	-1.8089	
Rb	mp-70	Im-3m	-0.9713	-0.9426	28.7	-0.1180	
Re	mp-8	P6 ₃ /mmc	-52.4070	-12.3891	40017.9	-1.3803	
Rh	mp-74	Fm-3m	-24.1362	-7.2571	16879.2	-0.7081	
Ru	mp-33	P6 ₃ /mmc	-25.4245	-9.2284	16196.1	-0.9478	

Element	mp-id	Space group	MP E /atom (eV)	Campaign E /atom (eV)	Δ (meV/at.)	ICOHP (eV)	ICOBI
S	mp-77	Fddd	-8.0340	-4.1186	3915.4	-5.9246	
Sb	mp-104	R-3m	-25.7063	-4.1450	21561.3	-1.7202	
Sc	mp-67	P6 ₃ /mmc	-11.3985	-6.2416	5156.9	-0.6596	
Se	mp-14	P3 ₁ 21	-14.3105	-3.4990	10811.5	-4.1496	
Si	mp-149	—	-8.7738	-5.4220	3351.8	-4.5418	
Sn	mp-623511	I4/mmm	-24.6848	-3.7899	20894.8	-0.8803	
Sr	mp-139	P6 ₃ /mmc	-15.1457	-1.6305	13515.2	-0.0171	
Ta	mp-50	Im-3m	-48.5827	-11.8166	36766.1	-1.2794	
Tc	mp-113	P6 ₃ /mmc	-25.9647	-10.3497	15615.0	-1.2061	
Te	mp-19	P3 ₁ 21	-25.4172	-3.1382	22279.0	-1.1233	
Ti	mp-46	P6 ₃ /mmc	-12.9226	-7.8002	5122.4	-0.7536	
Tl	mp-82	P6 ₃ /mmc	-53.1793	-2.2073	50972.0	-1.0463	
V	mp-146	Im-3m	-14.0896	-8.9616	5127.9	-0.8777	
W	mp-91	Im-3m	-51.3678	-12.9998	38368.0	-2.5600	
Y	mp-112	P6 ₃ /mmc	-20.3136	-6.4179	13895.7	-0.8682	
Zn	mp-79	P6 ₃ /mmc	-8.9106	-1.0876	7823.0	-1.8892	
Zr	mp-131	P6 ₃ /mmc	-22.7498	-8.5227	14227.1	-1.3170	

B Compound and Element List

Complete list of the 588 compounds and elements with usable ICOBI data (of 597 total; 9 excluded: Cu4Au for a dataset-wide LOBSTER k-point orthonormalization failure, 2 COD-sourced compounds (S4N2, S4N4) lacking `is_metal`/ICOBI, and 6 diatomic/reference structures built for the manuscript-validation gas-phase references (Cl₂, F₂, H₂, N₂, O₂ dimer boxes, and a CaO sphalerite polymorph) that were never run through this classification, §5), organized by section according to the independent ICOHP/ICOBI classification of §5 (`is_metal` for metallic character; otherwise the ICOBI of the **dominant first-coordination-shell** species pair — not a mean over every neighbor LOBSTER reports within the wider cutoff — compared against a threshold of 0.5052 to separate ionic from covalent; a **mixed** (Zintl-type) category groups compounds where a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide ion N₃[−] ionically bound to its cation) — no “bond type” column; each section instead states its own observed ICOBI range. ICOBI and ICOHP are per-bond means (§2.4); the ICOHP/ICOBI antibonding populations are the normalized values at $\Delta E = 1.0$ eV (§6.2, §L).

B.1 Metallic

Table S2: Metallic compounds and elements (`is_metal=True`, observed ICOBI 0.0000 to 2.9506) — formula, mp-id, space group, dominant (first-shell) bond ICOBI, dominant bond ICOHP, normalized ICOHP and ICOBI antibonding populations ($\Delta E = 1.0$ eV).

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ag*	mp-124	Fm-3m	0.1545	-1.2496	0.0000	0.0000
Ag ₃ As	mp-1183193	P6 ₃ /mmc	0.2236	-1.3102	0.0000	0.0000
Ag ₃ Pt*	mp-12065	Pm-3m	0.2113	-1.5965	0.0033	0.0026
Ag ₃ Sb*	mp-2273	Pmm2	0.2406	-1.5221	0.0005	0.0000
AgI*	mp-22915	Pm-3m	0.2126	-1.1434	0.0699	0.0000
AgN*	mp-1091417	Cmce	0.4624	-3.4181	0.0088	0.0030
AgN	mp-1009766	F-43m	0.4854	-3.5762	0.0277	0.0173
AgN	mp-1424734	Ama2	0.8238	-5.9446	0.0107	0.0025

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
AgN	mp-1428312	Pm-3m	0.2177	-1.4991	0.0083	0.0041
AgPb ₃	mp-1183256	I4/mmm	0.2181	-1.2748	0.0000	0.0000
AgPd	mp-1183222	P-6m2	0.1929	-1.4704	0.0292	0.0199
Al*	mp-134	Fm-3m	0.2270	-1.4193	0.0000	0.0000
Al ₁₂ Mo*	mp-550	Im-3	0.3238	-1.5619	0.0000	0.0000
Al ₁₂ W*	mp-11227	Im-3	0.5027	-2.7492	0.0000	0.0000
Al ₂ W*	mp-12524	P6 ₂ /mm2	0.4364	-2.4880	0.0000	0.0000
Al ₃ Ag	mp-1183130	Pm-3m	0.2444	-1.4593	0.0000	0.0000
Al ₃ As	mp-1183129	Pm-3m	0.2253	-1.3902	0.0001	0.0000
Al ₃ Os ₂ *	mp-16521	I4/mmm	0.3022	-1.6476	0.0037	0.0000
Al ₄ Si	mp-1228120	R-3m	0.2614	-2.0375	0.0087	0.0030
AlAg	mp-1228901	P-6m2	0.2436	-1.7117	0.0000	0.0000
AlAg ₂	mp-985805	Cmcm	0.2723	-1.5388	0.0000	0.0000
AlNi	—	—	0.2676	-1.9848	0.0016	0.0023
AlSb	mp-1023918	Cmcm	0.2647	-1.2802	0.0000	0.0000
As*	mp-158	Cmce	0.8848	-3.9683	0.0103	0.0000
As ₃ W ₂ *	mp-11696	C2/m	0.7072	-4.3348	0.0000	0.0000
AsPd ₂ *	mp-1080831	P-62m	0.3410	-2.4185	0.0077	0.0157
AsRh*	mp-22079	Pnma	0.3323	-1.8639	0.0449	0.0158
AsRh ₂ *	mp-1102364	Pnma	0.2438	-1.3778	0.1158	0.0189
Au*	mp-81	Fm-3m	0.1962	-1.3381	0.0000	0.0000
B ₃ Ir ₂	mp-1228647	C2/m	0.4473	-3.3600	0.0047	0.0011
BPd ₅	mp-1228665	I4/mmm	0.4754	-3.9868	0.0971	0.0181
BPd ₅	mp-1228665	I4/mmm	0.4754	-3.9924	0.0973	0.0181
BW*	mp-7832	I4 ₁ /amd	0.4277	-3.9556	0.0000	0.0000
Ba*	mp-122	Im-3m	0.0002	-0.0122	8.2400	2.9573
Ba ₃ Ca	mp-1183342	I4/mmm	0.0000	0.0000	—	—
Ba ₃ N ₂	mp-1013528	Pm-3m	0.0763	-0.6612	0.1655	0.0007
BaAu ₂ *	mp-30363	P6 ₃ /mmm	0.6152	-3.0712	0.0144	0.0034
BaF ₃	mp-1183303	I4/mmm	0.0350	-0.5277	0.2805	0.2537
BaMg ₃	mp-1016218	Pmm2	0.0023	-0.0755	5.4796	1.1861
BaN ₂ *	mp-1102371	Cc	0.1605	-0.4471	0.1039	0.0214
BaSn ₅ *	mp-571353	P6 ₃ /mmm	0.2221	-0.8507	0.0456	0.0042
Be*	mp-87	P6 ₃ /mmc	0.2033	-1.8975	0.0000	0.0000
Be ₁₂ Mo*	mp-1475	I4/mmm	0.2043	-1.9103	0.0000	0.0000
Be ₁₂ Pd*	mp-12666	I4/mmm	0.2078	-1.9466	0.0000	0.0000
Be ₂ Cu*	mp-2031	Fd-3m	0.3616	-3.0863	0.0000	0.0001
Be ₂ Nb ₃ *	mp-11272	P4/mbm	0.4200	-3.5436	0.0000	0.0000
Be ₂ V*	mp-11281	P6 ₃ /mmc	0.3658	-2.7859	0.0000	0.0000
BeAu ₃	mp-983539	Pm-3m	0.2230	-1.9129	0.0000	0.0030
BeCu	—	—	0.1984	-1.8359	0.0000	0.0000
BeF ₂	mp-975644	P6 ₃ /mmm	0.1548	-1.6978	0.0863	0.0000
Bi ₂ O ₃ *	mp-558953	P2 ₁ /c	0.9517	-6.7444	0.0328	0.0038
Bi ₂ O ₃	mp-1383505	P1	0.9506	-7.1985	0.0609	0.0000
C*	mp-48	P6 ₃ /mmc	1.2195	-11.5387	0.0000	0.0000
Ca*	mp-21	Im-3m	0.0025	-0.0304	3.2206	3.3021
Ca ₃ N ₂	mp-1013524	Pm-3m	0.1107	-0.3623	0.0090	0.0000
Ca ₃ Zn	mp-1039229	I4/mmm	0.1222	-0.3874	0.2004	0.0086
CaAl ₄ *	mp-1749	I4/mmm	0.4595	-1.8701	0.0086	0.0002
CaGa ₂ *	mp-992	P6 ₃ /mmm	0.9791	-3.8887	0.0219	0.0000
CaGa ₄ *	mp-2461	C2/m	0.4641	-2.1536	0.0084	0.0001
CaGe ₂	mp-643542	C2/m	0.7379	-3.1645	0.0337	0.0038
CaN*	mp-1058549	Fm-3m	0.1201	-0.6934	0.0450	0.0656
CaNi*	mp-1059707	Pm-3m	0.0678	-0.1910	0.5038	0.0500
CaO	mp-1181903	Fd-3m	0.0725	-0.5516	0.1865	0.0756
CaSn*	mp-2450	Cmcm	0.7044	-2.1965	0.0629	0.0276

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Cd*	mp-94	P6 ₃ /mmc	0.2423	-1.6119	0.0000	0.0000
CdIn ₃	mp-1183674	Fm-3m	0.2417	-1.4735	0.0000	0.0000
CdPd*	mp-1696	P4/mmm	0.2022	-1.5904	0.0000	0.0004
CdSb ₃	mp-1183739	P6 ₃ /mmc	0.2091	-1.1803	0.0378	0.0058
Co*	mp-102	Fm-3m	0.2680	-1.9150	0.0189	0.0373
CoB*	mp-20857	Pnma	0.4568	-4.5928	0.0099	0.0000
CoGe*	mp-21237	C2/m	0.4261	-2.9354	0.0002	0.0022
CoRe	mp-1225995	P-6m2	0.3611	-2.2350	0.0086	0.0082
CoSe ₂	mp-1390196	Fd-3m	0.7048	-4.3267	0.0348	0.0079
CoSn*	mp-20536	P6 ₃ /mmm	0.4900	-2.7764	0.0035	0.0023
Cr*	mp-90	Im-3m	0.2487	-0.6245	0.0115	0.0049
Cr ₂ C	mp-1226378	P-3m1	0.5270	-3.0803	0.0084	0.0000
Cr ₅ Si ₃ *	mp-7506	I4/mcm	0.4031	-3.1831	0.0057	0.0003
CrMo	mp-1226200	Cmmm	0.3600	-1.3679	0.0081	0.0026
CrRe ₃	mp-1183703	Pm-3m	0.3358	-1.4730	0.0060	0.0000
CrW	mp-1226162	Cmmm	0.5827	-3.4455	0.0000	0.0005
Cs*	mp-1	Im-3m	0.0556	-0.1230	0.1696	0.0693
CsO ₂ *	mp-1441	I4/mmm	1.0942	-11.4627	0.1143	0.1436
CsO ₂	mp-1096936	Cmmm	1.1298	-11.0556	0.1197	0.0872
CsRb	mp-1225895	Cmmm	0.0628	-0.1628	0.1759	0.0961
Cu*	mp-30	Fm-3m	0.0456	-0.2488	0.1641	0.0946
Cu ₂ S*	mp-12087	Fm-3m	0.1495	-0.9496	0.1576	0.0012
Cu ₂ S*	mp-553942	P6 ₃ /mmc	0.0830	-0.4779	0.1504	0.0087
Cu ₃ Ge*	mp-19724	Pmmn	0.1369	-0.8340	0.0963	0.0410
Cu ₃ Te ₂	mp-1225676	P4mm	0.1480	-0.8199	0.1597	0.0663
Cu ₇ Te ₄ *	mp-624307	P3m1	0.1323	-0.6945	0.1434	0.0510
CuH	mp-1225705	P-3m1	0.0446	-0.1622	0.1097	0.0037
CuNi	mp-1184069	P-6m2	0.1198	-0.4298	0.4478	0.0732
Fe*	mp-13	Im-3m	0.1417	-0.5370	0.0922	0.0794
FeS*	mp-505531	P4/nmm	0.4967	-2.5883	0.0736	0.0024
FeTe ₂ *	mp-1102002	Pa-3	0.4279	-1.7861	0.0401	0.0000
Ga*	mp-142	Cmce	0.4308	-2.6378	0.0000	0.0000
Ga ₂ Au ₇ *	mp-571598	P-62m	0.2592	-1.8968	0.0000	0.0000
Ga ₃ Hg	mp-1184289	P6 ₃ /mmc	0.2216	-1.4162	0.0000	0.0000
Ga ₃ Pb	mp-1184425	I4/mmm	0.1929	-1.0646	0.0042	0.0000
Ga ₃ Re*	mp-1188393	P4 ₂ /mnm	0.4857	-2.3451	0.0000	0.0000
GaSb	mp-1224785	I-4m2	0.2847	-1.4795	0.0030	0.0000
Ge*	mp-32	Fd-3m	0.8956	-4.6450	0.0015	0.0000
Ge ₂ Os*	mp-10032	C2/m	0.4042	-2.1066	0.0069	0.0003
Ge ₃ Mo ₅ *	mp-17094	I4/mcm	0.4229	-3.3674	0.0068	0.0009
Ge ₃ O	mp-1184665	I4/mmm	0.1453	-0.8722	0.0091	0.0027
Ge ₄ Rh*	mp-1104286	P3 ₁ 21	0.3355	-1.8036	0.0034	0.0006
GeAs*	mp-7591	I4mm	0.4634	-2.4402	0.0063	0.0014
GeSb	mp-9935	Fm-3m	0.4373	-2.1513	0.0101	0.0039
H ₃ Pd ₄	mp-1224413	R-3m	0.1466	-1.0855	0.1005	0.0202
H ₄ N	mp-1238814	P-43m	0.6754	-6.0262	0.0498	0.0006
HC ₂	mp-995184	C2/m	1.2737	-11.4891	0.0000	0.0025
HC ₃	mp-995200	P2 ₁ /m	1.2493	-11.5739	0.0000	0.0000
HC ₆	mp-995202	P-1	1.2320	-11.5624	0.0000	0.0000
Hf*	mp-103	P6 ₃ /mmc	0.3121	-1.1612	0.0000	0.0000
Hf ₃ Pb	mp-1184772	I4/mmm	0.3286	-1.3613	0.0000	0.0000
HfCd*	mp-1007758	P4/nmm	0.3741	-1.6812	0.0000	0.0000
HfCr ₂ *	mp-569624	P6 ₃ /mmc	0.4565	-1.1713	0.0000	0.0000
HfCu ₃	mp-1184704	I4/mmm	0.1762	-0.7182	0.0196	0.0000
HfMg ₁₅	mp-1023478	P-6m2	0.0440	-0.1164	1.0039	0.4374
HfMo	mp-1224314	Cmmm	0.3368	-1.2722	0.0000	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
HfTc ₂ *	mp-1095669	P6 ₃ /mmc	0.3775	-1.4429	0.0098	0.0012
HfW ₂	mp-1224230	Fmmm	0.4943	-2.5963	0.0000	0.0000
Hg*	mp-10861	P6 ₃ /mmm	0.1622	-0.9652	0.0000	0.0000
Hg ₃ Pb	mp-1184700	P6 ₃ /mmc	0.2256	-1.4191	0.0000	0.0000
Hg ₃ Sb	mp-977242	P6 ₃ /mmc	0.2146	-1.3610	0.0001	0.0003
Hg ₄ Pt*	mp-936	Im-3m	0.5477	-3.5588	0.0000	0.0005
Hg ₅ Pd ₂ *	mp-30721	P4/mbm	0.2364	-1.7271	0.0000	0.0002
IN ₂	mp-1097011	I4/mcm	2.9506	-23.1289	0.0177	0.0275
In*	mp-1055994	I4/mmm	0.2177	-1.1562	0.0000	0.0000
In ₂ Te ₃ *	mp-1105025	I4mm	0.6032	-2.6916	0.0224	0.0090
In ₃ Ir*	mp-630976	P4 ₂ /mnm	0.5910	-3.4701	0.0000	0.0000
In ₇ Cu ₁₀ *	mp-646039	C2 ₇ /m	0.1416	-0.8072	0.0230	0.0039
InNi*	mp-20997	Pm-3m	0.2785	-1.7672	0.0081	0.0156
InSe ₂	mp-1232036	P4/nmm	0.2771	-1.4780	0.0349	0.0038
InSn ₄	mp-1223653	Cmmm	0.3867	-2.1185	0.0000	0.0000
Ir*	mp-101	Fm-3m	0.3469	-2.4130	0.0317	0.0274
Ir ₃ Ru	mp-974358	I4/mmm	0.3767	-2.6659	0.0317	0.0198
Ir ₄ W	mp-1223666	R-3m	0.3980	-2.8035	0.0422	0.0434
IrPd	mp-1223675	R-3m	0.3508	-2.9933	0.0756	0.0384
K*	mp-10157	Fm-3m	0.0430	-0.0808	0.0923	0.0499
K ₂ O	mp-1223784	P4/mmm	0.1375	-0.3763	0.0831	0.0000
K ₃ Rb	mp-976115	P6 ₃ /mmc	0.0431	-0.0882	0.1130	0.0889
K ₅ As ₄	mp-684799	P-1	0.9859	-3.9306	0.0452	0.0189
KMg ₁₅	mp-1023498	P-6m2	0.0006	-0.0094	4.8845	1.9656
KN ₂ *	mp-1071868	Cmc2 ₁	1.0490	-7.9889	0.0202	0.0000
KN ₂	mp-1180757	Cmcm	1.0210	-7.4152	0.0411	0.0000
KN ₃	mp-636056	I4/mcm	2.8711	-22.8747	0.0090	0.0107
KP*	mp-1058315	R-3m	0.1254	-0.4533	0.0190	0.0442
KP	mp-1057787	Immm	0.1070	-0.4621	0.0170	0.0218
KS ₃	mp-1185151	P6 ₃ /mmc	0.4006	-1.5431	0.0179	0.0027
Li*	mp-1018134	R-3m	0.0886	-0.6931	0.0000	0.0000
Li ₁₃ Sn ₅ *	mp-30769	P-3m1	0.6069	-2.7881	0.0000	0.0000
Li ₃ Ag	mp-976408	I4/mmm	0.1084	-1.0444	0.0000	0.0000
Li ₃ Ca	mp-976075	I4/mmm	0.1652	-0.6223	0.0000	0.0000
Li ₃ Hg*	mp-1646	Fm-3m	0.1488	-1.2194	0.0000	0.0000
LiB*	mp-1001835	P6 ₃ /mmc	1.1571	-9.9815	0.0000	0.0000
LiRh ₃ *	mp-1001787	Imm2	0.1776	-1.5461	0.0814	0.0192
LiTl ₃	mp-1185253	I4/mmm	0.2370	-1.2989	0.0000	0.0000
Mg*	mp-153	P6 ₃ /mmc	0.0000	0.0000	—	—
Mg ₁₅ B	mp-1023515	P-6m2	0.0280	-0.1152	0.9423	0.5909
Mg ₁₅ Ni	mp-1023622	P-6m2	0.0020	-0.0213	3.2054	2.1119
Mg ₂ Ge*	mp-250	Fd-3m	0.1806	-0.5039	0.1144	0.0306
Mg ₂ Ni*	mp-2137	P6 ₂ 22	0.0818	-0.2478	0.6457	0.3329
Mg ₃ Al	mp-978271	Pm-3m	0.0812	-0.3224	0.1540	0.0681
Mg ₃ Ru ₂ *	mp-625	P4 ₁ 32	0.5409	-1.6132	0.1816	0.0936
Mg ₃ Zr	mp-1094466	P6 ₃ /mmc	0.2889	-0.3975	0.0887	0.0141
Mg ₅ Sc	mp-1185837	R32	0.1000	-0.1246	0.2961	0.0068
Mg ₅ Te	mp-1185828	R32	0.0108	-0.0917	1.0311	0.5406
MgAl ₂ *	mp-1102064	I4 ₁ /amd	0.3363	-1.6067	0.0037	0.0000
MgCd*	mp-11307	Pmma	0.4362	-2.0470	0.0026	0.0006
MgIn ₂ *	mp-1077016	Fd-3m	0.4220	-1.8995	0.0118	0.0002
MgPb ₂	mp-2016042	C2 ₇ /m	0.2720	-1.0252	0.0628	0.0086
MgPd ₃ *	mp-7753	Pm-3m	0.2542	-1.7837	0.0264	0.0304
MgSn	mp-1094669	Pm-3m	0.2238	-0.7050	0.1000	0.0334
MgSn	mp-1094801	P4/mmm	0.3134	-1.2760	0.1099	0.0281
MgTi	mp-1094346	P2 ₁ /m	0.4231	-0.7659	0.0009	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
MgTl ₃	mp-978262	P6 ₃ /mmc	0.2760	-1.3397	0.0002	0.0000
Mn*	mp-35	I-43m	—	-0.6082	0.0411	0.0047
Mn ₄ B	mp-1221769	F222	0.3919	-2.3186	0.0085	0.0007
Mn ₄ Ga	mp-1221812	R-3m	0.3290	-2.4448	0.0073	0.0017
MnAl ₁₂ *	mp-1104165	Im-3	0.2798	-1.6283	0.0000	0.0000
MnO ₂ *	mp-510408	P4 ₂ /mnm	0.4433	-3.0527	0.0548	0.0603
Mo*	mp-129	Im-3m	0.3113	-1.1059	0.0139	0.0005
Mo ₂ C	mp-1221498	P-3m1	0.5408	-3.5050	0.0057	0.0002
Mo ₃ Se ₄ *	mp-21021	R-3	0.4924	-2.4124	0.0151	0.0004
MoPd ₂	mp-1206704	Immm	0.5227	-1.7521	0.0013	0.0000
MoRu	mp-1221386	P-6m2	0.3772	-1.3952	0.0182	0.0004
N ₂	mp-1059834	Imma	1.3533	-15.9343	0.0159	0.0010
NO	mp-1180334	P2 ₁₂ ₁₂ ₁	1.8010	-21.3123	0.0427	0.0039
Na*	mp-10172	P6 ₃ /mmc	0.0470	-0.0839	0.1168	0.0901
Na ₂ Cl	mp-1077015	Cmmm	0.0377	-0.1951	0.0901	0.0029
Na ₂ Cl	mp-990084	R-3m	0.0479	-0.3527	0.0618	0.0000
Na ₃ Cl	mp-1064484	P4/mmm	0.0297	-0.2254	0.0859	0.0249
Na ₃ Hg	mp-1186050	I4/mmm	0.1243	-0.3788	0.0010	0.0000
Na ₃ In	mp-864758	I4/mmm	0.1784	-0.3865	0.0055	0.0000
Na ₃ Mg	mp-982637	I4/mmm	0.0436	-0.0860	0.1585	0.0598
Na ₃ Si ₃₄	mp-1221235	R-3m	0.8887	-4.3194	0.0033	0.0052
Nb*	mp-75	Im-3m	0.3268	-1.1439	0.0000	0.0000
Nb ₃ Ir*	mp-1458	Pm-3n	0.4144	-2.0103	0.0000	0.0002
Nb ₃ Ni	mp-999394	I4/mmm	0.4256	-1.5440	0.0000	0.0002
Nb ₄ Rh	mp-1220441	Fmmm	0.3405	-1.1843	0.0002	0.0000
NbCd ₃ *	mp-11309	Pm-3m	0.2671	-1.6744	0.0000	0.0000
NbGa ₃ *	mp-1973	I4/mmm	0.3104	-1.7262	0.0000	0.0000
NbO*	mp-2692	Fm-3m	0.2933	-2.2812	0.0292	0.0023
NbO	mp-634965	P4/mmm	0.9184	-5.5839	0.0000	0.0000
NbP*	mp-9830	I4 ₁ /amd	0.3299	-1.7803	0.0001	0.0000
NbP	mp-1220317	Cmmm	0.3238	-1.7805	0.0084	0.0004
NbRh ₂	mp-1220414	Pm	0.3585	-1.4459	0.0440	0.0023
NbS ₂	mp-774873	P6 ₃ /mmc	0.6390	-3.1946	0.0049	0.0000
NbZn ₁₆ *	mp-640041	Cmcm	0.2534	-1.6346	0.0000	0.0000
Ni*	mp-23	Fm-3m	0.0947	-0.4204	0.3799	0.2037
Ni ₃ Mo*	mp-11506	Pmmn	0.2915	-1.0002	0.1370	0.0141
NiB*	mp-14019	Cmcm	0.6996	-6.1857	0.0198	0.0093
NiHg ₄ *	mp-570835	Im-3m	0.3125	-2.2179	0.0000	0.0004
NiS*	mp-1547	R3m	0.2630	-1.5895	0.1140	0.0342
NiSb ₃ *	mp-1106023	Im-3	0.3169	-1.4693	0.0547	0.0298
Os*	mp-49	P6 ₃ /mmc	0.2567	-1.1579	0.0538	0.0135
OsC*	mp-7142	P-6m2	0.4688	-3.0941	0.0402	0.0068
OsC	mp-1009537	Pm-3m	0.3353	-2.0943	0.0446	0.0000
OsC	mp-1009539	Fm-3m	0.4618	-3.1760	0.0828	0.0058
OsC	mp-999308	P6 ₃ /mmc	0.4521	-3.1583	0.0525	0.0226
PW ₃	mp-1209828	I-42m	0.4114	-2.7903	0.0000	0.0000
Pb*	mp-20483	Fm-3m	0.1943	-0.8519	0.0276	0.0000
Pb ₂ Au*	mp-22795	I4/mcm	0.4524	-3.2812	0.0000	0.0000
PbAu ₂ *	mp-19871	Fd-3m	0.3048	-2.3586	0.0000	0.0000
PbAu ₃	mp-982246	Fm-3m	0.1941	-1.4564	0.0112	0.0022
Pd*	mp-2	Fm-3m	0.1631	-1.2162	0.1264	0.0877
Pt*	mp-126	Fm-3m	0.2367	-1.8089	0.0927	0.0819
Rb*	mp-70	Im-3m	0.0590	-0.1180	0.1307	0.0377
Rb ₂ Hg ₇ *	mp-31474	P-3m1	0.1920	-0.9499	0.0000	0.0000
Rb ₃ In	mp-974948	Pm-3m	0.1129	-0.2266	0.0341	0.0000
Rb ₃ Li	mp-974938	Pm-3m	0.0991	-0.3104	0.0185	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Rb ₃ P ₁₁	mp-1173444	C2/m	0.8390	-4.2144	0.0355	0.0027
RbP*	mp-1179688	C2/m	0.1031	-0.4288	0.0138	0.0467
RbP	mp-1058499	Immm	0.1277	-0.4304	0.0099	0.0324
Re*	mp-8	P6 ₃ /mmc	0.3162	-1.3803	0.0205	0.0030
Re ₂ Si*	mp-669547	P2 ₁ /c	0.3849	-1.8091	0.0190	0.0006
Re ₃ Ge ₇ *	mp-29141	Cmcm	0.4553	-2.2942	0.0022	0.0002
Re ₃ N ₅	mp-1080194	Imm2	0.8324	-5.2882	0.0269	0.0001
Re ₃ Pd	mp-862586	P6 ₃ /mmc	0.3788	-1.7449	0.0020	0.0011
Re ₅ Si ₃ *	mp-1105959	I4/mcm	0.3379	-2.7922	0.0251	0.0018
ReC*	mp-1079494	P6 ₃ /mmc	0.4741	-3.3056	0.0211	0.0022
ReC	mp-1009731	Pm-3m	0.3588	-2.2826	0.0149	0.0003
Rh*	mp-74	Fm-3m	0.1620	-0.7081	0.1852	0.0924
Ru*	mp-33	P6 ₃ /mmc	0.2375	-0.9478	0.0875	0.0211
RuC*	mp-10030	P-6m2	0.4770	-2.9147	0.0418	0.0078
RuC	mp-1009787	Fm-3m	0.4752	-2.9734	0.0862	0.0067
RuRh ₃	mp-974335	Pm-3m	0.2074	-0.8611	0.1382	0.0326
SN	mp-1078816	P2 ₁ /c	0.2823	-1.7903	0.0359	0.0006
SN	mp-1173453	P2 ₁	1.5195	-10.9244	0.0458	0.0000
SN	mp-684642	P2 ₁	1.3504	-13.4315	0.0333	0.0038
Sb*	mp-104	R-3m	0.4398	-1.7202	0.0425	0.0063
Sb ₂ S ₅	mp-1359089	P2 ₁ /c	1.0848	-6.8891	0.0690	0.0097
Sc*	mp-67	P6 ₃ /mmc	0.2110	-0.6596	0.0000	0.0000
Sc ₅ Ge ₃ *	mp-17190	P6 ₃ /mcm	0.3162	-1.5145	0.0000	0.0000
ScAg ₂ *	mp-1018128	I4/mmm	0.2014	-1.5516	0.0000	0.0000
ScOs ₃	mp-981384	Pm-3m	0.2618	-1.1903	0.0252	0.0106
ScTc ₃	mp-867262	Fm-3m	0.3043	-1.0739	0.0080	0.0010
ScTe*	mp-10026	P6 ₃ /mmc	0.3365	-1.5673	0.0000	0.0000
Si ₃ Pt	mp-972763	P6 ₃ /mmc	0.4284	-2.6580	0.0034	0.0000
Si ₃ W ₅ *	mp-31219	I4/mcm	0.4312	-2.5598	0.0000	0.0000
SiAu ₃	mp-1186998	Fm-3m	0.1974	-1.4121	0.0156	0.0000
SiO ₂	mp-556044	P-31c	0.7437	-6.4399	0.0117	0.0000
SiO ₂	mp-638049	F4 ₁₃₂	0.6594	-6.0556	0.0488	0.0000
SiRh*	mp-2287	P2 ₁ /c	0.2801	-1.6004	0.0243	0.0009
SiTc ₃	mp-972784	P6 ₃ /mmc	0.3385	-1.4573	0.0257	0.0020
SiTe ₂ *	mp-2755	P-3m1	0.5658	-2.7523	0.0695	0.0130
Sn*	mp-623511	I4/mmm	0.1989	-0.8803	0.0235	0.0006
Sn*	mp-117	Fd-3m	0.8707	-3.6930	0.0001	0.0004
Sn ₃ S	mp-1187038	Fm-3m	0.2241	-1.1171	0.0176	0.0000
Sn ₃ Se	mp-1187055	Fm-3m	0.2108	-1.0119	0.0214	0.0027
Sn ₇ Ir ₅ *	mp-20466	I4mm	0.5375	-3.2800	0.0000	0.0000
Sn ₇ Pd ₅	mp-1219006	C2	0.3765	-2.7926	0.0000	0.0004
SnP*	mp-475	Fm-3m	0.4347	-2.3226	0.0105	0.0058
SnPb ₃	mp-1187118	P6 ₃ /mmc	0.1988	-0.8714	0.0218	0.0000
SnPt*	mp-19856	P6 ₃ /mmc	0.4612	-3.2535	0.0000	0.0000
Si*	mp-139	P6 ₃ /mmc	0.0009	-0.0171	4.9050	5.3659
Sr ₂ As*	mp-15697	I4/mmm	0.0787	-0.2746	0.2097	0.0267
Sr ₃ N ₂	mp-1013529	Pm-3m	0.1064	-0.4148	0.0858	0.0000
Sr ₃ Sn	mp-1187136	Pm-3m	0.0501	-0.2374	0.4377	0.0368
Sr ₅ Sn ₃ *	mp-17720	I4/mcm	0.8182	-2.1022	0.1796	0.0000
SrAl ₂ *	mp-1638	Fd-3m	0.4263	-1.6649	0.0004	0.0000
SrB ₂	mp-1008755	P6 ₃ /mmm	0.9937	-5.2175	0.0052	0.0007
SrN*	mp-1058586	Fm-3m	0.1230	-0.6070	0.0462	0.0849
SrO ₁₀	mp-1208655	P4/mmm	0.8181	-6.3571	0.1157	0.0000
SrSi*	mp-2661	Cmcm	0.8077	-2.8883	0.0564	0.0097
SrTl ₃	mp-1206636	Pm-3m	0.2857	-1.2068	0.0201	0.0007
Ta*	mp-50	Im-3m	0.3226	-1.2794	0.0000	0.0000

Formula	mp-id	Space group	ICObI	ICOHP (eV)	ICOHP antibonding	ICObI antibonding
Ta ₄ C ₃	mp-1218000	R3m	0.5265	-3.5440	0.0000	0.0000
Ta ₅ Ge ₃ *	mp-17593	I4/mcm	0.3556	-2.6739	0.0000	0.0000
Ta ₇ Co ₆ *	mp-1104548	R-3m	0.4891	-3.0211	0.0018	0.0000
TaBe ₁₂ *	mp-567842	I4/mmm	0.2080	-1.8875	0.0000	0.0000
TaN*	mp-1497	P6/mmm	0.4241	-2.9806	0.0009	0.0000
TaN	mp-1009831	Pm-3m	0.3185	-2.1377	0.0002	0.0002
TaN	mp-1009833	F-43m	0.6451	-4.4476	0.0000	0.0006
TaP*	mp-1067587	I4_1md	0.4762	-2.7480	0.0000	0.0000
TaP	mp-1187244	P-6m2	0.4792	-2.7509	0.0000	0.0000
TaRe	mp-1217894	Cmmm	0.3311	-1.5114	0.0026	0.0003
TaSb ₂ *	mp-11697	C2/m	0.4899	-2.0334	0.0002	0.0000
Tc*	mp-113	P6_3/mmc	0.3130	-1.2061	0.0240	0.0006
Tc ₃ Mo	mp-1187395	Pm-3m	0.3332	-1.2727	0.0068	0.0014
Tc ₃ Os	mp-862779	P6_3/mmc	0.3182	-1.2186	0.0252	0.0052
Tc ₃ W	mp-1187398	P6_3/mmc	0.4723	-2.3705	0.0000	0.0000
TcW	mp-1217337	Cmmm	0.6035	-3.6836	0.0000	0.0002
Te ₂ Au*	mp-567525	C2/m	0.5036	-3.0627	0.0497	0.0327
Te ₂ Ir*	mp-1551	C2/m	0.7621	-4.4689	0.0230	0.0046
Te ₂ Pd*	mp-782	P-3m1	0.5813	-3.5060	0.0291	0.0200
Te ₂ Pd ₇	mp-1217443	R-3m	0.2769	-1.8731	0.0363	0.0145
Ti*	mp-46	P6_3/mmc	0.3010	-0.7536	0.0000	0.0000
Ti ₂ Cu*	mp-742	I4/mmm	0.3374	-0.7591	0.0016	0.0000
Ti ₃ Ge*	mp-1189044	I-4	0.3243	-1.6051	0.0000	0.0000
Ti ₃ In*	mp-866046	P6_3/mmc	0.3128	-1.1764	0.0000	0.0000
Ti ₃ Nb	mp-1217091	Amm2	0.3774	-1.2750	0.0000	0.0000
Ti ₃ Pt*	mp-2339	Pm-3n	0.3632	-1.7698	0.0000	0.0000
TiBe ₁₂ *	mp-1104067	I4/mmm	0.2186	-1.9904	0.0000	0.0000
TiMo ₂	mp-1216675	Fmmm	0.3395	-1.1740	0.0000	0.0000
TiMo ₂	mp-1216675	Fmmm	0.3395	-1.1740	0.0000	0.0000
TiNi*	mp-1048	P2_1/m	0.2818	-0.8685	0.0059	0.0000
TiNi	mp-1067248	Cmcm	0.2798	-0.8586	0.0060	0.0000
TiO*	mp-1071163	P-62m	0.3594	-2.4547	0.0015	0.0000
TiSi ₂ *	mp-1077503	Cmcm	0.4599	-3.0700	0.0000	0.0000
TiSi ₂	mp-570991	Immm	0.2659	-1.2939	0.0000	0.0000
TiW	mp-1216621	Cmmm	0.6568	-3.6207	0.0000	0.0000
Tl*	mp-82	P6_3/mmc	0.1905	-1.0463	0.0000	0.0000
Tl ₃ Au	mp-1080569	Pm-3n	0.2291	-1.2850	0.0000	0.0000
Tl ₃ Ga	mp-1187539	Fm-3m	0.2090	-1.0914	0.0000	0.0000
Tl ₄ Sn	mp-1216560	R-3m	0.2039	-1.3837	0.0010	0.0008
TlAs	mp-1007770	F-43m	0.7362	-3.5581	0.0000	0.0002
TlAu ₃	mp-1187508	Pm-3m	0.2130	-1.5377	0.0000	0.0000
TlC*	mp-567465	Fm-3m	0.3776	-2.4265	0.0000	0.0010
TlC	mp-1058887	P4/mmm	0.4837	-3.1370	0.0000	0.0003
TlN*	mp-1017982	P6_3mc	0.6066	-3.6824	0.0023	0.0000
TlN	mp-1002222	Fm-3m	0.3446	-2.1516	0.0029	0.0000
TlN	mp-1007778	F-43m	0.6105	-3.6815	0.0017	0.0000
V*	mp-146	Im-3m	0.3178	-0.8777	0.0000	0.0000
VC ₃	mp-1095057	P6_3/mmc	0.3730	-2.5141	0.0012	0.0023
VN	mp-1017532	P6_3/mmc	0.4293	-2.6954	0.0000	0.0000
VNi ₂ *	mp-11531	Immm	0.4769	-1.4029	0.0918	0.0008
W*	mp-91	Im-3m	0.4372	-2.5600	0.0000	0.0001
W ₉ C ₄	mp-684989	R32	0.6031	-4.5942	0.0000	0.0000
WBr ₅	mp-1436789	P-1	0.8263	-4.3695	0.0774	0.0061
WC*	mp-13136	Fm-3m	0.5508	-4.2240	0.0012	0.0009
WC	mp-1008630	Pm-3m	0.4127	-2.9246	0.0000	0.0010
WC	mp-1008635	F-43m	0.8710	-6.2790	0.0000	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
WO ₂ *	mp-1524394	P2 ₁ /c	0.6713	-5.7408	0.0039	0.0000
Y*	mp-112	P6 ₃ /mmc	0.2374	-0.8682	0.0000	0.0000
Y ₃ Co*	mp-1105598	Pnma	0.4720	-2.6094	0.0000	0.0000
Y ₃ Co	mp-1189329	P2 ₁	0.4718	-2.6045	0.0000	0.0000
YAg ₃	mp-1187811	Fm-3m	0.1780	-1.2749	0.0000	0.0000
Zn*	mp-79	P6 ₃ /mmc	0.2296	-1.8892	0.0000	0.0000
Zn	mp-2647117	Im-3m	0.1819	-1.3641	0.0000	0.0000
Zn ₁₃ Rh*	mp-13447	C2/m	0.1989	-1.5615	0.0000	0.0000
Zn ₂ Ag	mp-1093999	P6 ₃ /mmc	0.2898	-2.4762	0.0000	0.0000
Zn ₃ Au	mp-1188000	P6 ₃ /mmc	0.2275	-1.8469	0.0000	0.0000
Zn ₃ N ₂	mp-1524551	P6 ₃ /mmc	0.3888	-2.8616	0.0030	0.0000
Zn ₃ N ₂ *	mp-9460	Ia-3	0.4180	-3.1067	0.0036	0.0000
Zn ₇ Tc	mp-1206378	Fm-3m	0.2749	-1.7542	0.0000	0.0009
ZnAg*	mp-1912	Pm-3m	0.2035	-1.8265	0.0000	0.0000
ZnPb ₃	mp-1187949	Fm-3m	0.2392	-1.3227	0.0004	0.0004
ZnSn	—	P6 ₃ /mmc	0.4604	-3.0247	0.0000	0.0000
Zr*	mp-131	P6 ₃ /mmc	0.3223	-1.3170	0.0000	0.0000
Zr ₂ Au*	mp-2348	I4/mmm	0.3095	-2.0203	0.0000	0.0000
Zr ₃ Be	mp-1188016	I4/mmm	0.3655	-1.7390	0.0000	0.0000
Zr ₃ Fe*	mp-31205	Cmcm	0.4128	-1.4259	0.0000	0.0000
Zr ₃ Ge	mp-1188039	P6 ₃ /mmc	0.3460	-1.7196	0.0000	0.0000
Zr ₅ Pb ₃ *	mp-681992	P6 ₃ /mcm	0.3301	-1.9739	0.0000	0.0000
ZrBr*	mp-1065846	C2/m	0.3528	-2.0185	0.0000	0.0001
ZrGa ₃ *	mp-30685	I4/mmm	0.2933	-1.8532	0.0000	0.0000
ZrHg ₃ *	mp-1608	Pm-3m	0.2490	-1.9086	0.0000	0.0000
ZrMo ₂ *	mp-2049	Fd-3m	0.4671	-1.6863	0.0002	0.0000
ZrRh*	mp-2808	Pm-3m	0.3062	-1.4732	0.0094	0.0024
ZrTi	mp-1215236	P4/mmm	0.3777	-1.6592	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

B.2 Ionic

Table S3: Ionic compounds (non-metallic, ICOBI < 0.5315, observed range 0.0444 to 0.5305) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
AgH	mp-1096804	P6 ₃ mc	0.2039	-1.5451	0.0000	0.0000
AgO*	mp-1079720	P2 ₁ /c	0.2604	-2.0408	0.0297	0.0024
AlH ₃ *	mp-1106356	Fd-3m	0.4074	-2.4374	0.0000	0.0000
B*	mp-160	R-3m	0.5203	-4.7960	0.0000	0.0000
B ₆ Pb	mp-1096825	Pm-3m	0.2353	-1.9376	0.0000	0.0000
Ba ₄ Pd	mp-1214438	Fd-3m	0.4601	-3.3443	0.1949	0.0075
BaCl ₂ *	mp-568662	Fm-3m	0.0749	-0.5008	0.2998	0.0066
BaCl ₂ *	mp-23199	Pnma	0.0637	-0.4308	0.3027	0.0019
BaF ₃ *	mp-1080821	P2 ₁ /m	0.2279	-0.7883	0.3169	0.0437
BaH ₂ *	mp-23715	Pnma	0.0615	-0.2536	0.1495	0.0000
BaO*	mp-1008500	P6 ₃ /mmc	0.0788	-0.6380	0.1468	0.0016
BaO	mp-1950989	P6 ₃ /mmc	0.1070	-0.4367	0.2225	0.0000
Be ₃ N ₂ *	mp-6977	P6 ₃ /mmc	0.4361	-4.2580	0.0000	0.0000
Be ₃ N ₂ *	mp-18337	Ia-3	0.4759	-4.7015	0.0000	0.0000
Be ₃ N ₂	mp-1017550	P-3m1	0.3952	-3.8753	0.0000	0.0000
BeCl ₂	mp-1172909	I-43m	0.3697	-0.9454	0.8095	0.1761
BeF ₂ *	mp-15951	P3 ₁ 21	0.5269	-6.8585	0.0035	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
BeO*	mp-7599	P4_2/mnm	0.5148	-5.7846	0.0000	0.0000
BeO	mp-1778	F-43m	0.5232	-5.8507	0.0000	0.0000
Ca ₃ N ₂ *	mp-1047	R-3c	0.1781	-0.6015	0.1048	0.0000
Ca ₃ N ₂ *	mp-844	Ia-3	0.1820	-0.6211	0.0837	0.0000
CaCl ₂ *	mp-23214	Pnnm	0.1054	-0.7540	0.2048	0.0008
CaCl ₂	mp-1120739	Fm-3m	0.0814	-0.5708	0.3236	0.0068
CaF ₂ *	mp-10464	Pnma	0.0565	-0.5566	0.2424	0.0000
CaF ₂	mp-554355	P4/mmm	0.0591	-0.5440	0.1286	0.0000
CaO*	mp-1079707	Cmce	0.1008	-0.5793	0.1805	0.0000
CaO*	mp-2605	Fm-3m	0.1143	-0.7028	0.0897	0.0000
CdI ₂ *	mp-567259	P-3m1	0.4859	-2.4996	0.0508	0.0000
Cs ₂ S*	mp-1077814	P6_3/mmc	0.0804	-1.0011	0.0682	0.0000
Cs ₂ Se*	mp-1011696	Pnma	0.0811	-0.6189	0.0861	0.0000
CsCl*	mp-573697	Fm-3m	0.0595	-0.3453	0.0490	0.0000
CsF*	mp-8455	Pm-3m	0.1224	-1.5102	0.0523	0.0008
CsH*	mp-632319	Pm-3m	0.0935	-1.2305	0.0678	0.0002
GeS	mp-12910	Cmcm	0.4348	-2.2462	0.0516	0.0000
GeTe*	mp-938	R3m	0.4101	-1.7409	0.0714	0.0021
H ₂ O*	mp-2739191	C222_1	0.4372	-4.6426	0.0087	0.0000
HI*	mp-697025	C2/c	0.4915	-2.7099	0.0522	0.0000
HgF ₂ *	mp-8177	Fm-3m	0.2320	-1.8744	0.1660	0.0000
I*	mp-23153	Cmce	0.0444	-0.1504	0.4981	0.0900
IF ₇ *	mp-27988	Aea2	0.5305	-5.5662	0.0568	0.0000
InBr*	mp-22870	Cmcm	0.3010	-1.3292	0.0929	0.0024
InCl*	mp-571636	Cmc2_1	0.1698	-0.8368	0.0921	0.0006
K ₂ O*	mp-971	Fm-3m	0.1009	-0.8814	0.0059	0.0000
K ₂ S*	mp-559278	P6_3/mmc	0.1040	-0.4891	0.0071	0.0006
K ₂ S*	mp-1022	Fm-3m	0.1212	-0.4744	0.0000	0.0000
KCl*	mp-23289	Pm-3m	0.0510	-0.4078	0.1374	0.0000
KF ₃ *	mp-1544557	P2_12_12_1	0.3162	-0.9256	0.4749	0.3545
KF ₃	mp-2749823	P2_1	0.3119	-0.9244	0.4733	0.3514
KI*	mp-22898	Fm-3m	0.0893	-0.4549	0.0615	0.0000
Li ₂ O*	mp-1960	Fm-3m	0.2393	-2.0320	0.0000	0.0000
Li ₂ O	mp-755894	Pnma	0.2225	-2.0369	0.0000	0.0000
Li ₂ S*	mp-1125	Pnma	0.2847	-2.0194	0.0000	0.0000
Li ₂ S	mp-557142	P6_3/mmc	0.3327	-2.1900	0.0000	0.0000
Li ₂ Se*	mp-2286	Fm-3m	0.3399	-2.1702	0.0000	0.0000
Li ₃ N*	mp-2341	P6_3/mmc	0.2587	-1.8972	0.0000	0.0000
LiBr	—	—	0.2790	-2.0381	0.0000	0.0000
LiCl*	mp-22905	Fm-3m	0.2558	-2.0554	0.0000	0.0000
LiCl	mp-1185319	P6_3mc	0.3879	-2.9688	0.0000	0.0000
LiF*	mp-1138	Fm-3m	0.1692	-1.9334	0.0001	0.0000
LiF	mp-1009009	Pm-3m	0.1237	-1.3411	0.0249	0.0000
MgCl ₂ *	mp-570259	P-3m1	0.1312	-0.9334	0.2905	0.0028
MgCl ₂ *	mp-23210	R-3m	0.1313	-0.9344	0.2883	0.0009
MgCl ₂	mp-569626	Ama2	0.1475	-1.0096	0.1898	0.0001
MgF ₂ *	mp-1072956	Pnnm	0.0901	-1.0192	0.1330	0.0000
MgF ₂	mp-1062842	Amm2	0.0808	-0.8465	0.1522	0.0000
MgF ₂	mp-560236	C2/m	0.1007	-1.0813	0.0878	0.0000
MgH ₂ *	mp-23711	Pbcn	0.1156	-0.5595	0.0617	0.0000
MgO*	mp-549706	P6_3mc	0.1572	-1.2025	0.0812	0.0000
MgO	mp-1009127	Pm-3m	0.0813	-0.6164	0.2735	0.0000
MgS*	mp-1315	Fm-3m	0.1236	-0.7426	0.2438	0.0000
MgS	mp-13032	F-43m	0.2006	-1.1035	0.1624	0.0000
Na ₂ O*	mp-2352	Fm-3m	0.1191	-0.6209	0.0000	0.0000
Na ₂ O	mp-1975297	C2/m	0.1297	-0.6320	0.0003	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
NaCl	—	—	0.0859	-0.5953	0.1111	0.0000
NaS*	mp-409	P-62m	0.4976	-2.6241	0.1126	0.0000
NaS	mp-1180106	C2/m	0.2104	-0.9839	0.0000	0.0000
PbSe*	mp-22009	Fmm2	0.4169	-1.6445	0.0576	0.0000
PbSe*	mp-2201	Fm-3m	0.3457	-1.4608	0.0807	0.0039
Rb ₂ O*	mp-1394	Fm-3m	0.0867	-1.2998	0.0368	0.0000
Rb ₂ O	mp-2023482	P-3m1	0.1302	-0.4067	0.0636	0.0000
Rb ₃ P	mp-974822	Fm-3m	0.1253	-0.5327	0.0666	0.0000
RbCl*	mp-23299	Pm-3m	0.0478	-0.6732	0.1019	0.0000
RbF*	mp-2064	Pm-3m	0.0879	-1.3700	0.0447	0.0036
RbF	mp-11718	Fm-3m	0.0460	-0.4536	0.0398	0.0000
RbI*	mp-22903	Fm-3m	0.0840	-0.4162	0.0534	0.0013
RbS	mp-1179684	C2/m	0.1226	-0.5861	0.0850	0.0092
Sc ₂ O ₃	mp-755313	R-3c	0.4210	-2.7698	0.0026	0.0000
SiO ₂ *	mp-9258	Pa-3	0.5130	-5.5969	0.0264	0.0000
SiO ₂	mp-10064	Fm-3m	0.3974	-3.7939	0.0240	0.0000
SnBr ₂	mp-978985	P4 ₂ /mnm	0.3595	-1.6332	0.1289	0.0000
SrF ₂ *	mp-1102240	Pnma	0.0589	-0.5401	0.2090	0.0000
SrF ₂	mp-1102911	Pnma	0.0505	-0.4858	0.2166	0.0000
SrS*	mp-10627	Pm-3m	0.0762	-0.4641	0.2376	0.0000
Te*	mp-19	P3 ₂ 121	0.3172	-1.1233	0.1197	0.0127
Te ₂ Ru*	mp-267	Pnnm	0.4723	-2.2556	0.0486	0.0017
TiO ₂ *	mp-430	P2 ₁ /c	0.5087	-3.0819	0.0145	0.0000
TlCl*	mp-571079	Cmcm	0.1717	-0.8409	0.1079	0.0000
Y ₂ O ₃ *	mp-558573	C2/m	0.4214	-2.9843	0.0000	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

B.3 Covalent

Table S4: Covalent compounds (non-metallic, ICOBI ≥ 0.5315 , observed range 0.5501 to 1.9431) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
As ₂ S ₃ *	mp-1189572	P-1	0.9136	-4.7416	0.0538	0.0000
As ₄ S ₃ *	mp-27543	Pnma	0.9714	-5.2731	0.0137	0.0000
As ₄ S ₃ *	mp-557321	Pnma	0.9717	-5.2816	0.0101	0.0000
AsF ₃ *	mp-28027	Pna2 ₁	0.6951	-6.3410	0.1114	0.0000
Au ₂ S*	mp-947	Pn-3m	0.6297	-4.8036	0.0181	0.0000
BI ₃ *	mp-23189	P6 ₃ /m	1.2727	-6.8061	0.0232	0.0000
Be ₃ P ₂ *	mp-567841	Ia-3	0.5637	-3.8282	0.0000	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.5501	-3.7626	0.0000	0.0002
Be ₃ P ₂ *	mp-567841	Ia-3	0.5637	-3.8282	0.0000	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.5501	-3.7626	0.0000	0.0002
BeCl ₂ *	mp-23267	Ibam	0.7165	-5.6352	0.0021	0.0000
BeCl ₂	mp-1190080	I-43m	0.7528	-5.9613	0.0026	0.0000
Bi ₂ O ₃	mp-714859	P1	1.1110	-10.1326	0.0646	0.0001
Br*	mp-23154	Cmce	0.7827	-3.5362	0.4257	0.0097
C	mp-1080826	C2/m	0.9446	-9.5768	0.0000	0.0000
C	mp-1190171	Pnma	0.9445	-9.5798	0.0000	0.0000
C*	mp-66	Fd-3m	0.9497	-9.6602	0.0000	0.0000
C*	mp-47	P6 ₃ /mmc	0.9482	-9.7832	0.0000	0.0000
C	—	—	1.2173	-11.6891	0.0000	0.0000
C ₃ N ₄ *	mp-571653	P-43m	0.9531	-10.4956	0.0150	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
C ₃ N ₄	mp-567885	P-6m2	1.2498	-13.1075	0.0143	0.0000
C ₃ N ₄ *	mp-570572	R3m	1.2498	-13.0884	0.0134	0.0000
C ₃ N ₄	mp-971683	P3_1	1.2479	-13.2995	0.0096	0.0000
C ₃ N ₄	mp-1182484	P6_3/m	1.9179	-15.3586	0.0000	0.0000
C ₃ N ₄	mp-2852	I-43d	0.9575	-10.9099	0.0106	0.0000
C ₃ N ₄	mp-563	Fd-3m	0.7047	-7.1477	0.0086	0.0000
Cl ₂ *	mp-22848	Cmce	0.9254	-5.5728	0.4555	0.0003
ClF ₃ *	mp-556767	Pnma	0.6390	-5.2173	0.1347	0.0000
F ₂ *	mp-561203	C2/c	0.9731	-6.9536	0.4569	0.0000
Ga ₂ Se ₃	mp-1224809	Imm2	0.8324	-4.7105	0.0085	0.0000
GeSe ₂ *	mp-10074	I-42d	0.9011	-4.9682	0.0171	0.0000
H ₂ *	mp-730101	P2_12_12_1	0.9954	-5.1222	0.0000	0.0000
H ₂ CN ₂ *	mp-703527	Pbca	1.9431	-16.8585	0.0000	0.0000
H ₃ N*	mp-643432	P2_12_12_1	0.8816	-8.0639	0.0000	0.0000
HC ₂ N ₃	mp-1103408	Cmc2_1	0.9561	-10.8107	0.0045	0.0000
HCl	mp-684609	Imm2	0.9480	-7.3985	0.0478	0.0001
IrCl ₃	mp-729507	C2	0.7112	-4.9995	0.0957	0.0000
Mn ₂ O ₇ *	mp-28338	P2_1/c	1.3609	-6.7339	0.0257	0.0000
MnN	mp-1009130	F-43m	0.6682	-3.9792	0.0247	0.0062
MoSe ₂ *	mp-1634	P6_3/mmc	0.6411	-2.8467	0.0035	0.0000
Nb ₂ O ₅ *	mp-1595	C2/m	0.7548	-4.9296	0.0075	0.0000
O ₂ *	mp-1524462	C2/m	1.4816	-17.7956	0.1271	0.2186
P*	mp-568348	P2/c	0.9202	-5.0760	0.0209	0.0000
P ₄ Se ₅ *	mp-2447	Pna2_1	0.9440	-5.0941	0.0348	0.0000
PS*	mp-612	C2/c	0.9840	-5.8992	0.0301	0.0000
PtCl ₂	mp-1219748	C2/m	0.7183	-5.3013	0.0756	0.0000
S*	mp-77	Fddd	0.9692	-5.9246	0.0880	0.0000
SN*	mp-236	P2_1/c	1.1480	-8.7413	0.0440	0.0000
Se*	mp-14	P3_121	0.9053	-4.1496	0.0828	0.0000
Si	—	—	0.9243	-4.5418	0.0000	0.0000
SiO ₂ *	mp-546794	I-42d	0.7466	-7.9383	0.0071	0.0000
SiS ₂ *	mp-1095294	P2_1/c	0.9089	-5.7585	0.0044	0.0000
TeO ₂ *	mp-8377	P2_12_12_1	0.6215	-4.7653	0.0772	0.0000
TiO ₂ *	mp-1439	Pbcn	0.5838	-3.5796	0.0077	0.0000
TiO ₂ *	mp-9173	Pnma	0.5897	-3.5580	0.0123	0.0000
ZrO ₂	mp-775909	P4_2/mnm	0.5720	-4.1446	0.0017	0.0000

Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

B.4 Mixed (Zintl-type)

Table S5: Mixed-bonding (Zintl-type) compounds (a strongly covalent homoatomic pair coexists with a markedly weaker heteroatomic pair, e.g. the azide N₃[−] ion ionically bound to its cation; observed range 0.8490 to 2.7322) — same columns.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
Ba ₅ P ₄ *	mp-17566	Pnma	0.9040	-3.4927	0.2104	0.0000
BaP ₁₀ *	mp-28035	Cmc2_1	0.9058	-4.8297	0.0394	0.0000
BaS ₃ *	mp-556296	P2_12_12	0.9201	-5.2850	0.1084	0.0000
Bi ₂ O ₃	mp-674644	P1	0.9332	-7.5456	0.0131	0.0000
C(NO) ₂ *	mp-1197051	P2_12_12_1	1.7617	-16.7609	0.0043	0.0000
C ₂ N ₂ O	mp-1214250	P6_3/m	2.4576	-22.4067	0.0003	0.0000

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
CaN ₆	mp-656097	C2	1.8852	-17.7987	0.0004	0.0000
CaN ₆ *	mp-676	Fddd	1.8841	-18.1451	0.0003	0.0000
CdO ₂ *	mp-2310	Pa-3	0.9738	-7.1680	0.0540	0.0000
CdP ₄ *	mp-7904	P2 ₁ /c	0.9223	-5.1380	0.0094	0.0000
Cs ₂ As ₃ *	mp-15557	Fmmm	1.0431	-4.4248	0.0386	0.0000
CsN ₃ *	mp-510557	I4/mcm	1.8888	-17.7624	0.0000	0.0000
CsN ₃	mp-1225892	P4/mmm	1.8983	-17.5725	0.0000	0.0000
CsN ₃ *	mp-510557	I4/mcm	1.8888	-17.7624	0.0000	0.0000
CsN ₃	mp-1225892	P4/mmm	1.8983	-17.5725	0.0000	0.0000
CsP ₇ *	mp-1201790	Pbcm	0.9343	-4.9186	0.0062	0.0000
CsP ₇ *	mp-1201790	Pbcm	0.9343	-4.9186	0.0062	0.0000
CsP ₇	mp-1213206	Pca2 ₁	0.9336	-4.9318	0.0064	0.0000
CsP ₇	mp-1213206	Pca2 ₁	0.9336	-4.9318	0.0064	0.0000
H ₂ CN ₅	mp-1212563	C2/c	1.2926	-13.4719	0.0040	0.0000
H ₉ CN ₉ *	mp-1202455	P-1	1.2740	-12.5502	0.0042	0.0000
HC	mp-995197	C2/m	1.2662	-10.8409	0.0000	0.0000
HN	mp-1224501	P-1	0.8490	-8.0912	0.0231	0.0000
HN*	mp-23922	Pmna	1.8882	-19.1499	0.0000	0.0000
HN*	mp-720515	P2 ₁ /c	1.5928	-15.9550	0.0000	0.0000
KN ₃ *	mp-827	I4/mcm	1.8889	-18.4973	0.0000	0.0000
KN ₃ *	mp-827	I4/mcm	1.8889	-18.4973	0.0000	0.0000
KP ₁₅	mp-1211471	P-1	0.9260	-5.0473	0.0164	0.0000
LiN ₃ *	mp-2659	C2/m	1.8523	-20.2194	0.0000	0.0000
LiP ₅ *	mp-2412	Pna2 ₁	0.9037	-4.9376	0.0026	0.0000
LiP ₅ *	mp-2412	Pna2 ₁	0.9037	-4.9376	0.0026	0.0000
LiP ₅	mp-32760	Pna2 ₁	0.8771	-4.6811	0.0064	0.0000
LiP ₅	mp-32760	Pna2 ₁	0.8771	-4.6811	0.0064	0.0000
LiS ₄	mp-1143546	P1	0.9663	-5.6937	0.0304	0.0000
MgP ₄ *	mp-384	P2 ₁ /c	0.9196	-5.0216	0.0273	0.0000
N ₂ O*	mp-1180473	Pnma	2.3033	-22.3518	0.0013	0.0000
NaN ₃ *	mp-1066400	R3m	1.8853	-18.8113	0.0000	0.0000
NaN ₃ *	mp-22003	R-3m	1.8872	-19.1626	0.0002	0.0000
NaN ₃ *	mp-1066400	R3m	1.8853	-18.8113	0.0000	0.0000
NaN ₃ *	mp-22003	R-3m	1.8872	-19.1626	0.0002	0.0000
NaN ₃	mp-634410	C2	1.8691	-20.2972	0.0000	0.0000
NaN ₃	mp-1065265	C2/m	2.7322	-23.2128	0.0006	0.0215
NaP ₁₅ *	mp-1194653	P-1	0.9227	-5.0620	0.0186	0.0000
NaP ₅ *	mp-31086	Pnma	0.9111	-4.9558	0.0285	0.0000
NaS*	mp-2400	P6 ₃ /mmc	0.9510	-4.7236	0.1279	0.0000
PI ₂ *	mp-29443	P-1	0.9586	-4.8005	0.1081	0.0000
PNO	mp-675742	P-1	1.1751	-12.3013	0.0406	0.0000
PbN ₆ *	mp-620058	Pna2 ₁	1.8686	-19.1852	0.0008	0.0000
PbN ₆ *	mp-667338	Pnma	1.8687	-19.2227	0.0007	0.0000
RbN ₃ *	mp-581833	P4/mmm	1.8969	-18.3403	0.0000	0.0000
RbN ₃	mp-1219567	Amm2	1.3756	-9.8572	0.0216	0.0000
RbP ₁₅	mp-1209298	P-1	0.9270	-5.0385	0.0156	0.0000
RbS*	mp-558071	Immm	0.9920	-5.0554	0.1964	0.0019
Si ₃ H	mp-995193	P-3m1	0.9370	-4.6600	0.0000	0.0000
SiH ₂ *	mp-1192532	P2 ₁ /m	0.9857	-4.7720	0.0000	0.0000
Sr ₃ As ₄	mp-678007	P1	0.8522	-3.0556	0.1161	0.0000
Sr ₃ P ₄ *	mp-14288	Fdd2	0.8910	-4.1353	0.1224	0.0000
Sr ₃ P ₄	mp-682953	P1	0.8889	-4.0705	0.1250	0.0000
SrN ₆ *	mp-2131	Fddd	1.8857	-18.4334	0.0000	0.0000
SrO ₂ *	mp-2697	I4/mmm	0.9835	-7.0573	0.1841	0.0000
SrO ₂ *	mp-726705	Pnma	0.9821	-6.8700	0.2389	0.0000

SrO ₂	mp-2763179	Pnma	0.9805	-6.9738	0.2327	0.0000
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Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

B.5 Unclassified

Table S6: Unclassified compounds (no ICOBI data available) — same columns, ICOBI/antibonding shown as “–” where missing.

Formula	mp-id	Space group	ICOBI	ICOHP (eV)	ICOHP antibonding	ICOBI antibonding
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Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ICOBI/ICOHP: value of the dominant first-shell species pair (not a mean over every LOBSTER-reported neighbor, see §2.4).

C Complete Case-1 Reaction List

Every one of the project’s 517 case-1 (decomposition-to-elements) reactions (§6.2) — a superset of Appendices J–K, which cover only the 208 compounds of the “extension” campaign. For each: the compound (starred formula, mp-id, space group), the balanced reaction with the space group of **every** elemental reference in parentheses, the **ViabilityLabel** (`reaction_analysis.classify.classify_viability()`, §6.2), $\Delta(\text{ICOHP})/\text{atom}$ (mission 5, this project’s main reaction-ICOHP descriptor) and $\Delta(\text{ICOHP antibonding})/\text{atom}$ (mission 4b, §L — “–” where the compound or one of its element references lacks the full `COHPCAR.lobster` trace this descriptor needs, unlike the unfiltered `ICOHPLIST.lobster` sum the $\Delta(\text{ICOHP})$ column uses).

Table S7: The project’s complete 517-reaction case-1 (decomposition-to-elements) population (a superset of Tables H/I, which are restricted to the “extension” campaign, 208 compounds) — compound (starred formula, mp-id, space group), balanced reaction with each elemental reference’s space group in parentheses, viability label (`classify_viability()`), $\Delta(\text{ICOHP})/\text{atom}$ (mission 5) and $\Delta(\text{ICOHP antibonding})/\text{atom}$ (mission 4b, “–” where the full COHP trace is missing for the compound or one of its references).

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP antibonding})/\text{at.}$
Ag ₃ As	mp-1183193	P6 ₃ /mmc	Ag ₃ As → Ag (Fm-3m) + 0.3333 As (Cmce)	unstable/nonexistent	-6.0568	
Ag ₃ Pt*	mp-12065	Pm-3m	Ag ₃ Pt → Ag (Fm-3m) + 0.3333 Pt (Fm-3m)	stable	-9.5571	
Ag ₃ Sb*	mp-2273	Pmm2	Ag ₃ Sb → Ag (Fm-3m) + 0.3333 Sb (R-3m)	unstable/nonexistent	-4.7976	
AgH	mp-1096804	P6 ₃ mc	AgH → Ag (Fm-3m) + 0.5 H ₂ (P2 ₁ 2 ₁ 2 ₁)	unstable/nonexistent	-3.0026	
AgI*	mp-22915	Pm-3m	AgI → Ag (Fm-3m) + I (Cmce)	stable	-2.3171	
AgN*	mp-1091417	Cmce	AgN → Ag (Fm-3m) + 0.5 N ₂ (–)	unstable/nonexistent	-5.1257	
AgN	mp-1009766	F-43m	AgN → Ag (Fm-3m) + 0.5 N ₂ (–)	unstable/nonexistent	-4.0338	
AgN	mp-1424734	Ama2	AgN → Ag (Fm-3m) + 0.5 N ₂ (–)	unstable/nonexistent	-5.6289	
AgN	mp-1428312	Pm-3m	AgN → Ag (Fm-3m) + 0.5 N ₂ (–)	unstable/nonexistent	-2.8940	
AgO*	mp-1079720	P2 ₁ /c	AgO → Ag (Fm-3m) + 0.5 O ₂ (C2/m)	stable	-4.6619	
AgPb ₃	mp-1183256	I4/mmm	AgPb ₃ → Ag (Fm-3m) + 3 Pb (Fm-3m)	unstable/nonexistent	-4.9377	
AgPd	mp-1183222	P-6m2	AgPd → Ag (Fm-3m) + Pd (Fm-3m)	stable	-1.4597	
Al ₁₂ Mo*	mp-550	Im-3	Al ₁₂ Mo → Al (Fm-3m) + 0.08333 Mo (Im-3m)	stable	-9.1319	
Al ₁₂ W*	mp-11227	Im-3	Al ₁₂ W → Al (Fm-3m) + 0.08333 W (Im-3m)	stable	-9.9194	
Al ₂ W*	mp-12524	P6 ₃ /222	Al ₂ W → Al (Fm-3m) + 0.5 W (Im-3m)	stable	-12.0597	
Al ₃ Ag	mp-1183130	Pm-3m	Al ₃ Ag → Al (Fm-3m) + 0.3333 Ag (Fm-3m)	unstable/nonexistent	-8.0729	
Al ₃ As	mp-1183129	Pm-3m	Al ₃ As → Al (Fm-3m) + 0.3333 As (Cmce)	unstable/nonexistent	-6.5408	
Al ₃ Os ₂ *	mp-16521	I4/mmm	Al ₃ Os ₂ → Al (Fm-3m) + 0.6667 Os (P6 ₃ /mmc)	stable	-4.2790	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
Al ₄ Si	mp-1228120	R-3m	Al ₄ Si $\rightarrow$ Al (Fm-3m) + 0.25 Si (-)	metastable viable	0.6603	
AlAg	mp-1228901	P-6m2	AlAg $\rightarrow$ Al (Fm-3m) + Ag (Fm-3m)	stable	-4.5528	
AlAg ₂	mp-985805	Cmcm	AlAg ₂ $\rightarrow$ Al (Fm-3m) + 2 Ag (Fm-3m)	stable	-8.1396	
AlH ₃ *	mp-1106356	Fd-3m	AlH ₃ $\rightarrow$ Al (Fm-3m) + 1.5 H ₂ (P2 ₁ 2 ₁ 2 ₁)	stable	-2.2803	
AlNi	-	-	AlNi $\rightarrow$ Al (Fm-3m) + Ni (Fm-3m)	insufficient data	3.0197	
AlSb	mp-1023918	Cmcm	AlSb $\rightarrow$ Al (Fm-3m) + Sb (R-3m)	stable	-4.1664	
As ₂ S ₃ *	mp-1189572	P-1	As ₂ S ₃ $\rightarrow$ As (Cmce) + 1.5 S (Fddd)	stable	-0.3064	
As ₃ W ₂ *	mp-11696	C2/m	As ₃ W ₂ $\rightarrow$ As (Cmce) + 0.6667 W (Im-3m)	stable	-3.7140	
As ₄ S ₃ *	mp-27543	Pnma	As ₄ S ₃ $\rightarrow$ As (Cmce) + 0.75 S (Fddd)	stable	-0.1669	
As ₄ S ₃ *	mp-557321	Pnma	As ₄ S ₃ $\rightarrow$ As (Cmce) + 0.75 S (Fddd)	stable	-0.1538	
AsF ₃ *	mp-28027	Pna2 ₁	AsF ₃ $\rightarrow$ As (Cmce) + 1.5 F ₂ (C2/c)	stable	0.8953	
AsPd ₂ *	mp-1080831	P-62m	AsPd ₂ $\rightarrow$ As (Cmce) + 2 Pd (Fm-3m)	stable	-2.7380	
AsRh*	mp-22079	Pnma	AsRh $\rightarrow$ As (Cmce) + Rh (Fm-3m)	stable	-1.0028	
AsRh ₂ *	mp-1102364	Pnma	AsRh ₂ $\rightarrow$ As (Cmce) + 2 Rh (Fm-3m)	stable	-2.2416	
Au ₂ S*	mp-947	Pn-3m	Au ₂ S $\rightarrow$ Au (Fm-3m) + 0.5 S (Fddd)	stable	-9.9543	
B ₃ Ir ₂	mp-1228647	C2/m	B ₃ Ir ₂ $\rightarrow$ B (R-3m) + 0.6667 Ir (Fm-3m)	stable	-6.4391	
B ₆ Pb	mp-1096825	Pm-3m	B ₆ Pb $\rightarrow$ B (R-3m) + 0.1667 Pb (Fm-3m)	unstable/nonexistent	-2.1019	
Bl ₃ *	mp-23189	P6 ₃ /m	Bl ₃ $\rightarrow$ B (R-3m) + 3 I (Cmce)	stable	0.0006	
BPd ₅	mp-1228665	I4/mmm	BPd ₅ $\rightarrow$ B (R-3m) + 5 Pd (Fm-3m)	stable	-7.3287	
BPd ₅	mp-1228665	I4/mmm	BPd ₅ $\rightarrow$ B (R-3m) + 5 Pd (Fm-3m)	stable	-7.3244	
BW*	mp-7832	I4 ₁ /amd	BW $\rightarrow$ B (R-3m) + W (Im-3m)	stable	-7.5458	
Ba ₃ Ca	mp-1183342	I4/mmm	Ba ₃ Ca $\rightarrow$ Ba (Im-3m) + 0.3333 Ca (Im-3m)	unstable/nonexistent	-0.2097	
Ba ₃ N ₂	mp-1013528	Pm-3m	Ba ₃ N ₂ $\rightarrow$ Ba (Im-3m) + 0.3333 N ₂ (-)	unstable/nonexistent	-2.1233	
Ba ₄ Pd	mp-1214438	Fd-3m	Ba ₄ Pd $\rightarrow$ Ba (Im-3m) + 0.25 Pd (Fm-3m)	stable	-0.5460	
Ba ₅ P ₄ *	mp-17566	Pnma	Ba ₅ P ₄ $\rightarrow$ Ba (Im-3m) + 0.8 P (P2/c)	stable	-0.9590	
BaAu ₂ *	mp-30363	P6 ₃ /mmm	BaAu ₂ $\rightarrow$ Ba (Im-3m) + 2 Au (Fm-3m)	stable	-8.7392	
BaCl ₂ *	mp-568662	Fm-3m	BaCl ₂ $\rightarrow$ Ba (Im-3m) + Cl ₂ (Cmce)	stable	-0.1270	
BaCl ₂ *	mp-23199	Pnma	BaCl ₂ $\rightarrow$ Ba (Im-3m) + Cl ₂ (Cmce)	stable	-0.2244	
BaF ₃ *	mp-1080821	P2 ₁ /m	BaF ₃ $\rightarrow$ Ba (Im-3m) + 1.5 F ₂ (C2/c)	stable	-0.8680	
BaF ₃	mp-1183303	I4/mmm	BaF ₃ $\rightarrow$ Ba (Im-3m) + 1.5 F ₂ (C2/c)	stable	-0.2493	
BaH ₂ *	mp-23715	Pnma	BaH ₂ $\rightarrow$ Ba (Im-3m) + H ₂ (P2 ₁ 2 ₁ 2 ₁)	stable	-0.4935	
BaMg ₃	mp-1016218	Pmm2	BaMg ₃ $\rightarrow$ Ba (Im-3m) + 3 Mg (P6 ₃ /mmc)	metastable viable	0.0718	
BaN ₂ *	mp-1102371	Cc	BaN ₂ $\rightarrow$ Ba (Im-3m) + N ₂ (-)	unstable/nonexistent	-5.3424	
BaO*	mp-1008500	P6 ₃ /mmc	BaO $\rightarrow$ Ba (Im-3m) + 0.5 O ₂ (C2/m)	stable	-0.3221	
BaO	mp-1950989	P6 ₃ /mmc	BaO $\rightarrow$ Ba (Im-3m) + 0.5 O ₂ (C2/m)	stable	-1.6129	
BaP ₁₀ *	mp-28035	Cmc2 ₁	BaP ₁₀ $\rightarrow$ Ba (Im-3m) + 10 P (P2/c)	stable	-0.3674	
BaS ₃ *	mp-556296	P2 ₁ 2 ₁ 2 ₁	BaS ₃ $\rightarrow$ Ba (Im-3m) + 3 S (Fddd)	stable	-0.6233	
BaSn ₅ *	mp-571353	P6 ₃ /mmm	BaSn ₅ $\rightarrow$ Ba (Im-3m) + 5 Sn (I4/mmm)	stable	-3.9787	
Be ₁₂ Mo*	mp-1475	I4/mmm	Be ₁₂ Mo $\rightarrow$ Be (P6 ₃ /mmc) + 0.08333 Mo (Im-3m)	stable	-2.1027	
Be ₁₂ Pd*	mp-12666	I4/mmm	Be ₁₂ Pd $\rightarrow$ Be (P6 ₃ /mmc) + 0.08333 Pd (Fm-3m)	stable	-2.1177	
Be ₂ Cu*	mp-2031	Fd-3m	Be ₂ Cu $\rightarrow$ Be (P6 ₃ /mmc) + 0.5 Cu (Fm-3m)	stable	-1.1356	
Be ₂ Nb ₃ *	mp-11272	P4/mbm	Be ₂ Nb ₃ $\rightarrow$ Be (P6 ₃ /mmc) + 1.5 Nb (Im-3m)	stable	-5.6162	
Be ₂ V*	mp-11281	P6 ₃ /mmc	Be ₂ V $\rightarrow$ Be (P6 ₃ /mmc) + 0.5 V (Im-3m)	stable	-3.0021	
Be ₃ N ₂ *	mp-6977	P6 ₃ /mmc	Be ₃ N ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	2.2271	
Be ₃ N ₂ *	mp-18337	Ia-3	Be ₃ N ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	1.8948	
Be ₃ N ₂	mp-1017550	P-3m1	Be ₃ N ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	2.3655	
Be ₃ P ₂ *	mp-567841	Ia-3	Be ₃ P ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.6667 P (P2/c)	stable	-0.2111	
Be ₃ P ₂	mp-1084771	Pn-3m	Be ₃ P ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.6667 P (P2/c)	stable	-0.1795	
Be ₃ P ₂ *	mp-567841	Ia-3	Be ₃ P ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.6667 P (P2/c)	stable	-0.2111	
Be ₃ P ₂	mp-1084771	Pn-3m	Be ₃ P ₂ $\rightarrow$ Be (P6 ₃ /mmc) + 0.6667 P (P2/c)	stable	-0.1795	
BeAu ₃	mp-983539	Pm-3m	BeAu ₃ $\rightarrow$ Be (P6 ₃ /mmc) + 3 Au (Fm-3m)	unstable/nonexistent	-11.2624	
BeCl ₂ *	mp-23267	Ibam	BeCl ₂ $\rightarrow$ Be (P6 ₃ /mmc) + Cl ₂ (Cmce)	stable	1.8204	
BeCl ₂	mp-1172909	I-43m	BeCl ₂ $\rightarrow$ Be (P6 ₃ /mmc) + Cl ₂ (Cmce)	unstable/nonexistent	-5.7807	
BeCl ₂	mp-1190080	I-43m	BeCl ₂ $\rightarrow$ Be (P6 ₃ /mmc) + Cl ₂ (Cmce)	stable	1.9871	
BeCu	-	-	BeCu $\rightarrow$ Be (P6 ₃ /mmc) + Cu (Fm-3m)	insufficient data	5.0371	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
BeF ₂ *	mp-15951	P3_121	BeF ₂ → Be (P6_3/mmc) + F ₂ (C2/c)	stable	3.1514	
BeF ₂	mp-975644	P6/mmm	BeF ₂ → Be (P6_3/mmc) + F ₂ (C2/c)	stable	-0.6816	
BeO*	mp-7599	P4_2/mnm	BeO → Be (P6_3/mmc) + 0.5 O ₂ (C2/m)	stable	2.7864	
BeO	mp-1778	F-43m	BeO → Be (P6_3/mmc) + 0.5 O ₂ (C2/m)	stable	4.4391	
C(NO) ₂ *	mp-1197051	P2_12_12_1	C(NO) ₂ → C (P6_3/mmc) + N ₂ (-) + O ₂ (C2/m)	metastable viable	2.1720	
C ₂ N ₂ O	mp-1214250	P6_3/m	C ₂ N ₂ O → C (P6_3/mmc) + 0.5 N ₂ (-) + 0.25 O ₂ (C2/m)	stable	1.3607	
C ₃ N ₄ *	mp-571653	P-43m	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	metastable viable	3.9344	
C ₃ N ₄	mp-567885	P-6m2	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	metastable viable	2.9033	
C ₃ N ₄ *	mp-570572	R3m	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	metastable viable	2.8864	
C ₃ N ₄	mp-971683	P3_1	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	metastable viable	3.2292	
C ₃ N ₄	mp-1182484	P6_3/m	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	unstable/nonexistent	-0.8264	
C ₃ N ₄	mp-2852	I-43d	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	metastable viable	4.6592	
C ₃ N ₄	mp-563	Fd-3m	C ₃ N ₄ → C (P6_3/mmc) + 0.6667 N ₂ (-)	metastable viable	2.3637	
Ca ₃ N ₂ *	mp-1047	R-3c	Ca ₃ N ₂ → Ca (Im-3m) + 0.3333 N ₂ (-)	stable	-2.5186	
Ca ₃ N ₂ *	mp-844	Ia-3	Ca ₃ N ₂ → Ca (Im-3m) + 0.3333 N ₂ (-)	stable	-2.4679	
Ca ₃ N ₂	mp-1013524	Pm-3m	Ca ₃ N ₂ → Ca (Im-3m) + 0.3333 N ₂ (-)	stable	-2.5203	
Ca ₃ Zn	mp-1039229	I4/mmm	Ca ₃ Zn → Ca (Im-3m) + 0.3333 Zn (P6_3/mmc)	unstable/nonexistent	-3.1150	
CaAl ₄ *	mp-1749	I4/mmm	CaAl ₄ → Ca (Im-3m) + 4 Al (Fm-3m)	stable	-8.9809	
CaCl ₂ *	mp-23214	Pnmm	CaCl ₂ → Ca (Im-3m) + Cl ₂ (Cmce)	stable	-0.3199	
CaCl ₂	mp-1120739	Fm-3m	CaCl ₂ → Ca (Im-3m) + Cl ₂ (Cmce)	stable	-0.2208	
CaF ₂ *	mp-10464	Pnma	CaF ₂ → Ca (Im-3m) + F ₂ (C2/c)	stable	-0.2227	
CaF ₂	mp-554355	P4/mmm	CaF ₂ → Ca (Im-3m) + F ₂ (C2/c)	stable	-0.1814	
CaGa ₂ *	mp-992	P6/mmm	CaGa ₂ → Ca (Im-3m) + 2 Ga (Cmce)	stable	-1.0111	
CaGa ₄ *	mp-2461	C2/m	CaGa ₄ → Ca (Im-3m) + 4 Ga (Cmce)	stable	-1.0771	
CaGe ₂	mp-643542	C2/m	CaGe ₂ → Ca (Im-3m) + 2 Ge (Fd-3m)	stable	-1.4507	
CaN*	mp-1058549	Fm-3m	CaN → Ca (Im-3m) + 0.5 N ₂ (-)	stable	-2.4564	
CaN ₆	mp-656097	C2	CaN ₆ → Ca (Im-3m) + 3 N ₂ (-)	stable	1.4493	
CaN ₆ *	mp-676	Fddd	CaN ₆ → Ca (Im-3m) + 3 N ₂ (-)	stable	1.6982	
CaNi*	mp-1059707	Pm-3m	CaNi → Ca (Im-3m) + Ni (Fm-3m)	unstable/nonexistent	-1.9253	
CaO*	mp-1079707	Cmce	CaO → Ca (Im-3m) + 0.5 O ₂ (C2/m)	stable	-2.2127	
CaO*	mp-2605	Fm-3m	CaO → Ca (Im-3m) + 0.5 O ₂ (C2/m)	stable	-0.9525	
CaO	mp-1181903	Fd-3m	CaO → Ca (Im-3m) + 0.5 O ₂ (C2/m)	stable	-2.9971	
CaSn*	mp-2450	Cmcm	CaSn → Ca (Im-3m) + Sn (I4/mmm)	stable	-2.5569	
CdI ₂ *	mp-567259	P-3m1	CdI ₂ → Cd (P6_3/mmc) + 2 I (Cmce)	stable	-0.2263	
CdIn ₃	mp-1183674	Fm-3m	CdIn ₃ → Cd (P6_3/mmc) + 3 In (I4/mmm)	unstable/nonexistent	-6.8774	
CdO ₂ *	mp-2310	Pa-3	CdO ₂ → Cd (P6_3/mmc) + O ₂ (C2/m)	stable	-2.6279	
CdP ₄ *	mp-7904	P2_1/c	CdP ₄ → Cd (P6_3/mmc) + 4 P (P2/c)	stable	-0.5117	
CdPd*	mp-1696	P4/mmm	CdPd → Cd (P6_3/mmc) + Pd (Fm-3m)	stable	-6.1699	
CdSb ₃	mp-1183739	P6_3/mmc	CdSb ₃ → Cd (P6_3/mmc) + 3 Sb (R-3m)	unstable/nonexistent	-1.7559	
ClF ₃ *	mp-556767	Pnma	ClF ₃ → Cl ₂ (Cmce) + 3 F ₂ (C2/c)	stable	0.6884	
CoB*	mp-20857	Pnma	CoB → Co (Fm-3m) + B (R-3m)	stable	-4.7534	
CoGe*	mp-21237	C2/m	CoGe → Co (Fm-3m) + Ge (Fd-3m)	stable	-2.9619	
CoRe	mp-1225995	P-6m2	CoRe → Co (Fm-3m) + Re (P6_3/mmc)	unstable/nonexistent	-1.0394	
CoSe ₂	mp-1390196	Fd-3m	CoSe ₂ → Co (Fm-3m) + 2 Se (P3_121)	stable	-0.6308	
CoSn*	mp-20536	P6/mmm	CoSn → Co (Fm-3m) + Sn (I4/mmm)	stable	-5.0955	
Cr ₂ C	mp-1226378	P-3m1	Cr ₂ C → Cr (Im-3m) + 0.5 C (P6_3/mmc)	stable	-0.2590	
Cr ₅ Si ₃ *	mp-7506	I4/mcm	Cr ₅ Si ₃ → Cr (Im-3m) + 0.6 Si (-)	stable	0.7253	
CrMo	mp-1226200	Cmmm	CrMo → Cr (Im-3m) + Mo (Im-3m)	unstable/nonexistent	-1.3397	
CrRe ₃	mp-1183703	Pm-3m	CrRe ₃ → Cr (Im-3m) + 3 Re (P6_3/mmc)	unstable/nonexistent	-3.4325	
CrW	mp-1226162	Cmmm	CrW → Cr (Im-3m) + W (Im-3m)	unstable/nonexistent	-4.0598	
Cs ₂ As ₃ *	mp-15557	Fmmm	Cs ₂ As ₃ → Cs (Im-3m) + 1.5 As (Cmce)	stable	-0.1386	
Cs ₂ S*	mp-1077814	P6_3/mmc	Cs ₂ S → Cs (Im-3m) + 0.5 S (Fddd)	stable	0.9554	
Cs ₂ Se*	mp-1011696	Pnma	Cs ₂ Se → Cs (Im-3m) + 0.5 Se (P3_121)	stable	0.6734	
CsCl*	mp-573697	Fm-3m	CsCl → Cs (Im-3m) + 0.5 Cl ₂ (Cmce)	stable	0.8830	
CsF*	mp-8455	Pm-3m	CsF → Cs (Im-3m) + 0.5 F ₂ (C2/c)	stable	3.8464	
CsH*	mp-632319	Pm-3m	CsH → Cs (Im-3m) + 0.5 H ₂ (P2_12_12_1)	stable	3.1233	
CsN ₃ *	mp-510557	I4/mcm	CsN ₃ → Cs (Im-3m) + 1.5 N ₂ (-)	stable	1.4025	
CsN ₃	mp-1225892	P4/mmm	CsN ₃ → Cs (Im-3m) + 1.5 N ₂ (-)	stable	2.0264	
CsN ₃ *	mp-510557	I4/mcm	CsN ₃ → Cs (Im-3m) + 1.5 N ₂ (-)	stable	1.4025	
CsN ₃	mp-1225892	P4/mmm	CsN ₃ → Cs (Im-3m) + 1.5 N ₂ (-)	stable	2.0264	
CsO ₂ *	mp-1441	I4/mmm	CsO ₂ → Cs (Im-3m) + O ₂ (C2/m)	stable	0.0666	
CsO ₂	mp-1096936	Cmmm	CsO ₂ → Cs (Im-3m) + O ₂ (C2/m)	stable	0.7743	
CsP ₇ *	mp-1201790	Pbcm	CsP ₇ → Cs (Im-3m) + 7 P (P2/c)	stable	-0.2978	
CsP ₇ *	mp-1201790	Pbcm	CsP ₇ → Cs (Im-3m) + 7 P (P2/c)	stable	-0.2978	
CsP ₇	mp-1213206	Pca2_1	CsP ₇ → Cs (Im-3m) + 7 P (P2/c)	stable	-0.2667	
CsP ₇	mp-1213206	Pca2_1	CsP ₇ → Cs (Im-3m) + 7 P (P2/c)	stable	-0.2667	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
CsRb	mp-1225895	Cmmm	$\text{CsRb} \rightarrow \text{Cs (Im-3m)} + \text{Rb (Im-3m)}$	unstable/nonexistent	-0.0049	
Cu ₂ S*	mp-12087	Fm-3m	$\text{Cu}_2\text{S} \rightarrow \text{Cu (Fm-3m)} + 0.5 \text{ S (Fddd)}$	stable	-1.1548	
Cu ₂ S*	mp-553942	P6 ₃ /mmc	$\text{Cu}_2\text{S} \rightarrow \text{Cu (Fm-3m)} + 0.5 \text{ S (Fddd)}$	unstable/nonexistent	-1.3350	
Cu ₃ Ge*	mp-19724	Pmmn	$\text{Cu}_3\text{Ge} \rightarrow \text{Cu (Fm-3m)} + 0.3333 \text{ Ge (Fd-3m)}$	unstable/nonexistent	-1.1983	
Cu ₃ Te ₂	mp-1225676	P4mm	$\text{Cu}_3\text{Te}_2 \rightarrow \text{Cu (Fm-3m)} + 0.6667 \text{ Te (P3}_{121})$	stable	-0.4549	
Cu ₇ Te ₄ *	mp-624307	P3m1	$\text{Cu}_7\text{Te}_4 \rightarrow \text{Cu (Fm-3m)} + 0.5714 \text{ Te (P3}_{121})$	unstable/nonexistent	-0.8332	
CuH	mp-1225705	P-3m1	$\text{CuH} \rightarrow \text{Cu (Fm-3m)} + 0.5 \text{ H}_2 \text{ (P2}_{12121})$	unstable/nonexistent	-0.4439	
CuNi	mp-1184069	P-6m2	$\text{CuNi} \rightarrow \text{Cu (Fm-3m)} + \text{Ni (Fm-3m)}$	unstable/nonexistent	-1.2229	
FeS*	mp-505531	P4/nmm	$\text{FeS} \rightarrow \text{Fe (Im-3m)} + \text{S (Fddd)}$	stable	0.1833	
FeTe ₂ *	mp-1102002	Pa-3	$\text{FeTe}_2 \rightarrow \text{Fe (Im-3m)} + 2 \text{ Te (P3}_{121})$	stable	0.9187	
Ga ₂ Au ₇ *	mp-571598	P-62m	$\text{Ga}_2\text{Au}_7 \rightarrow \text{Ga (Cmce)} + 3.5 \text{ Au (Fm-3m)}$	stable	-11.0643	
Ga ₂ Se ₃	mp-1224809	Imm2	$\text{Ga}_2\text{Se}_3 \rightarrow \text{Ga (Cmce)} + 1.5 \text{ Se (P3}_{121})$	stable	1.4066	
Ga ₃ Hg	mp-1184289	P6 ₃ /mmc	$\text{Ga}_3\text{Hg} \rightarrow \text{Ga (Cmce)} + 0.3333 \text{ Hg (P6/mmm)}$	unstable/nonexistent	-1.4503	
Ga ₃ Pb	mp-1184425	I4/mmm	$\text{Ga}_3\text{Pb} \rightarrow \text{Ga (Cmce)} + 0.3333 \text{ Pb (Fm-3m)}$	unstable/nonexistent	-2.8017	
Ga ₃ Re*	mp-1188393	P4 ₂ /mnm	$\text{Ga}_3\text{Re} \rightarrow \text{Ga (Cmce)} + 0.3333 \text{ Re (P6}_{3}/\text{mmc)}$	stable	-1.2784	
GaSb	mp-1224785	I-4m2	$\text{GaSb} \rightarrow \text{Ga (Cmce)} + \text{Sb (R-3m)}$	stable	1.0895	
Ge ₂ Os*	mp-10032	C2/m	$\text{Ge}_2\text{Os} \rightarrow \text{Ge (Fd-3m)} + 0.5 \text{ Os (P6}_{3}/\text{mmc)}$	stable	0.1782	
Ge ₃ Mo ₅ *	mp-17094	I4/mcm	$\text{Ge}_3\text{Mo}_5 \rightarrow \text{Ge (Fd-3m)} + 1.667 \text{ Mo (Im-3m)}$	stable	-4.3783	
Ge ₃ O	mp-1184665	I4/mmm	$\text{Ge}_3\text{O} \rightarrow \text{Ge (Fd-3m)} + 0.1667 \text{ O}_2 \text{ (C2/m)}$	unstable/nonexistent	-2.2924	
Ge ₄ Rh*	mp-1104286	P3 ₁₂₁	$\text{Ge}_4\text{Rh} \rightarrow \text{Ge (Fd-3m)} + 0.25 \text{ Rh (Fm-3m)}$	stable	-1.0918	
GeAs*	mp-7591	I4mm	$\text{GeAs} \rightarrow \text{Ge (Fd-3m)} + \text{As (Cmce)}$	metastable viable	0.0075	
GeS	mp-12910	Cmcm	$\text{GeS} \rightarrow \text{Ge (Fd-3m)} + \text{S (Fddd)}$	stable	-1.2557	
GeSb	mp-9935	Fm-3m	$\text{GeSb} \rightarrow \text{Ge (Fd-3m)} + \text{Sb (R-3m)}$	unstable/nonexistent	-0.2111	
GeSe ₂ *	mp-10074	I-42d	$\text{GeSe}_2 \rightarrow \text{Ge (Fd-3m)} + 2 \text{ Se (P3}_{121})$	stable	0.7715	
GeTe*	mp-938	R3m	$\text{GeTe} \rightarrow \text{Ge (Fd-3m)} + \text{Te (P3}_{121})$	stable	-1.0625	
H ₂ CN ₂ *	mp-703527	Pbca	$\text{H}_2\text{CN}_2 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + \text{C (P6}_{3}/\text{mmc)} + \text{N}_2 \text{ (-)}$	stable	1.1404	
H ₂ CN ₅	mp-1212563	C2/c	$\text{H}_2\text{CN}_5 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + \text{C (P6}_{3}/\text{mmc)} + 2.5 \text{ N}_2 \text{ (-)}$	stable	3.0571	
H ₂ O*	mp-2739191	C222 ₁	$\text{H}_2\text{O} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 0.5 \text{ O}_2 \text{ (C2/m)}$	stable	1.6924	
H ₃ N*	mp-643432	P2 ₁₂₁₂₁	$\text{H}_3\text{N} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 0.3333 \text{ N}_2 \text{ (-)}$	stable	1.5823	
H ₃ Pd ₄	mp-1224413	R-3m	$\text{H}_3\text{Pd}_4 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 2.667 \text{ Pd (Fm-3m)}$	stable	-2.1881	
H ₄ N	mp-1238814	P-43m	$\text{H}_4\text{N} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 0.25 \text{ N}_2 \text{ (-)}$	metastable viable	0.9726	
H ₉ CN ₉ *	mp-1202455	P-1	$\text{H}_9\text{CN}_9 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 0.2222 \text{ C (P6}_{3}/\text{mmc)} + \text{N}_2 \text{ (-)}$	stable	2.1173	
HC	mp-995197	C2/m	$\text{HC} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 2 \text{ C (P6}_{3}/\text{mmc)}$	stable	-0.4698	
HC ₂	mp-995184	C2/m	$\text{HC}_2 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 4 \text{ C (P6}_{3}/\text{mmc)}$	unstable/nonexistent	-0.6802	
HC ₂ N ₃	mp-1103408	Cmc2 ₁	$\text{HC}_2\text{N}_3 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 4 \text{ C (P6}_{3}/\text{mmc)} + 3 \text{ N}_2 \text{ (-)}$	metastable viable	4.0815	
HC ₃	mp-995200	P2 ₁ /m	$\text{HC}_3 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 6 \text{ C (P6}_{3}/\text{mmc)}$	unstable/nonexistent	-0.3999	
HC ₆	mp-995202	P-1	$\text{HC}_6 \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 12 \text{ C (P6}_{3}/\text{mmc)}$	unstable/nonexistent	-0.3499	
HCl	mp-684609	Imm2	$\text{HCl} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + \text{Cl}_2 \text{ (Cmce)}$	stable	0.9905	
HI*	mp-697025	C2/c	$\text{HI} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + 2 \text{ I (Cmce)}$	stable	0.6749	
HN	mp-1224501	P-1	$\text{HN} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + \text{N}_2 \text{ (-)}$	stable	2.2686	
HN*	mp-23922	Pmna	$\text{HN} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + \text{N}_2 \text{ (-)}$	stable	2.5491	
HN*	mp-720515	P2 ₁ /c	$\text{HN} \rightarrow \text{H}_2 \text{ (P2}_{12121}) + \text{N}_2 \text{ (-)}$	stable	2.3390	
Hf ₃ Pb	mp-1184772	I4/mmm	$\text{Hf}_3\text{Pb} \rightarrow \text{Hf (P6}_{3}/\text{mmc)} + 0.3333 \text{ Pb (Fm-3m)}$	stable	-2.7347	
HfCd*	mp-1007758	P4/nmm	$\text{HfCd} \rightarrow \text{Hf (P6}_{3}/\text{mmc)} + \text{Cd (P6}_{3}/\text{mmc)}$	stable	-0.9038	
HfCr ₂ *	mp-569624	P6 ₃ /mmc	$\text{HfCr}_2 \rightarrow \text{Hf (P6}_{3}/\text{mmc)} + 2 \text{ Cr (Im-3m)}$	unstable/nonexistent	-1.2985	
HfCu ₃	mp-1184704	I4/mmm	$\text{HfCu}_3 \rightarrow \text{Hf (P6}_{3}/\text{mmc)} + 3 \text{ Cu (Fm-3m)}$	stable	-1.5767	
HfMg ₁₅	mp-1023478	P-6m2	$\text{HfMg}_{15} \rightarrow \text{Hf (P6}_{3}/\text{mmc)} + 15 \text{ Mg (P6}_{3}/\text{mmc)}$	unstable/nonexistent	-0.4425	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
HfMo	mp-1224314	Cmmm	HfMo $\rightarrow$ Hf (P6 ₃ /mmc) + Mo (Im-3m)	unstable/nonexistent	-2.3154	
HfTc ₂ *	mp-1095669	P6 ₃ /mmc	HfTc ₂ $\rightarrow$ Hf (P6 ₃ /mmc) + 2 Tc (P6 ₃ /mmc)	stable	-3.5919	
HfW ₂	mp-1224230	Fmmm	HfW ₂ $\rightarrow$ Hf (P6 ₃ /mmc) + 2 W (Im-3m)	unstable/nonexistent	-7.2853	
Hg ₃ Pb	mp-1184700	P6 ₃ /mmc	Hg ₃ Pb $\rightarrow$ Hg (P6 ₃ /mmc) + 0.3333 Pb (Fm-3m)	unstable/nonexistent	-1.4571	
Hg ₃ Sb	mp-977242	P6 ₃ /mmc	Hg ₃ Sb $\rightarrow$ Hg (P6 ₃ /mmc) + 0.3333 Sb (R-3m)	unstable/nonexistent	-0.5802	
Hg ₄ Pt*	mp-936	Im-3m	Hg ₄ Pt $\rightarrow$ Hg (P6 ₃ /mmc) + 0.25 Pt (Fm-3m)	stable	-2.5908	
Hg ₅ Pd ₂ *	mp-30721	P4/mbm	Hg ₅ Pd ₂ $\rightarrow$ Hg (P6 ₃ /mmc) + 0.4 Pd (Fm-3m)	stable	0.0109	
HgF ₂ *	mp-8177	Fm-3m	HgF ₂ $\rightarrow$ Hg (P6 ₃ /mmc) + F ₂ (C2/c)	stable	0.9841	
IF ₇ *	mp-27988	Aea2	IF ₇ $\rightarrow$ I (Cmce) + 3.5 F ₂ (C2/c)	stable	1.7141	
IN ₂	mp-1097011	I4/mcm	IN ₂ $\rightarrow$ I (Cmce) + N ₂ (-)	stable	-0.0786	
IN ₂ Te ₃ *	mp-1105025	I4mm	IN ₂ Te ₃ $\rightarrow$ In (I4/mmm) + 1.5 Te (P3 ₁ 21)	stable	-2.3667	
IN ₃ Ir*	mp-630976	P4 ₂ /mnm	IN ₃ Ir $\rightarrow$ In (I4/mmm) + 0.3333 Ir (Fm-3m)	stable	-8.0307	
IN ₇ Cu ₁₀ *	mp-646039	C2/m	IN ₇ Cu ₁₀ $\rightarrow$ In (I4/mmm) + 1.429 Cu (Fm-3m)	stable	-3.0677	
InBr*	mp-22870	Cmcm	InBr $\rightarrow$ In (I4/mmm) + Br (Cmce)	stable	-3.9955	
InCl*	mp-571636	Cmc2 ₁	InCl $\rightarrow$ In (I4/mmm) + 0.5 Cl ₂ (Cmce)	stable	-4.5869	
InNi*	mp-20997	Pm-3m	InNi $\rightarrow$ In (I4/mmm) + Ni (Fm-3m)	stable	-0.2237	
InSe ₂	mp-1232036	P4/nmm	InSe ₂ $\rightarrow$ In (I4/mmm) + 2 Se (P3 ₁ 21)	stable	-1.4409	
InSn ₄	mp-1223653	Cmmm	InSn ₄ $\rightarrow$ In (I4/mmm) + 4 Sn (I4/mmm)	unstable/nonexistent	-0.4245	
Ir ₃ Ru	mp-974358	I4/mmm	Ir ₃ Ru $\rightarrow$ Ir (Fm-3m) + 0.3333 Ru (P6 ₃ /mmc)	stable	-11.8964	
Ir ₄ W	mp-1223666	R-3m	Ir ₄ W $\rightarrow$ Ir (Fm-3m) + 0.25 W (Im-3m)	stable	-14.2828	
IrCl ₃	mp-729507	C2	IrCl ₃ $\rightarrow$ Ir (Fm-3m) + 1.5 Cl ₂ (Cmce)	stable	-1.8955	
IrPd	mp-1223675	R-3m	IrPd $\rightarrow$ Ir (Fm-3m) + Pd (Fm-3m)	unstable/nonexistent	-6.1817	
K ₂ O*	mp-971	Fm-3m	K ₂ O $\rightarrow$ K (Fm-3m) + 0.25 O ₂ (C2/m)	stable	-0.4357	
K ₂ O	mp-1223784	P4/mmm	K ₂ O $\rightarrow$ K (Fm-3m) + 0.25 O ₂ (C2/m)	stable	-1.5638	
K ₂ S*	mp-559278	P6 ₃ /mmc	K ₂ S $\rightarrow$ K (Fm-3m) + 0.5 S (Fddd)	stable	-0.2290	
K ₂ S*	mp-1022	Fm-3m	K ₂ S $\rightarrow$ K (Fm-3m) + 0.5 S (Fddd)	stable	-0.3359	
K ₃ Rb	mp-976115	P6 ₃ /mmc	K ₃ Rb $\rightarrow$ K (Fm-3m) + 0.3333 Rb (Im-3m)	unstable/nonexistent	-0.3765	
K ₅ As ₄	mp-684799	P-1	K ₅ As ₄ $\rightarrow$ K (Fm-3m) + 0.8 As (Cmce)	stable	-0.4923	
KCl*	mp-23289	Pm-3m	KCl $\rightarrow$ K (Fm-3m) + 0.5 Cl ₂ (Cmce)	stable	0.7616	
KF ₃ *	mp-1544557	P2 ₁ 2 ₁ 2 ₁	KF ₃ $\rightarrow$ K (Fm-3m) + 1.5 F ₂ (C2/c)	stable	-1.1511	
KF ₃	mp-2749823	P2 ₁	KF ₃ $\rightarrow$ K (Fm-3m) + 1.5 F ₂ (C2/c)	stable	-1.1505	
KI*	mp-22898	Fm-3m	KI $\rightarrow$ K (Fm-3m) + I (Cmce)	stable	0.5498	
KMg ₁₅	mp-1023498	P-6m2	KMg ₁₅ $\rightarrow$ K (Fm-3m) + 15 Mg (P6 ₃ /mmc)	metastable viable	0.0081	
KN ₂ *	mp-1071868	Cmc2 ₁	KN ₂ $\rightarrow$ K (Fm-3m) + N ₂ (-)	unstable/nonexistent	-0.1420	
KN ₂	mp-1180757	Cmcm	KN ₂ $\rightarrow$ K (Fm-3m) + N ₂ (-)	unstable/nonexistent	-1.0040	
KN ₃ *	mp-827	I4/mcm	KN ₃ $\rightarrow$ K (Fm-3m) + 1.5 N ₂ (-)	stable	1.8210	
KN ₃ *	mp-827	I4/mcm	KN ₃ $\rightarrow$ K (Fm-3m) + 1.5 N ₂ (-)	stable	1.8210	
KN ₃	mp-636056	I4/mcm	KN ₃ $\rightarrow$ K (Fm-3m) + 1.5 N ₂ (-)	unstable/nonexistent	-2.3063	
KP*	mp-1058315	R-3m	KP $\rightarrow$ K (Fm-3m) + P (P2/c)	unstable/nonexistent	-2.1600	
KP	mp-1057787	Immm	KP $\rightarrow$ K (Fm-3m) + P (P2/c)	unstable/nonexistent	-2.1366	
KP ₁₅	mp-1211471	P-1	KP ₁₅ $\rightarrow$ K (Fm-3m) + 15 P (P2/c)	stable	-0.1161	
KSn ₃	mp-1185151	P6 ₃ /mmc	KSn ₃ $\rightarrow$ K (Fm-3m) + 3 Sn (I4/mmm)	stable	-3.7798	
Li ₁₃ Sn ₅ *	mp-30769	P-3m1	Li ₁₃ Sn ₅ $\rightarrow$ Li (R-3m) + 0.3846 Sn (I4/mmm)	stable	-0.2124	
Li ₂ O*	mp-1960	Fm-3m	Li ₂ O $\rightarrow$ Li (R-3m) + 0.25 O ₂ (C2/m)	stable	2.3366	
Li ₂ O	mp-755894	Pnma	Li ₂ O $\rightarrow$ Li (R-3m) + 0.25 O ₂ (C2/m)	stable	1.9481	
Li ₂ S*	mp-1125	Pnma	Li ₂ S $\rightarrow$ Li (R-3m) + 0.5 S (Fddd)	stable	1.8505	
Li ₂ S	mp-557142	P6 ₃ /mmc	Li ₂ S $\rightarrow$ Li (R-3m) + 0.5 S (Fddd)	stable	1.5052	
Li ₂ Se*	mp-2286	Fm-3m	Li ₂ Se $\rightarrow$ Li (R-3m) + 0.5 Se (P3 ₁ 21)	stable	2.4217	
Li ₃ Ag	mp-976408	I4/mmm	Li ₃ Ag $\rightarrow$ Li (R-3m) + 0.3333 Ag (Fm-3m)	stable	-0.6532	
Li ₃ Ca	mp-976075	I4/mmm	Li ₃ Ca $\rightarrow$ Li (R-3m) + 0.3333 Ca (Im-3m)	unstable/nonexistent	-0.8859	
Li ₃ Hg*	mp-1646	Fm-3m	Li ₃ Hg $\rightarrow$ Li (R-3m) + 0.3333 Hg (P6 ₃ /mmc)	stable	0.5975	
Li ₃ N*	mp-2341	P6 ₃ /mmc	Li ₃ N $\rightarrow$ Li (R-3m) + 0.1667 N ₂ (-)	stable	1.8474	
LiB*	mp-1001835	P6 ₃ /mmc	LiB $\rightarrow$ Li (R-3m) + B (R-3m)	stable	-0.9046	
LiBr	-	-	LiBr $\rightarrow$ Li (R-3m) + Br (Cmce)	insufficient data	4.0582	
LiCl*	mp-22905	Fm-3m	LiCl $\rightarrow$ Li (R-3m) + 0.5 Cl ₂ (Cmce)	stable	2.8030	
LiCl	mp-1185319	P6 ₃ /mmc	LiCl $\rightarrow$ Li (R-3m) + 0.5 Cl ₂ (Cmce)	stable	2.5005	
LiF*	mp-1138	Fm-3m	LiF $\rightarrow$ Li (R-3m) + 0.5 F ₂ (C2/c)	stable	4.4673	
LiF	mp-1009009	Pm-3m	LiF $\rightarrow$ Li (R-3m) + 0.5 F ₂ (C2/c)	stable	4.5131	
LiN ₃ *	mp-2659	C2/m	LiN ₃ $\rightarrow$ Li (R-3m) + 1.5 N ₂ (-)	stable	4.8242	
LiP ₅ *	mp-2412	Pna2 ₁	LiP ₅ $\rightarrow$ Li (R-3m) + 5 P (P2/c)	stable	0.9043	
LiP ₅ *	mp-2412	Pna2 ₁	LiP ₅ $\rightarrow$ Li (R-3m) + 5 P (P2/c)	stable	0.9043	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
LiP ₅	mp-32760	Pna2 ₁	LiP ₅ → Li (R-3m) + 5 P (P2/c)	stable	0.8692	
LiP ₅	mp-32760	Pna2 ₁	LiP ₅ → Li (R-3m) + 5 P (P2/c)	stable	0.8692	
LiRh ₃ *	mp-1001787	Imm2	LiRh ₃ → Li (R-3m) + 3 Rh (Fm-3m)	stable	-1.9151	
LiS ₄	mp-1143546	P1	LiS ₄ → Li (R-3m) + 4 S (Fddd)	stable	-0.0897	
LiTl ₃	mp-1185253	I4/mmm	LiTl ₃ → Li (R-3m) + 3 Tl (P6 ₃ /mmc)	stable	-1.4424	
Mg ₁₅ B	mp-1023515	P-6m2	Mg ₁₅ B → Mg (P6 ₃ /mmc) + 0.06667 B (R-3m)	unstable/nonexistent	-0.7179	
Mg ₁₅ Ni	mp-1023622	P-6m2	Mg ₁₅ Ni → Mg (P6 ₃ /mmc) + 0.06667 Ni (Fm-3m)	unstable/nonexistent	-0.2808	
Mg ₂ Ge*	mp-250	Fd-3m	Mg ₂ Ge → Mg (P6 ₃ /mmc) + 0.5 Ge (Fd-3m)	unstable/nonexistent	-1.3453	
Mg ₂ Ni*	mp-2137	P6 ₂ 22	Mg ₂ Ni → Mg (P6 ₃ /mmc) + 0.5 Ni (Fm-3m)	stable	-1.3906	
Mg ₃ Al	mp-978271	Pm-3m	Mg ₃ Al → Mg (P6 ₃ /mmc) + 0.3333 Al (Fm-3m)	unstable/nonexistent	-2.9844	
Mg ₃ Ru ₂ *	mp-625	P4 ₁ 32	Mg ₃ Ru ₂ → Mg (P6 ₃ /mmc) + 0.6667 Ru (P6 ₃ /mmc)	stable	-1.9276	
Mg ₃ Zr	mp-1094466	P6 ₃ /mmc	Mg ₃ Zr → Mg (P6 ₃ /mmc) + 0.3333 Zr (P6 ₃ /mmc)	unstable/nonexistent	-1.6947	
Mg ₅ Sc	mp-1185837	R32	Mg ₅ Sc → Mg (P6 ₃ /mmc) + 0.2 Sc (P6 ₃ /mmc)	unstable/nonexistent	-0.4705	
Mg ₅ Te	mp-1185828	R32	Mg ₅ Te → Mg (P6 ₃ /mmc) + 0.2 Te (P3 ₁ 21)	stable	-0.1671	
MgAl ₂ *	mp-1102064	I4 ₁ /amd	MgAl ₂ → Mg (P6 ₃ /mmc) + 2 Al (Fm-3m)	unstable/nonexistent	-6.8963	
MgCd*	mp-11307	Pmma	MgCd → Mg (P6 ₃ /mmc) + Cd (P6 ₃ /mmc)	stable	-2.1365	
MgCl ₂ *	mp-570259	P-3m1	MgCl ₂ → Mg (P6 ₃ /mmc) + Cl ₂ (Cmce)	stable	0.1545	
MgCl ₂ *	mp-23210	R-3m	MgCl ₂ → Mg (P6 ₃ /mmc) + Cl ₂ (Cmce)	stable	0.1568	
MgCl ₂	mp-569626	Ama2	MgCl ₂ → Mg (P6 ₃ /mmc) + Cl ₂ (Cmce)	stable	-0.0862	
MgF ₂ *	mp-1072956	Pnnm	MgF ₂ → Mg (P6 ₃ /mmc) + F ₂ (C2/c)	stable	0.0015	
MgF ₂	mp-1062842	Amm2	MgF ₂ → Mg (P6 ₃ /mmc) + F ₂ (C2/c)	stable	0.0229	
MgF ₂	mp-560236	C2/m	MgF ₂ → Mg (P6 ₃ /mmc) + F ₂ (C2/c)	stable	-0.3906	
MgH ₂ *	mp-23711	Pbcn	MgH ₂ → Mg (P6 ₃ /mmc) + H ₂ (P2 ₁ 2 ₁ 2 ₁)	stable	-0.4411	
MgIn ₂ *	mp-1077016	Fd-3m	MgIn ₂ → Mg (P6 ₃ /mmc) + 2 In (I4/mmm)	unstable/nonexistent	-4.7015	
MgO*	mp-549706	P6 ₃ mc	MgO → Mg (P6 ₃ /mmc) + 0.5 O ₂ (C2/m)	stable	-1.4179	
MgO	mp-1009127	Pm-3m	MgO → Mg (P6 ₃ /mmc) + 0.5 O ₂ (C2/m)	stable	-1.0763	
MgP ₄ *	mp-384	P2 ₁ /c	MgP ₄ → Mg (P6 ₃ /mmc) + 4 P (P2/c)	stable	-0.1068	
MgPb ₂	mp-2016042	C2/m	MgPb ₂ → Mg (P6 ₃ /mmc) + 2 Pb (Fm-3m)	stable	-2.7253	
MgPd ₃ *	mp-7753	Pm-3m	MgPd ₃ → Mg (P6 ₃ /mmc) + 3 Pd (Fm-3m)	stable	-5.5997	
MgS*	mp-1315	Fm-3m	MgS → Mg (P6 ₃ /mmc) + S (Fddd)	stable	-0.3401	
MgS	mp-13032	F-43m	MgS → Mg (P6 ₃ /mmc) + S (Fddd)	stable	-0.4163	
MgSn	mp-1094669	Pm-3m	MgSn → Mg (P6 ₃ /mmc) + Sn (I4/mmm)	stable	-1.3854	
MgSn	mp-1094801	P4/mmm	MgSn → Mg (P6 ₃ /mmc) + Sn (I4/mmm)	stable	-0.8961	
MgTi	mp-1094346	P2 ₁ /m	MgTi → Mg (P6 ₃ /mmc) + Ti (P6 ₃ /mmc)	unstable/nonexistent	-1.5994	
MgTl ₃	mp-978262	P6 ₃ /mmc	MgTl ₃ → Mg (P6 ₃ /mmc) + 3 Tl (P6 ₃ /mmc)	unstable/nonexistent	-2.1860	
Mn ₂ O ₇ *	mp-28338	P2 ₁ /c	Mn ₂ O ₇ → Mn (I-43m) + 1.75 O ₂ (C2/m)	stable	-1.4687	
Mn ₄ B	mp-1221769	F222	Mn ₄ B → Mn (I-43m) + 0.25 B (R-3m)	stable	1.2987	
Mn ₄ Ga	mp-1221812	R-3m	Mn ₄ Ga → Mn (I-43m) + 0.25 Ga (Cmce)	metastable viable	5.2757	
MnAl ₁₂ *	mp-1104165	Im-3	MnAl ₁₂ → Mn (I-43m) + 12 Al (Fm-3m)	stable	-8.1748	
MnN	mp-1009130	F-43m	MnN → Mn (I-43m) + 0.5 N ₂ (-)	stable	2.1223	
MnO ₂ *	mp-510408	P4 ₂ /mnm	MnO ₂ → Mn (I-43m) + O ₂ (C2/m)	stable	-0.5274	
Mo ₂ C	mp-1221498	P-3m1	Mo ₂ C → Mo (Im-3m) + 0.5 C (P6 ₃ /mmc)	stable	-4.9953	
Mo ₃ Se ₄ *	mp-21021	R-3	Mo ₃ Se ₄ → Mo (Im-3m) + 1.333 Se (P3 ₁ 21)	stable	-2.8097	
MoPd ₂	mp-1206704	Immm	MoPd ₂ → Mo (Im-3m) + 2 Pd (Fm-3m)	stable	-5.5400	
MoRu	mp-1221386	P-6m2	MoRu → Mo (Im-3m) + Ru (P6 ₃ /mmc)	unstable/nonexistent	-2.2801	
MoSe ₂ *	mp-1634	P6 ₃ /mmc	MoSe ₂ → Mo (Im-3m) + 2 Se (P3 ₁ 21)	stable	-0.9475	
N ₂ O*	mp-1180473	Pnma	N ₂ O → N ₂ (-) + 0.5 O ₂ (C2/m)	stable	3.6383	
NO	mp-1180334	P2 ₁ 2 ₁ 2 ₁	NO → N ₂ (-) + O ₂ (C2/m)	metastable viable	2.1371	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
Na ₂ Cl	mp-1077015	Cmmm	Na ₂ Cl $\rightarrow$ Na (P6 ₃ /mmc) + 0.25 Cl ₂ (Cmce)	stable	-0.3965	
Na ₂ Cl	mp-990084	R-3m	Na ₂ Cl $\rightarrow$ Na (P6 ₃ /mmc) + 0.25 Cl ₂ (Cmce)	stable	-0.2472	
Na ₂ O*	mp-2352	Fm-3m	Na ₂ O $\rightarrow$ Na (P6 ₃ /mmc) + 0.25 O ₂ (C2/m)	stable	-1.3057	
Na ₂ O	mp-1975297	C2/m	Na ₂ O $\rightarrow$ Na (P6 ₃ /mmc) + 0.25 O ₂ (C2/m)	stable	-1.5619	
Na ₃ Cl	mp-1064484	P4/mmm	Na ₃ Cl $\rightarrow$ Na (P6 ₃ /mmc) + 0.1667 Cl ₂ (Cmce)	stable	-0.1382	
Na ₃ Hg	mp-1186050	I4/mmm	Na ₃ Hg $\rightarrow$ Na (P6 ₃ /mmc) + 0.3333 Hg (P6 ₃ /mmm)	stable	-1.0242	
Na ₃ In	mp-864758	I4/mmm	Na ₃ In $\rightarrow$ Na (P6 ₃ /mmc) + 0.3333 In (I4/mmm)	stable	-2.4504	
Na ₃ Mg	mp-982637	I4/mmm	Na ₃ Mg $\rightarrow$ Na (P6 ₃ /mmc) + 0.3333 Mg (P6 ₃ /mmc)	unstable/nonexistent	-0.0772	
Na ₃ Si ₃₄	mp-1221235	R-3m	Na ₃ Si ₃₄ $\rightarrow$ Na (P6 ₃ /mmc) + 11.33 Si (-)	unstable/nonexistent	-0.4270	
NaCl	-	-	NaCl $\rightarrow$ Na (P6 ₃ /mmc) + 0.5 Cl ₂ (Cmce)	insufficient data	0.3735	
NaN ₃ *	mp-1066400	R3m	NaN ₃ $\rightarrow$ Na (P6 ₃ /mmc) + 1.5 N ₂ (-)	stable	1.8936	
NaN ₃ *	mp-22003	R-3m	NaN ₃ $\rightarrow$ Na (P6 ₃ /mmc) + 1.5 N ₂ (-)	stable	2.1726	
NaN ₃ *	mp-1066400	R3m	NaN ₃ $\rightarrow$ Na (P6 ₃ /mmc) + 1.5 N ₂ (-)	stable	1.8936	
NaN ₃ *	mp-22003	R-3m	NaN ₃ $\rightarrow$ Na (P6 ₃ /mmc) + 1.5 N ₂ (-)	stable	2.1726	
NaN ₃	mp-634410	C2	NaN ₃ $\rightarrow$ Na (P6 ₃ /mmc) + 1.5 N ₂ (-)	metastable viable	2.9023	
NaN ₃	mp-1065265	C2/m	NaN ₃ $\rightarrow$ Na (P6 ₃ /mmc) + 1.5 N ₂ (-)	unstable/nonexistent	-1.7320	
NaP ₁₅ *	mp-1194653	P-1	NaP ₁₅ $\rightarrow$ Na (P6 ₃ /mmc) + 15 P (P2/c)	stable	-0.0672	
NaP ₅ *	mp-31086	Pnma	NaP ₅ $\rightarrow$ Na (P6 ₃ /mmc) + 5 P (P2/c)	stable	-0.0199	
NaS*	mp-2400	P6 ₃ /mmc	NaS $\rightarrow$ Na (P6 ₃ /mmc) + S (Fddd)	stable	-0.4953	
NaS*	mp-409	P-62m	NaS $\rightarrow$ Na (P6 ₃ /mmc) + S (Fddd)	stable	-0.4911	
NaS	mp-1180106	C2/m	NaS $\rightarrow$ Na (P6 ₃ /mmc) + S (Fddd)	unstable/nonexistent	-2.3999	
Nb ₂ O ₅ *	mp-1595	C2/m	Nb ₂ O ₅ $\rightarrow$ Nb (Im-3m) + 1.25 O ₂ (C2/m)	stable	-2.0689	
Nb ₃ Ir*	mp-1458	Pm-3n	Nb ₃ Ir $\rightarrow$ Nb (Im-3m) + 0.3333 Ir (Fm-3m)	stable	-10.1223	
Nb ₃ Ni	mp-999394	I4/mmm	Nb ₃ Ni $\rightarrow$ Nb (Im-3m) + 0.3333 Ni (Fm-3m)	unstable/nonexistent	-6.8505	
Nb ₄ Rh	mp-1220441	Fmmm	Nb ₄ Rh $\rightarrow$ Nb (Im-3m) + 0.25 Rh (Fm-3m)	stable	-4.8761	
NbCd ₃ *	mp-11309	Pm-3m	NbCd ₃ $\rightarrow$ Nb (Im-3m) + 3 Cd (P6 ₃ /mmc)	unstable/nonexistent	-6.4651	
NbGa ₃ *	mp-1973	I4/mmm	NbGa ₃ $\rightarrow$ Nb (Im-3m) + 3 Ga (Cmce)	stable	-0.7573	
NbO*	mp-2692	Fm-3m	NbO $\rightarrow$ Nb (Im-3m) + 0.5 O ₂ (C2/m)	stable	-2.3179	
NbO	mp-634965	P4/mmm	NbO $\rightarrow$ Nb (Im-3m) + 0.5 O ₂ (C2/m)	stable	-3.1495	
NbP*	mp-9830	I4 ₁ /amd	NbP $\rightarrow$ Nb (Im-3m) + P (P2/c)	stable	-2.4724	
NbP	mp-1220317	Cmmm	NbP $\rightarrow$ Nb (Im-3m) + P (P2/c)	stable	-1.1731	
NbRh ₂	mp-1220414	Pm	NbRh ₂ $\rightarrow$ Nb (Im-3m) + 2 Rh (Fm-3m)	stable	-5.2734	
NbS ₂	mp-774873	P6 ₃ /mmc	NbS ₂ $\rightarrow$ Nb (Im-3m) + 2 S (Fddd)	stable	-1.9968	
NbZn ₁₆ *	mp-640041	Cmcm	NbZn ₁₆ $\rightarrow$ Nb (Im-3m) + 16 Zn (P6 ₃ /mmc)	stable	-7.1223	
Ni ₃ Mo*	mp-11506	Pmmn	Ni ₃ Mo $\rightarrow$ Ni (Fm-3m) + 0.3333 Mo (Im-3m)	stable	-4.2954	
NiB*	mp-14019	Cmcm	NiB $\rightarrow$ Ni (Fm-3m) + B (R-3m)	stable	-0.8092	
NiHg ₄ *	mp-570835	Im-3m	NiHg ₄ $\rightarrow$ Ni (Fm-3m) + 4 Hg (P6 ₃ /mmm)	unstable/nonexistent	-0.3609	
NiS*	mp-1547	R3m	NiS $\rightarrow$ Ni (Fm-3m) + S (Fddd)	stable	-1.1476	
NiSb ₃ *	mp-1106023	Im-3	NiSb ₃ $\rightarrow$ Ni (Fm-3m) + 3 Sb (R-3m)	stable	-0.2136	
OsC*	mp-7142	P-6m2	OsC $\rightarrow$ Os (P6 ₃ /mmc) + C (P6 ₃ /mmc)	unstable/nonexistent	-2.0168	
OsC	mp-1009537	Pm-3m	OsC $\rightarrow$ Os (P6 ₃ /mmc) + C (P6 ₃ /mmc)	unstable/nonexistent	-2.3184	
OsC	mp-1009539	Fm-3m	OsC $\rightarrow$ Os (P6 ₃ /mmc) + C (P6 ₃ /mmc)	unstable/nonexistent	-3.1141	
OsC	mp-999308	P6 ₃ /mmc	OsC $\rightarrow$ Os (P6 ₃ /mmc) + C (P6 ₃ /mmc)	unstable/nonexistent	-2.9572	
P ₄ Se ₅ *	mp-2447	Pna2 ₁	P ₄ Se ₅ $\rightarrow$ P (P2/c) + 1.25 Se (P3 ₁ 21)	stable	0.3563	
PI ₂ *	mp-29443	P-1	PI ₂ $\rightarrow$ P (P2/c) + 2 I (Cmce)	stable	-0.0857	
PNO	mp-675742	P-1	PNO $\rightarrow$ P (P2/c) + 0.5 N ₂ (-) + 0.5 O ₂ (C2/m)	metastable viable	0.6744	
PS*	mp-612	C2/c	PS $\rightarrow$ P (P2/c) + S (Fddd)	stable	-0.0254	
PW ₃	mp-1209828	I-42m	PW ₃ $\rightarrow$ P (P2/c) + 3 W (Im-3m)	stable	-11.8793	
Pb ₂ Au*	mp-22795	I4/mcm	Pb ₂ Au $\rightarrow$ Pb (Fm-3m) + 0.5 Au (Fm-3m)	stable	-6.8118	
PbAu ₂ *	mp-19871	Fd-3m	PbAu ₂ $\rightarrow$ Pb (Fm-3m) + 2 Au (Fm-3m)	stable	-10.9426	
PbAu ₃	mp-982246	Fm-3m	PbAu ₃ $\rightarrow$ Pb (Fm-3m) + 3 Au (Fm-3m)	unstable/nonexistent	-10.8037	
PbN ₆ *	mp-620058	Pna2 ₁	PbN ₆ $\rightarrow$ Pb (Fm-3m) + 3 N ₂ (-)	metastable viable	1.6855	
PbN ₆ *	mp-667338	Pnma	PbN ₆ $\rightarrow$ Pb (Fm-3m) + 3 N ₂ (-)	metastable viable	1.6946	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
PbSe*	mp-22009	Fmm2	PbSe $\rightarrow$ Pb (Fm-3m) + Se (P3_121)	stable	-2.9524	
PbSe*	mp-2201	Fm-3m	PbSe $\rightarrow$ Pb (Fm-3m) + Se (P3_121)	stable	-2.4818	
PtCl ₂	mp-1219748	C2/m	PtCl ₂ $\rightarrow$ Pt (Fm-3m) + Cl ₂ (Cmce)	stable	-1.8411	
Rb ₂ Hg ₇ *	mp-31474	P-3m1	Rb ₂ Hg ₇ $\rightarrow$ Rb (Im-3m) + 3.5 Hg (P6/mmm)	stable	-1.1841	
Rb ₂ O*	mp-1394	Fm-3m	Rb ₂ O $\rightarrow$ Rb (Im-3m) + 0.25 O ₂ (C2/m)	stable	0.7235	
Rb ₂ O	mp-2023482	P-3m1	Rb ₂ O $\rightarrow$ Rb (Im-3m) + 0.25 O ₂ (C2/m)	stable	-0.7592	
Rb ₃ In	mp-974948	Pm-3m	Rb ₃ In $\rightarrow$ Rb (Im-3m) + 0.3333 In (I4/mmm)	unstable/nonexistent	-2.5986	
Rb ₃ Li	mp-974938	Pm-3m	Rb ₃ Li $\rightarrow$ Rb (Im-3m) + 0.3333 Li (R-3m)	unstable/nonexistent	-0.7560	
Rb ₃ P	mp-974822	Fm-3m	Rb ₃ P $\rightarrow$ Rb (Im-3m) + 0.3333 P (P2/c)	stable	0.9067	
Rb ₃ P ₁₁	mp-1173444	C2/m	Rb ₃ P ₁₁ $\rightarrow$ Rb (Im-3m) + 3.667 P (P2/c)	stable	-0.0005	
RbCl*	mp-23299	Pm-3m	RbCl $\rightarrow$ Rb (Im-3m) + 0.5 Cl ₂ (Cmce)	stable	1.7924	
RbF*	mp-2064	Pm-3m	RbF $\rightarrow$ Rb (Im-3m) + 0.5 F ₂ (C2/c)	stable	3.9244	
RbF	mp-11718	Fm-3m	RbF $\rightarrow$ Rb (Im-3m) + 0.5 F ₂ (C2/c)	stable	1.9497	
RbI*	mp-22903	Fm-3m	RbI $\rightarrow$ Rb (Im-3m) + I (Cmce)	stable	0.8192	
RbN ₃ *	mp-581833	P4/mmm	RbN ₃ $\rightarrow$ Rb (Im-3m) + 1.5 N ₂ (-)	stable	2.3450	
RbN ₃	mp-1219567	Amm2	RbN ₃ $\rightarrow$ Rb (Im-3m) + 1.5 N ₂ (-)	metastable viable	0.1292	
RbP*	mp-1179688	C2/m	RbP $\rightarrow$ Rb (Im-3m) + P (P2/c)	unstable/nonexistent	-1.9315	
RbP	mp-1058499	Immm	RbP $\rightarrow$ Rb (Im-3m) + P (P2/c)	unstable/nonexistent	-1.9104	
RbP ₁₅	mp-1209298	P-1	RbP ₁₅ $\rightarrow$ Rb (Im-3m) + 15 P (P2/c)	stable	-0.1066	
RbS*	mp-558071	Immm	RbS $\rightarrow$ Rb (Im-3m) + S (Fddd)	stable	-0.3217	
RbS	mp-1179684	C2/m	RbS $\rightarrow$ Rb (Im-3m) + S (Fddd)	stable	-2.7347	
Re ₂ Si*	mp-669547	P2_1/c	Re ₂ Si $\rightarrow$ Re (P6_3/mmc) + 0.5 Si (-)	unstable/nonexistent	-2.6469	
Re ₃ Ge ₇ *	mp-29141	Cmcm	Re ₃ Ge ₇ $\rightarrow$ Re (P6_3/mmc) + 2.333 Ge (Fd-3m)	unstable/nonexistent	-0.2457	
Re ₃ N ₅	mp-1080194	Imm2	Re ₃ N ₅ $\rightarrow$ Re (P6_3/mmc) + 0.8333 N ₂ (-)	unstable/nonexistent	-2.0545	
Re ₃ Pd	mp-862586	P6_3/mmc	Re ₃ Pd $\rightarrow$ Re (P6_3/mmc) + 0.3333 Pd (Fm-3m)	stable	-3.8495	
Re ₅ Si ₃ *	mp-1105959	I4/mcm	Re ₅ Si ₃ $\rightarrow$ Re (P6_3/mmc) + 0.6 Si (-)	stable	-1.9136	
ReC*	mp-1079494	P6_3/mmc	ReC $\rightarrow$ Re (P6_3/mmc) + C (P6_3/mmc)	unstable/nonexistent	-2.6625	
ReC	mp-1009731	Pm-3m	ReC $\rightarrow$ Re (P6_3/mmc) + C (P6_3/mmc)	unstable/nonexistent	-1.9955	
RuC*	mp-10030	P-6m2	RuC $\rightarrow$ Ru (P6_3/mmc) + C (P6_3/mmc)	unstable/nonexistent	-2.1301	
RuC	mp-1009787	Fm-3m	RuC $\rightarrow$ Ru (P6_3/mmc) + C (P6_3/mmc)	unstable/nonexistent	-2.8605	
RuRh ₃	mp-974335	Pm-3m	RuRh ₃ $\rightarrow$ Ru (P6_3/mmc) + 3 Rh (Fm-3m)	unstable/nonexistent	-4.1363	
S ₂ N	-	P4_2nm	S ₂ N $\rightarrow$ S (Fddd) + 0.25 N ₂ (-)	insufficient data	0.6645	
SN	-	P2_1/c	SN $\rightarrow$ S (Fddd) + 0.5 N ₂ (-)	insufficient data	1.0655	
SN*	mp-236	P2_1/c	SN $\rightarrow$ S (Fddd) + 0.5 N ₂ (-)	metastable viable	0.1440	
SN	mp-1078816	P2_1/c	SN $\rightarrow$ S (Fddd) + 0.5 N ₂ (-)	unstable/nonexistent	-1.2463	
SN	mp-1173453	P2_1	SN $\rightarrow$ S (Fddd) + 0.5 N ₂ (-)	unstable/nonexistent	-0.7054	
SN	mp-684642	P2_1	SN $\rightarrow$ S (Fddd) + 0.5 N ₂ (-)	metastable viable	0.7256	
Sb ₂ S ₅	mp-1359089	P2_1/c	Sb ₂ S ₅ $\rightarrow$ Sb (R-3m) + 2.5 S (Fddd)	stable	-0.4568	
Sc ₂ O ₃	mp-755313	R-3c	Sc ₂ O ₃ $\rightarrow$ Sc (P6_3/mmc) + 0.75 O ₂ (C2/m)	stable	0.0830	
Sc ₅ Ge ₃ *	mp-17190	P6_3/mcm	Sc ₅ Ge ₃ $\rightarrow$ Sc (P6_3/mmc) + 0.6 Ge (Fd-3m)	stable	-0.5250	
ScAg ₂ *	mp-1018128	I4/mmm	ScAg ₂ $\rightarrow$ Sc (P6_3/mmc) + 2 Ag (Fm-3m)	stable	-2.6842	
ScOs ₃	mp-981384	Pm-3m	ScOs ₃ $\rightarrow$ Sc (P6_3/mmc) + 3 Os (P6_3/mmc)	unstable/nonexistent	-2.6732	
ScTc ₃	mp-867262	Fm-3m	ScTc ₃ $\rightarrow$ Sc (P6_3/mmc) + 3 Tc (P6_3/mmc)	stable	-2.4447	
ScTe*	mp-10026	P6_3/mmc	ScTe $\rightarrow$ Sc (P6_3/mmc) + Te (P3_121)	stable	0.9020	
Si ₃ H	mp-995193	P-3m1	Si ₃ H $\rightarrow$ Si (-) + 0.1667 H ₂ (P2_12_12_1)	unstable/nonexistent	-0.0398	
Si ₃ Pt	mp-972763	P6_3/mmc	Si ₃ Pt $\rightarrow$ Si (-) + 0.3333 Pt (Fm-3m)	stable	-0.4837	
Si ₃ W ₅ *	mp-31219	I4/mcm	Si ₃ W ₅ $\rightarrow$ Si (-) + 1.667 W (Im-3m)	stable	-10.1672	
SiAu ₃	mp-1186998	Fm-3m	SiAu ₃ $\rightarrow$ Si (-) + 3 Au (Fm-3m)	unstable/nonexistent	-10.8734	
SiH ₂ *	mp-1192532	P2_1/m	SiH ₂ $\rightarrow$ Si (-) + H ₂ (P2_12_12_1)	unstable/nonexistent	-0.0980	
SiO ₂ *	mp-546794	I-42d	SiO ₂ $\rightarrow$ Si (-) + O ₂ (C2/m)	stable	2.1420	
SiO ₂ *	mp-9258	Pa-3	SiO ₂ $\rightarrow$ Si (-) + O ₂ (C2/m)	stable	3.3332	
SiO ₂	mp-10064	Fm-3m	SiO ₂ $\rightarrow$ Si (-) + O ₂ (C2/m)	stable	2.1735	
SiO ₂	mp-556044	P-31c	SiO ₂ $\rightarrow$ Si (-) + O ₂ (C2/m)	stable	0.0958	
SiO ₂	mp-638049	F4_132	SiO ₂ $\rightarrow$ Si (-) + O ₂ (C2/m)	stable	0.3849	
SiRh*	mp-2287	P2_1/c	SiRh $\rightarrow$ Si (-) + Rh (Fm-3m)	stable	-1.0382	
SiS ₂ *	mp-1095294	P2_1/c	SiS ₂ $\rightarrow$ Si (-) + 2 S (Fddd)	stable	0.9960	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
SiTc ₃	mp-972784	P6 ₃ /mmc	SiTc ₃ → Si (-) + 3 Tc (P6 ₃ /mmc)	stable	-2.5252	
SiTe ₂ *	mp-2755	P-3m1	SiTe ₂ → Si (-) + 2 Te (P3 ₁ 121)	metastable viable	0.5603	
Sn ₃ S	mp-1187038	Fm-3m	Sn ₃ S → Sn (I4/mmm) + 0.3333 S (Fddd)	unstable/nonexistent	-4.0453	
Sn ₃ Se	mp-1187055	Fm-3m	Sn ₃ Se → Sn (I4/mmm) + 0.3333 Se (P3 ₁ 121)	unstable/nonexistent	-3.8904	
Sn ₇ Ir ₅ *	mp-20466	I4mm	Sn ₇ Ir ₅ → Sn (I4/mmm) + 0.7143 Ir (Fm-3m)	stable	-6.9737	
Sn ₇ Pd ₅	mp-1219006	C2	Sn ₇ Pd ₅ → Sn (I4/mmm) + 0.7143 Pd (Fm-3m)	stable	-2.7505	
SnBr ₂	mp-978985	P4 ₂ /mnm	SnBr ₂ → Sn (I4/mmm) + 2 Br (Cmce)	stable	-1.5288	
SnP*	mp-475	Fm-3m	SnP → Sn (I4/mmm) + P (P2/c)	unstable/nonexistent	-1.2021	
SnPb ₃	mp-1187118	P6 ₃ /mmc	SnPb ₃ → Sn (I4/mmm) + 3 Pb (Fm-3m)	unstable/nonexistent	-5.1956	
SnPt*	mp-19856	P6 ₃ /mmc	SnPt → Sn (I4/mmm) + Pt (Fm-3m)	stable	-5.1803	
Sr ₂ As*	mp-15697	I4/mmm	Sr ₂ As → Sr (P6 ₃ /mmc) + 0.5 As (Cmce)	stable	-0.6669	
Sr ₃ As ₄	mp-678007	P1	Sr ₃ As ₄ → Sr (P6 ₃ /mmc) + 1.333 As (Cmce)	stable	-0.7460	
Sr ₃ N ₂	mp-1013529	Pm-3m	Sr ₃ N ₂ → Sr (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	-2.4326	
Sr ₃ P ₄ *	mp-14288	Fdd2	Sr ₃ P ₄ → Sr (P6 ₃ /mmc) + 1.333 P (P2/c)	stable	-1.0472	
Sr ₃ P ₄	mp-682953	P1	Sr ₃ P ₄ → Sr (P6 ₃ /mmc) + 1.333 P (P2/c)	stable	-1.0793	
Sr ₃ Sn	mp-1187136	Pm-3m	Sr ₃ Sn → Sr (P6 ₃ /mmc) + 0.3333 Sn (I4/mmm)	stable	-1.6014	
Sr ₅ Sn ₃ *	mp-17720	I4/mcm	Sr ₅ Sn ₃ → Sr (P6 ₃ /mmc) + 0.6 Sn (I4/mmm)	stable	-2.1978	
SrAl ₂ *	mp-1638	Fd-3m	SrAl ₂ → Sr (P6 ₃ /mmc) + 2 Al (Fm-3m)	stable	-7.3220	
SrB ₂	mp-1008755	P6/mmm	SrB ₂ → Sr (P6 ₃ /mmc) + 2 B (R-3m)	unstable/nonexistent	-2.7885	
SrF ₂ *	mp-1102240	Pnma	SrF ₂ → Sr (P6 ₃ /mmc) + F ₂ (C2/c)	stable	-0.5421	
SrF ₂	mp-1102911	Pnma	SrF ₂ → Sr (P6 ₃ /mmc) + F ₂ (C2/c)	stable	-0.1481	
SrN*	mp-1058586	Fm-3m	SrN → Sr (P6 ₃ /mmc) + 0.5 N ₂ (-)	stable	-2.0777	
SrN ₆ *	mp-2131	Fddd	SrN ₆ → Sr (P6 ₃ /mmc) + 3 N ₂ (-)	stable	1.8720	
SrO ₁₀	mp-1208655	P4/mmm	SrO ₁₀ → Sr (P6 ₃ /mmc) + 5 O ₂ (C2/m)	unstable/nonexistent	-2.6795	
SrO ₂ *	mp-2697	I4/mmm	SrO ₂ → Sr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.8542	
SrO ₂ *	mp-726705	Pnma	SrO ₂ → Sr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-1.9205	
SrO ₂	mp-2763179	Pnma	SrO ₂ → Sr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-1.7522	
SrS*	mp-10627	Pm-3m	SrS → Sr (P6 ₃ /mmc) + S (Fddd)	stable	-0.1088	
SrSi*	mp-2661	Cmcm	SrSi → Sr (P6 ₃ /mmc) + Si (-)	stable	-1.3415	
SrTl ₃	mp-1206636	Pm-3m	SrTl ₃ → Sr (P6 ₃ /mmc) + 3 Tl (P6 ₃ /mmc)	stable	-2.2057	
Ta ₄ C ₃	mp-1218000	R3m	Ta ₄ C ₃ → Ta (Im-3m) + 0.75 C (P6 ₃ /mmc)	stable	-5.5562	
Ta ₅ Ge ₃ *	mp-17593	I4/mcm	Ta ₅ Ge ₃ → Ta (Im-3m) + 0.6 Ge (Fd-3m)	stable	-5.0868	
Ta ₇ Co ₆ *	mp-1104548	R-3m	Ta ₇ Co ₆ → Ta (Im-3m) + 0.8571 Co (Fm-3m)	stable	-9.4065	
TaBe ₁₂ *	mp-567842	I4/mmm	TaBe ₁₂ → Ta (Im-3m) + 12 Be (P6 ₃ /mmc)	stable	-2.2282	
TaN*	mp-1497	P6/mmm	TaN → Ta (Im-3m) + 0.5 N ₂ (-)	stable	-3.4690	
TaN	mp-1009831	Pm-3m	TaN → Ta (Im-3m) + 0.5 N ₂ (-)	stable	-1.5922	
TaN	mp-1009833	F-43m	TaN → Ta (Im-3m) + 0.5 N ₂ (-)	stable	-2.8766	
TaP*	mp-1067587	I4 ₁ md	TaP → Ta (Im-3m) + P (P2/c)	stable	-2.2732	
TaP	mp-1187244	P-6m2	TaP → Ta (Im-3m) + P (P2/c)	stable	-1.2040	
TaRe	mp-1217894	Cmmm	TaRe → Ta (Im-3m) + Re (P6 ₃ /mmc)	stable	-2.6081	
TaSb ₂ *	mp-11697	C2/m	TaSb ₂ → Ta (Im-3m) + 2 Sb (R-3m)	stable	-1.7847	
Tc ₃ Mo	mp-1187395	Pm-3m	Tc ₃ Mo → Tc (P6 ₃ /mmc) + 0.3333 Mo (Im-3m)	unstable/nonexistent	-4.9437	
Tc ₃ Os	mp-862779	P6 ₃ /mmc	Tc ₃ Os → Tc (P6 ₃ /mmc) + 0.3333 Os (P6 ₃ /mmc)	stable	-3.7993	
Tc ₃ W	mp-1187398	P6 ₃ /mmc	Tc ₃ W → Tc (P6 ₃ /mmc) + 0.3333 W (Im-3m)	unstable/nonexistent	-7.0792	
TcW	mp-1217337	Cmmm	TcW → Tc (P6 ₃ /mmc) + W (Im-3m)	unstable/nonexistent	-3.7117	
Te ₂ Au*	mp-567525	C2/m	Te ₂ Au → Te (P3 ₁ 121) + 0.5 Au (Fm-3m)	stable	-2.2308	
Te ₂ Ir*	mp-1551	C2/m	Te ₂ Ir → Te (P3 ₁ 121) + 0.5 Ir (Fm-3m)	stable	-2.0299	
Te ₂ Pd*	mp-782	P-3m1	Te ₂ Pd → Te (P3 ₁ 121) + 0.5 Pd (Fm-3m)	stable	1.3781	
Te ₂ Pd ₇	mp-1217443	R-3m	Te ₂ Pd ₇ → Te (P3 ₁ 121) + 3.5 Pd (Fm-3m)	stable	-4.9451	
Te ₂ Ru*	mp-267	Pnnm	Te ₂ Ru → Te (P3 ₁ 121) + 0.5 Ru (P6 ₃ /mmc)	stable	0.6569	
TeO ₂ *	mp-8377	P2 ₁ 2 ₁ 2 ₁ -1	TeO ₂ → Te (P3 ₁ 121) + O ₂ (C2/m)	stable	-0.1489	
Ti ₂ Cu*	mp-742	I4/mmm	Ti ₂ Cu → Ti (P6 ₃ /mmc) + 0.5 Cu (Fm-3m)	stable	-0.7183	
Ti ₃ Ge*	mp-1189044	I-4	Ti ₃ Ge → Ti (P6 ₃ /mmc) + 0.3333 Ge (Fd-3m)	stable	-1.3616	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})/\text{at.}$
Ti ₃ In*	mp-866046	P6 ₃ /mmc	Ti ₃ In $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 In (I4/mmm)	stable	-2.5967	
Ti ₃ Nb	mp-1217091	Amm2	Ti ₃ Nb $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 Nb (Im-3m)	unstable/nonexistent	-1.4185	
Ti ₃ Pt*	mp-2339	Pm-3n	Ti ₃ Pt $\rightarrow$ Ti (P6 ₃ /mmc) + 0.3333 Pt (Fm-3m)	stable	-3.6405	
TiBe ₁₂ *	mp-1104067	I4/mmm	TiBe ₁₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 12 Be (P6 ₃ /mmc)	stable	-1.7424	
TiMo ₂	mp-1216675	Fmmm	TiMo ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Mo (Im-3m)	stable	-3.3756	
TiMo ₂	mp-1216675	Fmmm	TiMo ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Mo (Im-3m)	stable	-3.3756	
TiNi*	mp-1048	P2 ₁ /m	TiNi $\rightarrow$ Ti (P6 ₃ /mmc) + Ni (Fm-3m)	stable	-1.5890	
TiNi	mp-1067248	Cmcm	TiNi $\rightarrow$ Ti (P6 ₃ /mmc) + Ni (Fm-3m)	stable	-1.6587	
TiO*	mp-1071163	P-62m	TiO $\rightarrow$ Ti (P6 ₃ /mmc) + 0.5 O ₂ (C2/m)	stable	-0.1411	
TiO ₂ *	mp-1439	Pbcn	TiO ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.6461	
TiO ₂ *	mp-9173	Pnma	TiO ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.6509	
TiO ₂ *	mp-430	P2 ₁ /c	TiO ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.6267	
TiSi ₂ *	mp-1077503	Cmcm	TiSi ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Si (-)	stable	0.8802	
TiSi ₂	mp-570991	Immm	TiSi ₂ $\rightarrow$ Ti (P6 ₃ /mmc) + 2 Si (-)	stable	1.3340	
TiW	mp-1216621	Cmmm	TiW $\rightarrow$ Ti (P6 ₃ /mmc) + W (Im-3m)	unstable/nonexistent	-3.4601	
Tl ₃ Au	mp-1080569	Pm-3n	Tl ₃ Au $\rightarrow$ Tl (P6 ₃ /mmc) + 0.3333 Au (Fm-3m)	unstable/nonexistent	-6.1367	
Tl ₃ Ga	mp-1187539	Fm-3m	Tl ₃ Ga $\rightarrow$ Tl (P6 ₃ /mmc) + 0.3333 Ga (Cmce)	unstable/nonexistent	-3.3767	
Tl ₄ Sn	mp-1216560	R-3m	Tl ₄ Sn $\rightarrow$ Tl (P6 ₃ /mmc) + 0.25 Sn (I4/mmm)	metastable viable	1.8363	
TlAs	mp-1007770	F-43m	TlAs $\rightarrow$ Tl (P6 ₃ /mmc) + As (Cmce)	unstable/nonexistent	-0.5143	
TlAu ₃	mp-1187508	Pm-3m	TlAu ₃ $\rightarrow$ Tl (P6 ₃ /mmc) + 3 Au (Fm-3m)	unstable/nonexistent	-11.1423	
TlC*	mp-567465	Fm-3m	TlC $\rightarrow$ Tl (P6 ₃ /mmc) + C (P6 ₃ /mmc)	unstable/nonexistent	-3.1404	
TlC	mp-1058887	P4/mmm	TlC $\rightarrow$ Tl (P6 ₃ /mmc) + C (P6 ₃ /mmc)	unstable/nonexistent	-3.3638	
TlCl*	mp-571079	Cmcm	TlCl $\rightarrow$ Tl (P6 ₃ /mmc) + 0.5 Cl ₂ (Cmce)	stable	-2.2967	
TlN*	mp-1017982	P6 ₃ /mmc	TlN $\rightarrow$ Tl (P6 ₃ /mmc) + 0.5 N ₂ (-)	unstable/nonexistent	-1.2880	
TlN	mp-1002222	Fm-3m	TlN $\rightarrow$ Tl (P6 ₃ /mmc) + 0.5 N ₂ (-)	metastable viable	0.1390	
TlN	mp-1007778	F-43m	TlN $\rightarrow$ Tl (P6 ₃ /mmc) + 0.5 N ₂ (-)	unstable/nonexistent	-0.7330	
VCo ₃	mp-1095057	P6 ₃ /mmc	VCo ₃ $\rightarrow$ V (Im-3m) + 3 Co (Fm-3m)	unstable/nonexistent	-8.8379	
VN	mp-1017532	P6 ₃ /mmc	VN $\rightarrow$ V (Im-3m) + 0.5 N ₂ (-)	stable	-1.6291	
VNi ₂ *	mp-11531	Immm	VNi ₂ $\rightarrow$ V (Im-3m) + 2 Ni (Fm-3m)	stable	-3.1047	
W ₉ C ₄	mp-684989	R32	W ₉ C ₄ $\rightarrow$ W (Im-3m) + 0.4444 C (P6 ₃ /mmc)	stable	-13.7265	
WBr ₅	mp-1436789	P-1	WBr ₅ $\rightarrow$ W (Im-3m) + 5 Br (Cmce)	unstable/nonexistent	-1.8227	
WC*	mp-13136	Fm-3m	WC $\rightarrow$ W (Im-3m) + C (P6 ₃ /mmc)	unstable/nonexistent	-6.2684	
WC	mp-1008630	Pm-3m	WC $\rightarrow$ W (Im-3m) + C (P6 ₃ /mmc)	unstable/nonexistent	-5.9400	
WC	mp-1008635	F-43m	WC $\rightarrow$ W (Im-3m) + C (P6 ₃ /mmc)	unstable/nonexistent	-7.7315	
WO ₂ *	mp-1524394	P2 ₁ /c	WO ₂ $\rightarrow$ W (Im-3m) + O ₂ (C2/m)	stable	-4.5973	
Y ₂ O ₃ *	mp-558573	C2/m	Y ₂ O ₃ $\rightarrow$ Y (P6 ₃ /mmc) + 0.75 O ₂ (C2/m)	stable	1.1430	
Y ₃ Co*	mp-1105598	Pnma	Y ₃ Co $\rightarrow$ Y (P6 ₃ /mmc) + 0.3333 Co (Fm-3m)	stable	-2.9097	
Y ₃ Co	mp-1189329	P2 ₁	Y ₃ Co $\rightarrow$ Y (P6 ₃ /mmc) + 0.3333 Co (Fm-3m)	stable	-2.8835	
YAg ₃	mp-1187811	Fm-3m	YAg ₃ $\rightarrow$ Y (P6 ₃ /mmc) + 3 Ag (Fm-3m)	stable	-5.2080	
Zn ₁₃ Rh*	mp-13447	C2/m	Zn ₁₃ Rh $\rightarrow$ Zn (P6 ₃ /mmc) + 0.07692 Rh (Fm-3m)	stable	-6.8206	
Zn ₂ Ag	mp-1093999	P6 ₃ /mmc	Zn ₂ Ag $\rightarrow$ Zn (P6 ₃ /mmc) + 0.5 Ag (Fm-3m)	stable	-7.1508	
Zn ₃ Au	mp-1188000	P6 ₃ /mmc	Zn ₃ Au $\rightarrow$ Zn (P6 ₃ /mmc) + 0.3333 Au (Fm-3m)	stable	-8.6653	
Zn ₃ N ₂	mp-1524551	P6 ₃ /mmc	Zn ₃ N ₂ $\rightarrow$ Zn (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	-4.2961	
Zn ₃ N ₂ *	mp-9460	Ia-3	Zn ₃ N ₂ $\rightarrow$ Zn (P6 ₃ /mmc) + 0.3333 N ₂ (-)	stable	-4.4279	
Zn ₇ Tc	mp-1206378	Fm-3m	Zn ₇ Tc $\rightarrow$ Zn (P6 ₃ /mmc) + 0.1429 Tc (P6 ₃ /mmc)	stable	-6.2328	
ZnAg*	mp-1912	Pm-3m	ZnAg $\rightarrow$ Zn (P6 ₃ /mmc) + Ag (Fm-3m)	stable	-4.5783	
ZnPb ₃	mp-1187949	Fm-3m	ZnPb ₃ $\rightarrow$ Zn (P6 ₃ /mmc) + 3 Pb (Fm-3m)	unstable/nonexistent	-5.8264	
ZnSn	-	P6 ₃ /mmc	ZnSn $\rightarrow$ Zn (P6 ₃ /mmc) + Sn (I4/mmm)	insufficient data	-5.0137	
Zr ₂ Au*	mp-2348	I4/mmm	Zr ₂ Au $\rightarrow$ Zr (P6 ₃ /mmc) + 0.5 Au (Fm-3m)	stable	-2.1792	

Compound	mp-id	Space group	Reaction (products and their space group)	Viability	$\Delta(\text{ICOHP})/\text{at.}$	$\Delta(\text{ICOHP})$
Zr ₃ Be	mp-1188016	I4/mmm	Zr ₃ Be $\rightarrow$ Zr (P6 ₃ /mmc) + 0.3333 Be (P6 ₃ /mmc)	unstable/nonexistent	-1.8795	
Zr ₃ Fe*	mp-31205	Cmcm	Zr ₃ Fe $\rightarrow$ Zr (P6 ₃ /mmc) + 0.3333 Fe (Im-3m)	stable	-1.5836	
Zr ₃ Ge	mp-1188039	P6 ₃ /mmc	Zr ₃ Ge $\rightarrow$ Zr (P6 ₃ /mmc) + 0.3333 Ge (Fd-3m)	stable	-1.6674	
Zr ₅ Pb ₃ *	mp-681992	P6 ₃ /mcm	Zr ₅ Pb ₃ $\rightarrow$ Zr (P6 ₃ /mmc) + 0.6 Pb (Fm-3m)	stable	-2.3350	
ZrBr*	mp-1065846	C2/m	ZrBr $\rightarrow$ Zr (P6 ₃ /mmc) + Br (Cmce)	stable	-0.2489	
ZrGa ₃ *	mp-30685	I4/mmm	ZrGa ₃ $\rightarrow$ Zr (P6 ₃ /mmc) + 3 Ga (Cmce)	stable	1.2248	
ZrHg ₃ *	mp-1608	Pm-3m	ZrHg ₃ $\rightarrow$ Zr (P6 ₃ /mmc) + 3 Hg (P6 ₃ /mmm)	stable	1.1077	
ZrMo ₂ *	mp-2049	Fd-3m	ZrMo ₂ $\rightarrow$ Zr (P6 ₃ /mmc) + 2 Mo (Im-3m)	stable	-6.8696	
ZrO ₂	mp-775909	P4 ₂ /mnm	ZrO ₂ $\rightarrow$ Zr (P6 ₃ /mmc) + O ₂ (C2/m)	stable	-0.5517	
ZrRh*	mp-2808	Pm-3m	ZrRh $\rightarrow$ Zr (P6 ₃ /mmc) + Rh (Fm-3m)	stable	-1.1561	
ZrTi	mp-1215236	P4/mmm	ZrTi $\rightarrow$ Zr (P6 ₃ /mmc) + Ti (P6 ₃ /mmc)	unstable/nonexistent	-0.2533	

D VASP Calculation Parameters

Fixed INCAR parameters, identical across the whole dataset (static, LOBSTER-compatible run) — only NBANDS (appendix F) and the k-point grid vary by compound.

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (-1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMEAR	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (-1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

E PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen’s `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table S8: PAW-PBE potentials used, one per element present in the dataset (pymatgen’s MPRelaxSet library, with one documented exception: W substituted for W_pv/W_sv, see §7).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)
B	B (06Sep2000)
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Bi	Bi (08Apr2002)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)
Hg	Hg (06Sep2000)
I	I (08Apr2002)
In	In_d (06Sep2000)
Ir	Ir (06Sep2000)
K	K_sv (06Sep2000)
Li	Li_sv (10Sep2004)
Mg	Mg_pv (13Apr2007)
Mn	Mn_pv (02Aug2007)
Mo	Mo_pv (04Feb2005)
N	N (08Apr2002)
Na	Na_pv (19Sep2006)
Nb	Nb_pv (08Apr2002)
Ni	Ni_pv (06Sep2000)
O	O (08Apr2002)
Os	Os_pv (20Jan2003)
P	P (06Sep2000)
Pb	Pb_d (06Sep2000)
Pd	Pd (04Jan2005)
Pt	Pt (04Feb2005)
Rb	Rb_sv (06Sep2000)
Re	Re_pv (06Sep2000)
Rh	Rh_pv (25Jan2005)
Ru	Ru_pv (28Jan2005)
S	S (06Sep2000)
Sb	Sb (06Sep2000)
Sc	Sc_sv (07Sep2000)
Se	Se (06Sep2000)
Si	Si (05Jan2001)
Sn	Sn_d (06Sep2000)

Element	PAW-PBE potential
Sr	Sr_sv (07Sep2000)
Ta	Ta_pv (07Sep2000)
Tc	Tc_pv (04Feb2005)
Te	Te (08Apr2002)
Ti	Ti_pv (07Sep2000)
Tl	Tl_d (06Sep2000)
V	V_pv (07Sep2000)
W	W (08Apr2002)
Y	Y_sv (25May2007)
Zn	Zn (06Sep2000)
Zr	Zr_sv (04Jan2005)

F k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ -centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table S9: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As ₂ S ₃	mp-1189572	5 3 2 (G)	80
AsF ₃	mp-28027	5 4 4 (G)	64
BI ₃	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu ₂	mp-30363	6 6 7 (G)	23
BaCl ₂	mp-568662	6 6 6 (G)	13
BaH ₂	mp-23715	6 4 3 (G)	28
Be ₂ Cu	mp-2031	7 7 7 (G)	38
Be ₂ Nb ₃	mp-11272	8 4 4 (M)	74
Be ₂ V	mp-11281	7 7 4 (G)	76
CaAl ₄	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs ₂ As ₃	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge ₂ Os	mp-10032	9 6 4 (G)	54
HfTc ₂	mp-1095669	5 5 3 (G)	108
Hg ₄ Pt	mp-936	5 5 5 (G)	45
HgF ₂	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN ₃	mp-827	5 5 5 (G)	34
Li ₂ Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg ₂ Ni	mp-2137	5 5 2 (G)	102
MgP ₄	mp-384	5 5 3 (G)	40

Table S9 (continued)

Formula	mp-id	k-point grid	NBANDS
MnAl ₁₂	mp-1104165	4 4 4 (M)	57
MoSe ₂	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb ₃ Ir	mp-1458	5 5 5 (G)	72
NbGa ₃	mp-1973	8 8 6 (M)	36
PI ₂	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc ₅ Ge ₃	mp-17190	3 3 5 (G)	154
ScAg ₂	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO ₂	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta ₅ Ge ₃	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb ₂	mp-11697	8 5 4 (G)	34
Te ₂ Pd	mp-782	7 7 5 (G)	17
Te ₂ Ru	mp-267	7 5 4 (G)	34
TeO ₂	mp-8377	6 5 3 (G)	48
Ti ₃ In	mp-866046	5 5 6 (G)	72
Ti ₃ Pt	mp-2339	5 5 5 (G)	72
TiBe ₁₂	mp-1104067	7 5 5 (G)	69
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi ₂	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi ₂	mp-11531	12 8 7 (G)	27
WO ₂	mp-1524394	6 5 5 (G)	68
Zr ₂ Au	mp-2348	9 9 4 (G)	29
Zr ₅ Pb ₃	mp-681992	3 3 5 (G)	154
ZrMo ₂	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI ₂	mp-567259	7 7 4 (G)	17
MgCl ₂	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li ₃ Hg	mp-1646	7 7 7 (G)	24
Ta ₇ Co ₆	mp-1104548	6 6 3 (G)	117
CaGa ₄	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH ₂	mp-23711	6 5 5 (G)	24
Sn ₇ Ir ₅	mp-20466	4 4 4 (M)	108
NaN ₃	mp-1066400	7 7 7 (G)	16
ClF ₃	mp-556767	6 4 3 (G)	64
SiS ₂	mp-1095294	5 4 3 (G)	48
Sr ₅ Sn ₃	mp-17720	3 3 3 (G)	104
Te ₂ Ir	mp-1551	7 5 2 (G)	51
Li ₁₃ Sn ₅	mp-30769	6 6 1 (G)	110

Table S9 (continued)

Formula	mp-id	k-point grid	NBANDS
H ₃ N	mp-643432	8 5 5 (G)	28
Cs ₂ Se	mp-1011696	5 3 2 (G)	56
In ₃ Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb ₂ Hg ₇	mp-31474	4 4 4 (G)	73
PbAu ₂	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO ₂	mp-546794	6 6 6 (M)	24
IF ₇	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al ₃ Os ₂	mp-16521	9 9 3 (G)	30
Nb ₂ O ₅	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd ₃	mp-7753	7 7 7 (G)	31
AsPd ₂	mp-1080831	8 4 4 (G)	66
GeSe ₂	mp-10074	5 5 5 (G)	34
Re ₃ Ge ₇	mp-29141	9 6 1 (G)	180
Cu ₃ Ge	mp-19724	6 6 5 (G)	72
Ti ₃ Ge	mp-1189044	4 4 4 (M)	144
ZrGa ₃	mp-30685	7 7 5 (G)	37
SrAl ₂	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al ₁₂ W	mp-11227	4 4 4 (M)	57
CsH	mp-632319	7 7 7 (G)	6
CaGa ₂	mp-992	7 7 6 (G)	23
NiHg ₄	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn ₅	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te ₂ Au	mp-567525	7 7 5 (G)	17
Be ₃ N ₂	mp-6977	10 10 3 (G)	46
Y ₂ O ₃	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe ₂	mp-1102002	4 4 4 (M)	68
H ₂ O	mp-2739191	7 7 4 (G)	24
CdO ₂	mp-2310	5 5 5 (G)	68
Mo ₃ Se ₄	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be ₁₂ Pd	mp-12666	6 6 6 (M)	69
Sr ₂ As	mp-15697	6 6 3 (G)	28
AsRh ₂	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K ₂ S	mp-559278	5 5 4 (G)	28
Ge ₄ Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li ₃ Ag	mp-976408	7 7 7 (G)	24
Y ₃ Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr ₃ As ₄	mp-678007	3 3 4 (G)	62
Ga ₂ Se ₃	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13

Table S9 (continued)

Formula	mp-id	k-point grid	NBANDS
BeCl ₂	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTi ₃	mp-1206636	5 5 5 (G)	32
IrCl ₃	mp-729507	3 3 6 (G)	84
Ta ₄ C ₃	mp-1218000	6 6 6 (M)	48
Sc ₂ O ₃	mp-755313	6 5 6 (G)	64
SnBr ₂	mp-978985	4 4 6 (M)	34
Sn ₇ Pd ₅	mp-1219006	5 5 3 (G)	108
Tl ₄ Sn	mp-1216560	5 5 5 (G)	45
LiTi ₃	mp-1185253	6 6 6 (M)	32
PtCl ₂	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K ₅ As ₄	mp-684799	4 4 3 (G)	41
KSn ₃	mp-1185151	4 4 5 (G)	64
NbS ₂	mp-774873	10 6 2 (M)	34
YAg ₃	mp-1187811	6 6 6 (G)	37
B ₃ Ir ₂	mp-1228647	9 5 4 (G)	60
TiMo ₂	mp-1216675	5 6 14 (G)	27
Cr ₂ C	mp-1226378	10 10 7 (G)	22
ZrO ₂	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN ₂	mp-1097011	5 5 5 (G)	24
CoSe ₂	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo ₃	mp-1095057	7 6 6 (G)	72
Tl ₃ Ga	mp-1187539	5 5 5 (G)	36
CaGe ₂	mp-643542	7 7 6 (G)	23
NbRh ₂	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF ₂	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na ₂ Cl	mp-990084	6 6 6 (M)	12
ZnPb ₃	mp-1187949	5 5 5 (G)	36
Sr ₃ Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al ₄ Si	mp-1228120	6 6 6 (M)	20
B ₆ Pb	mp-1096825	6 6 6 (M)	33
ScTc ₃	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd ₅	mp-1228665	6 6 6 (M)	49
Zr ₃ Be	mp-1188016	6 6 6 (M)	35
Na ₃ Cl	mp-1064484	8 8 3 (G)	16
SiAu ₃	mp-1186998	7 7 7 (G)	31
InSe ₂	mp-1232036	7 7 3 (G)	34
TiSi ₂	mp-570991	8 8 8 (M)	17
BeAu ₃	mp-983539	7 7 7 (G)	32
Zr ₃ Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg ₁₅ B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

G Elements and Allotropes

Every single-element structure with no matched polymorph in the dataset: the elemental references (**ELEMENT_REFERENCE**) used to decompose multi-element compounds. An experimental compound’s formula is marked with a superscript star (black if it is the stable phase, red if it is a metastable polymorph/allotrope); unmarked formulas are theoretical. Auto-generated by `report/gen_appendix_extension.py`.

Table S10: Elements and allotropes with no matched polymorph in the **extension** campaign: references used to decompose multi-element compounds. Allotropes with one or more other allotropes computed in this dataset (e.g. carbon) are grouped in Table S11 instead. Black * = stable experimental reference (thermodynamic phase); red * = metastable experimental reference; unmarked = theoretical. No Δ ICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6 ₃ /mmc	0.0000
Br*	mp-23154	Cmce	0.0000
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6 ₃ /mmc	0.0000
Cl ₂ *	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F ₂ *	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H ₂ *	mp-730101	P2 ₁₂ ₁₂ ₁	0.0000
Hf*	mp-103	P6 ₃ /mmc	0.0000
Hg*	mp-10861	P6/mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4/mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6 ₃ /mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N ₂	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O ₂ *	mp-1524462	C2/m	0.0000
Os*	mp-49	P6 ₃ /mmc	0.0000
P*	mp-568348	P2/c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Re*	mp-8	P6 ₃ /mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6 ₃ /mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000
Sc*	mp-67	P6 ₃ /mmc	0.0000
Se*	mp-14	P3 ₁ 21	0.0011
Sn*	mp-623511	I4/mmm	0.0614
Sr*	mp-139	P6 ₃ /mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6 ₃ /mmc	0.0000
Te*	mp-19	P3 ₁ 21	0.0000
Ti*	mp-46	P6 ₃ /mmc	0.0152
Tl*	mp-82	P6 ₃ /mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000
Y*	mp-112	P6 ₃ /mmc	0.0000
Zn*	mp-79	P6 ₃ /mmc	0.0000
Zr*	mp-131	P6 ₃ /mmc	0.0000

H Polymorphs and Allotropes

Elements and compounds with several entries sharing a formula in the dataset, grouped here rather than scattered across the elements/compounds tables (see the table caption for detail).

Table S11: Elements (allotropes) and compounds (polymorphs) with **several entries sharing a formula** in the **extension** campaign (54 groups, 130 entries) — grouped here rather than scattered across the elements/compounds tables, since the natural comparison for a polymorph group is polymorph-to-polymorph, not the decomposition-to-elements reaction (hence no ΔICOHP column). A rule separates each group; within a group, sorted by increasing E_{hull} . Black * = stable experimental; red * = metastable experimental; unmarked = theoretical.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
AgN*	mp-1091417	Cmce	1.5201
AgN	mp-1009766	F-43m	1.5495
AgN	mp-1428312	Pm-3m	1.6519
AgN	mp-1424734	Ama2	1.9196
BaF ₃ *	mp-1080821	P2 ₁ /m	0.0000
BaF ₃	mp-1183303	I4/mmm	0.0612
BaO*	mp-1008500	P6 ₃ /mmc	0.0189
BaO	mp-1950989	P6 ₃ /mmc	0.0434
Be ₃ N ₂ *	mp-18337	Ia-3	0.0000
Be ₃ N ₂	mp-1017550	P-3m1	0.2081
Be ₃ P ₂ *	mp-567841	Ia-3	0.0000
Be ₃ P ₂	mp-1084771	Pn-3m	0.1567
BeCl ₂ *	mp-23267	Ibam	0.0087
BeCl ₂	mp-1172909	I-43m	3.1688
BeF ₂ *	mp-15951	P3 ₁ 21	0.0004

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
BeF ₂	mp-975644	P6/mmm	2.5181
BeO	mp-1778	F-43m	0.0070
BeO*	mp-7599	P4 ₂ /mnm	0.0153
Bi ₂ O ₃ *	mp-558953	P2 ₁ /c	0.6000
Bi ₂ O ₃	mp-674644	P1	0.6199
Bi ₂ O ₃	mp-1383505	P1	0.9068
Bi ₂ O ₃	mp-714859	P1	2.4029
C*	mp-48	P6 ₃ /mmc	0.0031
C*	mp-66	Fd-3m	0.1123
C*	mp-47	P6 ₃ /mmc	0.1395
C	mp-1190171	Pnma	0.2927
C	mp-1080826	C2/m	0.3009
C ₃ N ₄ *	mp-571653	P-43m	0.4981
C ₃ N ₄	mp-2852	I-43d	0.5033
C ₃ N ₄	mp-1182484	P6 ₃ /m	0.5848
C ₃ N ₄	mp-563	Fd-3m	1.5119
Ca ₃ N ₂ *	mp-844	Ia-3	0.0000
Ca ₃ N ₂ *	mp-1047	R-3c	0.0167
Ca ₃ N ₂	mp-1013524	Pm-3m	0.5112
CaCl ₂ *	mp-23214	Pnnm	0.0011
CaCl ₂	mp-1120739	Fm-3m	0.0422
CaF ₂ *	mp-10464	Pnma	0.0346
CaF ₂	mp-554355	P4/mmm	0.2403
CaO*	mp-2605	Fm-3m	0.0000
CaO*	mp-1079707	Cmce	0.2098
CaO	mp-1181903	Fd-3m	2.7281
CsN ₃ *	mp-510557	I4/mcm	0.0000
CsN ₃	mp-1225892	P4/mmm	0.0113
CsO ₂ *	mp-1441	I4/mmm	0.0349
CsO ₂	mp-1096936	Cmmm	0.0940
CsP ₇ *	mp-1201790	Pbcm	0.0000
CsP ₇	mp-1213206	Pca2 ₁	0.0001
K ₂ O*	mp-971	Fm-3m	0.0000
K ₂ O	mp-1223784	P4/mmm	0.2000
KF ₃	mp-2749823	P2 ₁	0.0119
KF ₃ *	mp-1544557	P2 ₁₂ ₁₂ ₁	0.0125
KN ₂ *	mp-1071868	Cmc2 ₁	1.1630
KN ₂	mp-1180757	Cmcm	1.2293
KP*	mp-1058315	R-3m	1.0080
KP	mp-1057787	Immm	1.0092
Li ₂ O*	mp-1960	Fm-3m	0.0000
Li ₂ O	mp-755894	Pnma	0.0844
Li ₂ S*	mp-1125	Pnma	0.0615
Li ₂ S	mp-557142	P6 ₃ /mmc	0.1750
LiCl	mp-1185319	P6 ₃ mc	0.0039
LiCl*	mp-22905	Fm-3m	0.0243
LiF*	mp-1138	Fm-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
LiF	mp-1009009	Pm-3m	0.2875
LiP ₅ *	mp-2412	Pna2_1	0.0077
LiP ₅	mp-32760	Pna2_1	0.0963
MgCl ₂ *	mp-23210	R-3m	0.0000
MgCl ₂	mp-569626	Ama2	0.1991
MgF ₂ *	mp-1072956	Pnnm	0.0025
MgF ₂	mp-1062842	Amm2	0.2636
MgO*	mp-549706	P6_3mc	0.1180
MgO	mp-1009127	Pm-3m	0.7484
MgS*	mp-1315	Fm-3m	0.0000
MgS	mp-13032	F-43m	0.0343
Na ₂ O*	mp-2352	Fm-3m	0.0000
Na ₂ O	mp-1975297	C2/m	0.0981
NaN ₃ *	mp-22003	R-3m	0.0017
NaN ₃	mp-1065265	C2/m	1.0162
NaS*	mp-409	P-62m	0.0023
NaS	mp-1180106	C2/m	1.0061
NbO*	mp-2692	Fm-3m	0.7888
NbO	mp-634965	P4/mmm	0.8240
NbP*	mp-9830	I4_1/amd	0.6021
NbP	mp-1220317	Cmmm	0.6209
OsC*	mp-7142	P-6m2	0.9161
OsC	mp-999308	P6_3/mmc	0.9278
OsC	mp-1009539	Fm-3m	1.3918
OsC	mp-1009537	Pm-3m	1.4795
Rb ₂ O*	mp-1394	Fm-3m	0.0032
Rb ₂ O	mp-2023482	P-3m1	0.0479
RbF	mp-11718	Fm-3m	0.0000
RbF*	mp-2064	Pm-3m	0.0516
RbN ₃ *	mp-581833	P4/mmm	0.0193
RbN ₃	mp-1219567	Amm2	0.7966
RbP*	mp-1179688	C2/m	0.9879
RbP	mp-1058499	Immm	0.9882
RbS*	mp-558071	Immm	0.0019
RbS	mp-1179684	C2/m	0.9975
ReC*	mp-1079494	P6_3/mmc	0.6362
ReC	mp-1009731	Pm-3m	1.1150
RuC*	mp-10030	P-6m2	0.6494
RuC	mp-1009787	Fm-3m	0.9097
SN*	COD:7017102	P2_1/c	—
SN*	mp-236	P2_1/c	0.5087
SN	mp-684642	P2_1	0.5498
SN	mp-1173453	P2_1	0.7225
SN	mp-1078816	P2_1/c	0.8090
SiO ₂ *	mp-9258	Pa-3	0.5770
SiO ₂	mp-10064	Fm-3m	1.2779
SiO ₂	mp-556044	P-31c	1.4574

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
SiO ₂	mp-638049	F4_132	1.7448
Sr ₃ P ₄ *	mp-14288	Fdd2	0.0000
Sr ₃ P ₄	mp-682953	P1	0.0081
SrF ₂	mp-1102911	Pnma	0.0267
SrF ₂ *	mp-1102240	Pnma	0.0646
SrO ₂ *	mp-726705	Pnma	0.0096
SrO ₂	mp-2763179	Pnma	0.0110
TaN*	mp-1497	P6/mmm	0.4516
TaN	mp-1009833	F-43m	0.4917
TaN	mp-1009831	Pm-3m	0.8163
TiO ₂ *	mp-1439	Pbcn	0.0045
TiO ₂ *	mp-430	P2_1/c	0.0395
TiO ₂ *	mp-9173	Pnma	0.0575
TiC*	mp-567465	Fm-3m	2.1080
TiC	mp-1058887	P4/mmm	2.3757
TiN*	mp-1017982	P6_3mc	0.7334
TiN	mp-1007778	F-43m	0.7337
TiN	mp-1002222	Fm-3m	0.9712
WC*	mp-13136	Fm-3m	0.4477
WC	mp-1008635	F-43m	0.6729
WC	mp-1008630	Pm-3m	0.8990

I Target Compounds

Every multi-element target compound with no matched polymorph in the dataset, with Δ ICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_case1_full.csv` by `report/gen_appendix_extension.py`.

Table S12: Target compounds with no matched polymorph in the **extension** campaign. Formulas with several entries in this dataset are grouped in Table S11 instead. Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. Δ ICOHP per atom, products – reactants convention (`reaction_analysis`); “–” where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	Δ ICOHP/at. (eV)	Label
Ba ₅ P ₄ *	mp-17566	Pnma	0.0000	-0.9590	exobondic
BaCl ₂ *	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaN ₂ *	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaS ₃ *	mp-556296	P2_12_12	0.0436	-0.6233	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
CS ₂ S*	mp-1077814	P6_3/mmc	0.0667	0.9554	endobondic
CSCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CSF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
K ₂ S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KN ₃	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
Li ₃ N*	mp-2341	P6_3/mmc	0.0039	0.6165	endobondic

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)	Label
Mn_2O_7^*	mp-28338	P2_1/c	0.3346	-1.4687	exobondic
MnO_2^*	mp-510408	P4_2/mnm	0.0464	-0.5274	exobondic
Na_2Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
PbN_6^*	mp-667338	Pnma	0.1093	-2.5257	exobondic
RbCl^*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
S_2N^*	COD:4031496	P4_2nm	—	-0.9767	exobondic
SrN^*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrS^*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic

J Endobondic Reactions

The subset of case-1 decomposition reactions ($A_a B_b \rightarrow a A + b B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products — reactants convention — note this is the sign *opposite* `reaction_icohp.py`’s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`’s docstring).

Table S13: Decomposition-into-elements (case 1) reactions labeled **endobondic** ($\Delta\text{ICOHP} \geq 0$, products — reactants) among the 208 compounds of the **extension** campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP}/\text{at.}$ (eV)
0.3333 $\text{Be}_3\text{N}_2 \rightarrow \text{Be} + 0.3333 \text{N}_2$	mp-1017550	0.2081	0.3960
$\text{BeCl}_2^* \rightarrow \text{Be} + \text{Cl}_2$	mp-23267	0.0087	1.8204
$\text{BeF}_2^* \rightarrow \text{Be} + \text{F}_2$	mp-15951	0.0004	3.1514
$\text{BeO}^* \rightarrow \text{Be} + 0.5 \text{O}_2$	mp-7599	0.0153	2.7864
$\text{BeO} \rightarrow \text{Be} + 0.5 \text{O}_2$	mp-1778	0.0070	4.4391
0.3333 $\text{C}_3\text{N}_4^* \rightarrow \text{C} + 0.6667 \text{N}_2$	mp-571653	0.4981	1.1208
0.3333 $\text{C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{N}_2$	mp-2852	0.5033	1.8456
0.5 $\text{Cs}_2\text{S}^* \rightarrow \text{Cs} + 0.5 \text{S}$	mp-1077814	0.0667	0.9554
$\text{CsCl}^* \rightarrow \text{Cs} + 0.5 \text{Cl}_2$	mp-573697	0.0071	0.8830
$\text{CsF}^* \rightarrow \text{Cs} + 0.5 \text{F}_2$	mp-8455	0.1088	3.8464
$\text{CsO}_2^* \rightarrow \text{Cs} + \text{O}_2$	mp-1441	0.0349	0.0666
$\text{CsO}_2 \rightarrow \text{Cs} + \text{O}_2$	mp-1096936	0.0940	0.7743
$\text{KCl}^* \rightarrow \text{K} + 0.5 \text{Cl}_2$	mp-23289	0.0776	0.7616
0.5 $\text{Li}_2\text{O}^* \rightarrow \text{Li} + 0.25 \text{O}_2$	mp-1960	0.0000	2.3366
0.5 $\text{Li}_2\text{O} \rightarrow \text{Li} + 0.25 \text{O}_2$	mp-755894	0.0844	1.9481
0.5 $\text{Li}_2\text{S}^* \rightarrow \text{Li} + 0.5 \text{S}$	mp-1125	0.0615	1.8505
0.5 $\text{Li}_2\text{S} \rightarrow \text{Li} + 0.5 \text{S}$	mp-557142	0.1750	1.5052
0.3333 $\text{Li}_3\text{N}^* \rightarrow \text{Li} + 0.1667 \text{N}_2$	mp-2341	0.0039	0.6165
$\text{LiCl}^* \rightarrow \text{Li} + 0.5 \text{Cl}_2$	mp-22905	0.0243	2.8030
$\text{LiCl} \rightarrow \text{Li} + 0.5 \text{Cl}_2$	mp-1185319	0.0039	2.5005
$\text{LiF}^* \rightarrow \text{Li} + 0.5 \text{F}_2$	mp-1138	0.0000	4.4673
$\text{LiF} \rightarrow \text{Li} + 0.5 \text{F}_2$	mp-1009009	0.2875	4.5131
$\text{LiP}_5^* \rightarrow \text{Li} + 5 \text{P}$	mp-2412	0.0077	0.9043
$\text{LiP}_5 \rightarrow \text{Li} + 5 \text{P}$	mp-32760	0.0963	0.8692
$\text{MgCl}_2^* \rightarrow \text{Mg} + \text{Cl}_2$	mp-23210	0.0000	0.1568
$\text{MgF}_2^* \rightarrow \text{Mg} + \text{F}_2$	mp-1072956	0.0025	0.0015
$\text{MgF}_2 \rightarrow \text{Mg} + \text{F}_2$	mp-1062842	0.2636	0.0229
0.5 $\text{Rb}_2\text{O}^* \rightarrow \text{Rb} + 0.25 \text{O}_2$	mp-1394	0.0032	0.7235
$\text{RbCl}^* \rightarrow \text{Rb} + 0.5 \text{Cl}_2$	mp-23299	0.0298	1.7924
$\text{RbF}^* \rightarrow \text{Rb} + 0.5 \text{F}_2$	mp-2064	0.0516	3.9244
$\text{RbF} \rightarrow \text{Rb} + 0.5 \text{F}_2$	mp-11718	0.0000	1.9497
$\text{SiO}_2^* \rightarrow \text{Si} + \text{O}_2$	mp-9258	0.5770	3.3332

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{SiO}_2 \rightarrow \text{Si} + \text{O}_2$	mp-10064	1.2779	2.1735
$\text{SiO}_2 \rightarrow \text{Si} + \text{O}_2$	mp-556044	1.4574	0.0958
$\text{SiO}_2 \rightarrow \text{Si} + \text{O}_2$	mp-638049	1.7448	0.3849

K Exobondic Reactions

The complementary **exobondic** subset ($\Delta\text{ICOHP} < 0$), same convention and provenance as Appendix J.

Table S14: Decomposition-into-elements (case 1) reactions labeled **exobondic** ($\Delta\text{ICOHP} < 0$, products – reactants) among the 208 compounds of the **extension** campaign. * = experimental starting compound (unmarked = theoretical).

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{AgN}^* \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1091417	1.5201	-7.5875
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1009766	1.5495	-6.4957
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1424734	1.9196	-8.0907
$\text{AgN} \rightarrow \text{Ag} + 0.5 \text{ N}_2$	mp-1428312	1.6519	-5.3559
$0.2 \text{ Ba}_5\text{P}_4^* \rightarrow \text{Ba} + 0.8 \text{ P}$	mp-17566	0.0000	-0.9590
$\text{BaCl}_2^* \rightarrow \text{Ba} + \text{Cl}_2$	mp-23199	0.0068	-0.2244
$\text{BaF}_3^* \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1080821	0.0000	-0.8680
$\text{BaF}_3 \rightarrow \text{Ba} + 1.5 \text{ F}_2$	mp-1183303	0.0612	-0.2493
$\text{BaN}_2^* \rightarrow \text{Ba} + \text{N}_2$	mp-1102371	2.0568	-8.6248
$\text{BaO}^* \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1008500	0.0189	-0.3221
$\text{BaO} \rightarrow \text{Ba} + 0.5 \text{ O}_2$	mp-1950989	0.0434	-1.6129
$\text{BaS}_3^* \rightarrow \text{Ba} + 3 \text{ S}$	mp-556296	0.0436	-0.6233
$0.3333 \text{ Be}_3\text{N}_2^* \rightarrow \text{Be} + 0.3333 \text{ N}_2$	mp-18337	0.0000	-0.0746
$0.3333 \text{ Be}_3\text{P}_2^* \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-567841	0.0000	-0.2111
$0.3333 \text{ Be}_3\text{P}_2 \rightarrow \text{Be} + 0.6667 \text{ P}$	mp-1084771	0.1567	-0.1795
$\text{BeCl}_2 \rightarrow \text{Be} + \text{Cl}_2$	mp-1172909	3.1688	-5.7807
$\text{BeF}_2 \rightarrow \text{Be} + \text{F}_2$	mp-975644	2.5181	-0.6816
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-1182484	0.5848	-3.6399
$0.3333 \text{ C}_3\text{N}_4 \rightarrow \text{C} + 0.6667 \text{ N}_2$	mp-563	1.5119	-0.4498
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1047	0.0167	-4.4881
$0.3333 \text{ Ca}_3\text{N}_2^* \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-844	0.0000	-4.4374
$0.3333 \text{ Ca}_3\text{N}_2 \rightarrow \text{Ca} + 0.3333 \text{ N}_2$	mp-1013524	0.5112	-4.4898
$\text{CaCl}_2^* \rightarrow \text{Ca} + \text{Cl}_2$	mp-23214	0.0011	-0.3199
$\text{CaCl}_2 \rightarrow \text{Ca} + \text{Cl}_2$	mp-1120739	0.0422	-0.2208
$\text{CaF}_2^* \rightarrow \text{Ca} + \text{F}_2$	mp-10464	0.0346	-0.2227
$\text{CaF}_2 \rightarrow \text{Ca} + \text{F}_2$	mp-554355	0.2403	-0.1814
$\text{CaN}^* \rightarrow \text{Ca} + 0.5 \text{ N}_2$	mp-1058549	0.3982	-4.9183
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1079707	0.2098	-2.2127
$\text{CaO}^* \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-2605	0.0000	-0.9525
$\text{CaO} \rightarrow \text{Ca} + 0.5 \text{ O}_2$	mp-1181903	2.7281	-2.9971
$\text{CsN}_3^* \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-510557	0.0000	-2.2903
$\text{CsN}_3 \rightarrow \text{Cs} + 1.5 \text{ N}_2$	mp-1225892	0.0113	-1.6664
$\text{CsP}_7^* \rightarrow \text{Cs} + 7 \text{ P}$	mp-1201790	0.0000	-0.2978
$\text{CsP}_7 \rightarrow \text{Cs} + 7 \text{ P}$	mp-1213206	0.0001	-0.2667
$0.5 \text{ K}_2\text{O}^* \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-971	0.0000	-0.4357
$0.5 \text{ K}_2\text{O} \rightarrow \text{K} + 0.25 \text{ O}_2$	mp-1223784	0.2000	-1.5638
$0.5 \text{ K}_2\text{S}^* \rightarrow \text{K} + 0.5 \text{ S}$	mp-1022	0.0000	-0.3359
$\text{KF}_3^* \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-1544557	0.0125	-1.1511
$\text{KF}_3 \rightarrow \text{K} + 1.5 \text{ F}_2$	mp-2749823	0.0119	-1.1505

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	Δ ICOHP/at. (eV)
$\text{KN}_2^* \rightarrow \text{K} + \text{N}_2$	mp-1071868	1.1630	-3.4244
$\text{KN}_2 \rightarrow \text{K} + \text{N}_2$	mp-1180757	1.2293	-4.2865
$\text{KN}_3 \rightarrow \text{K} + 1.5 \text{ N}_2$	mp-636056	1.0370	-5.9991
$\text{KP}^* \rightarrow \text{K} + \text{P}$	mp-1058315	1.0080	-2.1600
$\text{KP} \rightarrow \text{K} + \text{P}$	mp-1057787	1.0092	-2.1366
$\text{MgCl}_2 \rightarrow \text{Mg} + \text{Cl}_2$	mp-569626	0.1991	-0.0862
$\text{MgO}^* \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-549706	0.1180	-1.4179
$\text{MgO} \rightarrow \text{Mg} + 0.5 \text{ O}_2$	mp-1009127	0.7484	-1.0763
$\text{MgS}^* \rightarrow \text{Mg} + \text{S}$	mp-1315	0.0000	-0.3401
$\text{MgS} \rightarrow \text{Mg} + \text{S}$	mp-13032	0.0343	-0.4163
$0.5 \text{ Mn}_2\text{O}_7^* \rightarrow \text{Mn} + 1.75 \text{ O}_2$	mp-28338	0.3346	-1.4687
$\text{MnO}_2^* \rightarrow \text{Mn} + \text{O}_2$	mp-510408	0.0464	-0.5274
$0.5 \text{ Na}_2\text{Cl} \rightarrow \text{Na} + 0.25 \text{ Cl}_2$	mp-1077015	0.2391	-0.3965
$0.5 \text{ Na}_2\text{O}^* \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-2352	0.0000	-1.3057
$0.5 \text{ Na}_2\text{O} \rightarrow \text{Na} + 0.25 \text{ O}_2$	mp-1975297	0.0981	-1.5619
$\text{NaN}_3^* \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-22003	0.0017	-1.5201
$\text{NaN}_3 \rightarrow \text{Na} + 1.5 \text{ N}_2$	mp-1065265	1.0162	-5.4248
$\text{NaS}^* \rightarrow \text{Na} + \text{S}$	mp-409	0.0023	-0.4911
$\text{NaS} \rightarrow \text{Na} + \text{S}$	mp-1180106	1.0061	-2.3999
$\text{NbO}^* \rightarrow \text{Nb} + 0.5 \text{ O}_2$	mp-2692	0.7888	-2.3179
$\text{NbO} \rightarrow \text{Nb} + 0.5 \text{ O}_2$	mp-634965	0.8240	-3.1495
$\text{NbP}^* \rightarrow \text{Nb} + \text{P}$	mp-9830	0.6021	-2.4724
$\text{NbP} \rightarrow \text{Nb} + \text{P}$	mp-1220317	0.6209	-1.1731
$\text{OsC}^* \rightarrow \text{Os} + \text{C}$	mp-7142	0.9161	-2.0168
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-1009537	1.4795	-2.3184
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-1009539	1.3918	-3.1141
$\text{OsC} \rightarrow \text{Os} + \text{C}$	mp-999308	0.9278	-2.9572
$\text{PbN}_6^* \rightarrow \text{Pb} + 3 \text{ N}_2$	mp-667338	0.1093	-2.5257
$0.5 \text{ Rb}_2\text{O} \rightarrow \text{Rb} + 0.25 \text{ O}_2$	mp-2023482	0.0479	-0.7592
$\text{RbN}_3^* \rightarrow \text{Rb} + 1.5 \text{ N}_2$	mp-581833	0.0193	-1.3478
$\text{RbN}_3 \rightarrow \text{Rb} + 1.5 \text{ N}_2$	mp-1219567	0.7966	-3.5636
$\text{RbP}^* \rightarrow \text{Rb} + \text{P}$	mp-1179688	0.9879	-1.9315
$\text{RbP} \rightarrow \text{Rb} + \text{P}$	mp-1058499	0.9882	-1.9104
$\text{RbS}^* \rightarrow \text{Rb} + \text{S}$	mp-558071	0.0019	-0.3217
$\text{RbS} \rightarrow \text{Rb} + \text{S}$	mp-1179684	0.9975	-2.7347
$\text{ReC}^* \rightarrow \text{Re} + \text{C}$	mp-1079494	0.6362	-2.6625
$\text{ReC} \rightarrow \text{Re} + \text{C}$	mp-1009731	1.1150	-1.9955
$\text{RuC}^* \rightarrow \text{Ru} + \text{C}$	mp-10030	0.6494	-2.1301
$\text{RuC} \rightarrow \text{Ru} + \text{C}$	mp-1009787	0.9097	-2.8605
$0.5 \text{ S}_2\text{N}^* \rightarrow \text{S} + 0.25 \text{ N}_2$	COD:4031496	—	-0.9767
$\text{SN}^* \rightarrow \text{S} + 0.5 \text{ N}_2$	COD:7017102	—	-1.3963
$\text{SN}^* \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-236	0.5087	-2.3179
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-1078816	0.8090	-3.7082
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-1173453	0.7225	-3.1673
$\text{SN} \rightarrow \text{S} + 0.5 \text{ N}_2$	mp-684642	0.5498	-1.7362
$0.3333 \text{ Sr}_3\text{P}_4^* \rightarrow \text{Sr} + 1.333 \text{ P}$	mp-14288	0.0000	-1.0472
$0.3333 \text{ Sr}_3\text{P}_4 \rightarrow \text{Sr} + 1.333 \text{ P}$	mp-682953	0.0081	-1.0793
$\text{SrF}_2^* \rightarrow \text{Sr} + \text{F}_2$	mp-1102240	0.0646	-0.5421
$\text{SrF}_2 \rightarrow \text{Sr} + \text{F}_2$	mp-1102911	0.0267	-0.1481
$\text{SrN}^* \rightarrow \text{Sr} + 0.5 \text{ N}_2$	mp-1058586	0.4343	-4.5395
$\text{SrO}_2^* \rightarrow \text{Sr} + \text{O}_2$	mp-726705	0.0096	-1.9205
$\text{SrO}_2 \rightarrow \text{Sr} + \text{O}_2$	mp-2763179	0.0110	-1.7522
$\text{SrS}^* \rightarrow \text{Sr} + \text{S}$	mp-10627	0.2996	-0.1088
$\text{TaN}^* \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1497	0.4516	-5.9309
$\text{TaN} \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1009831	0.8163	-4.0540
$\text{TaN} \rightarrow \text{Ta} + 0.5 \text{ N}_2$	mp-1009833	0.4917	-5.3385

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	$\Delta\text{ICOHP/at.}$ (eV)
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-1439	0.0045	-0.6461
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-9173	0.0575	-0.6509
$\text{TiO}_2^* \rightarrow \text{Ti} + \text{O}_2$	mp-430	0.0395	-0.6267
$\text{TlC}^* \rightarrow \text{Tl} + \text{C}$	mp-567465	2.1080	-3.1404
$\text{TlC} \rightarrow \text{Tl} + \text{C}$	mp-1058887	2.3757	-3.3638
$\text{TlN}^* \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1017982	0.7334	-3.7498
$\text{TlN} \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1002222	0.9712	-2.3228
$\text{TlN} \rightarrow \text{Tl} + 0.5 \text{ N}_2$	mp-1007778	0.7337	-3.1948
$\text{WC}^* \rightarrow \text{W} + \text{C}$	mp-13136	0.4477	-6.2684
$\text{WC} \rightarrow \text{W} + \text{C}$	mp-1008630	0.8990	-5.9400
$\text{WC} \rightarrow \text{W} + \text{C}$	mp-1008635	0.6729	-7.7315

L ICOHP and ICOBI Antibonding Descriptors

L.1 Implementation

The ICOHP near-Fermi-level antibonding-population descriptor (§6.2, `cohp_extraction.py`) is extended here to ICOBI: same one-sided window ($E_{\text{ref}} - \Delta E$, E_{ref}], same E_{ref} (Fermi level for metals, VBM for gapped compounds — logic entirely independent of the integrated quantity), read this time from `COBICAR.lobster/ICOBILIST.lobster` rather than `COHPCAR.lobster/ICOHPLIST.lobster` (`pymatgen.io.lobster.outputs.Cohpcar(are_cobis=True)`).

Critical point, empirically verified, not assumed: ICOBI does **not** follow ICOHP’s sign convention. Whereas “negative ICOHP = bonding” (§2.4), ICOBI follows the opposite convention, “positive ICOBI = bonding” (like COOP, not COHP). Verified twice against real data: (1) exact match with `ICOBILIST.lobster` (maximum deviation 0.0 across 66 bond labels, `tests/test_cohp_extraction.py`) — `pymatgen` does not silently renormalize the sign when moving from COHP to COBI; (2) on the strongest Si–O bond of `extension_SiO2_anchor` (ICOHP = -5.599 , ICOBI = $+0.513$), the deep bonding region (-10 to -3 eV, far from any boundary artifact) gives a mean COHP(E) of -0.338 but a mean COBI(E) of $+0.041$ — opposite signs in the same physically-bonding region. The antibonding part of ICOBI is therefore its *negative* part, not its positive part — the truncation direction (`integrate_icobi_antibonding_in_window()`) is deliberately the mirror image of the one used for ICOHP, not a copy of it.

Computed for the 588 compounds with both a `COBICAR.lobster` and a `COHPCAR.lobster`, `is_metal` available for all of them (`analysis/compute_icobi_antibonding_all.py`, `analysis/compute_antibonding_all_full.py` — one compound, Cu₄Au, excluded for a catastrophic LOBSTER-projection failure, §9).

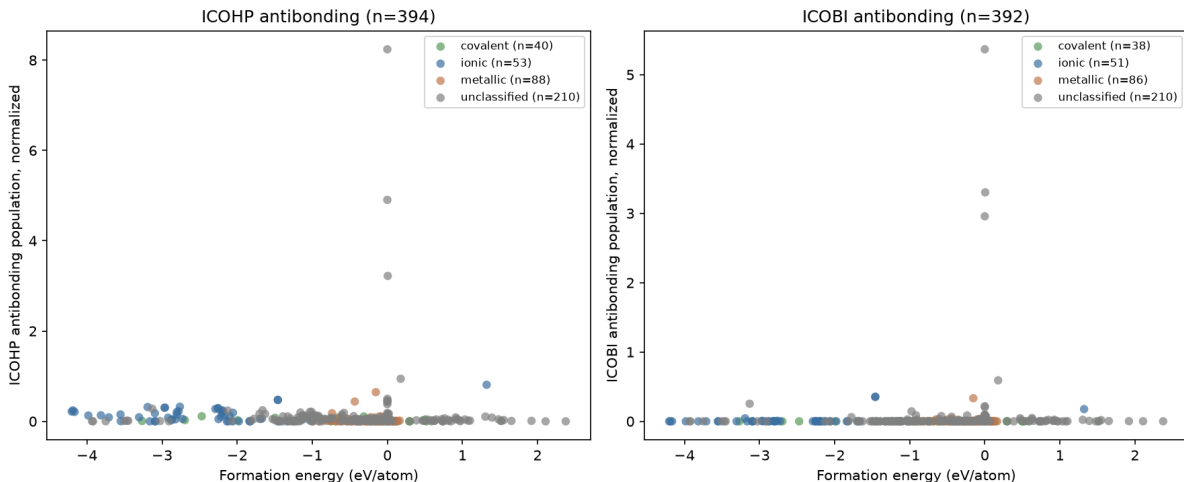
Table S15: Spearman correlations (normalized, $\Delta E = 1.0$ eV) against `formation_energy_per_atom` — ICOHP and ICOBI antibonding, same population ($n = 588$).

Group	n (ICOHP)	ρ (ICOHP)	n (ICOBI)	ρ (ICOBI)
all	579	−0.1550***	579	+0.2308***
covalent	54	−0.0464	54	+0.1129
ionic	96	−0.1958	96	+0.1615
metallic	367	+0.0563	367	+0.1197*
mixed	62	−0.5376***	62	+0.0251
<code>is_metal=False</code>	212	−0.3334***	212	−0.0108
<code>is_metal=True</code>	367	+0.0563	367	+0.1197*

* $p < 0.05$; *** $p < 0.001$; no star = not significant. Full detail (raw + normalized, all three ΔE choices):

`analysis/stats_summary_antibonding_joint.json`.

L.2 Correlation to formation energy



The two descriptors still behave differently, and *neither* keeps its former flagship stratified result. The overall sign of the ICOBI antibonding population remains *opposite* to the ICOHP antibonding population (+0.23 vs. −0.16) — for ICOHP, more occupied antibonding character near E_F is associated with a *more* stable formation (already noted as counter-intuitive in §6.2); for ICOBI, the relationship runs the naively-expected way. **The covalent subgroup of the ICOHP antibonding population, long the project’s most robust stratified result** ($\rho = -0.551$ at $n = 33$), **no longer survives at $n = 54$** ($\rho = -0.046$, $p = 0.74$) — retracted, §6.2. **New at this scale:** `bond_type=mixed` (Zintl) dominates the ICOHP antibonding population ($\rho = -0.538$, $p \approx 0$), a category that barely existed when the covalent result was established; `is_metal=False` remains the most consistently significant stratum for the ICOHP descriptor across this project’s successive scales. For ICOBI, only `metallic` reaches significance (weak, $p = 0.022$); no other ICOBI subgroup survives, as before. The overall ICOBI correlation therefore remains, as with the full reaction-ICOHP descriptor (§6.2), most likely a between-bond-type clustering effect rather than a robust within-group relationship.

L.3 $\Delta(\text{antibonding})$ of the decomposition-into-elements reaction

For a case-1 reaction (compound $\rightarrow$ elements), $\Delta(\text{antibonding})$ is **not** computed by `reaction_analysis/delta.py` (which combines *extensive* per-formula-unit ICOHP/ICOBI sums with integer coefficients): the near-frontier antibonding population is built from LOBSTER’s “average” trace, an *intensive* quantity (averaged over all bonds, not summed) with

no natural extensive decomposition. Convention adopted here, documented as an explicit methodological choice rather than a silent reuse: Δ = (atomic-fraction-weighted average of the elements’ antibonding descriptors) – (the compound’s own antibonding descriptor), always in the products – reactants convention. Computed for 439 case-1 reactions (`analysis/compute_delta_antibonding_case1.py`).

Table S16: Spearman correlations, Δ (antibonding) of the decomposition reaction (case 1, $n = 438$ –439; see §7 for the 434→438 update).

Descriptor	Target	n	ρ
Δ (ICOHP antibonding)	<code>formation_energy_per_atom</code>	438	-0.6225^{***}
Δ (ICOB1 antibonding)	<code>formation_energy_per_atom</code>	438	-0.4292^{***}
Δ (ICOHP antibonding)	<code>energy_above_hull</code>	438	-0.1734^{***}
Δ (ICOB1 antibonding)	<code>energy_above_hull</code>	438	-0.1215^*

* $p < 0.05$; *** $p < 0.001$.

Δ (ICOHP antibonding) remains, at this scale, the strongest continuous correlation in the entire project ($\rho = -0.623$, stronger than the full reaction-ICOHP descriptor of §6.2, $\rho = 0.295$) — weaker than the $\rho = -0.644$ reported at the earlier, smaller scale, but still the strongest. Notable, unchanged: unlike the raw values (opposite signs, above), **both Δ quantities run the same direction** (both negative). The two Δ quantities are themselves correlated with each other ($\rho = 0.588$, $p < 0.001$, $n = 439$): related but not redundant.

L.4 Stratification by bond type

Table S17: Spearman correlations, Δ (antibonding) against `formation_energy_per_atom`, by subgroup (case 1, $n = 288$ –438; 4 more compounds than earlier printings, see §7’s note on the `material_id/mp_id` fix).

Subgroup	n	ρ (Δ ICOHP)	n	ρ (Δ ICOB1)
all	438	-0.6225^{***}	438	-0.4292^{***}
covalent	29	-0.6956^{***}	29	-0.7750^{***}
ionic	89	-0.6301^{***}	89	-0.2258^*
metallic	288	-0.4359^{***}	288	-0.3602^{***}
mixed	32	-0.6394^{***}	32	-0.8397^{***}
<code>is_metal=False</code>	150	-0.6466^{***}	150	-0.4128^{***}
<code>is_metal=True</code>	288	-0.4359^{***}	288	-0.3602^{***}

* $p < 0.05$; *** $p < 0.001$; no star = not significant.

At this expanded scale, Δ (ICOHP antibonding) remains the first descriptor in the whole project whose overall correlation survives every stratification tested — covalent, ionic, metallic, mixed, `is_metal` on both sides, all significant at $p < 0.001$. Δ (ICOB1 antibonding) has also improved: it survives on *all* strata as well (versus only metallic and `is_metal` before), with the strongest result on *mixed* ($\rho = -0.840$). No other descriptor in this report matches that: the raw antibonding population (§6.2) lost its only surviving stratum (covalent, retracted) and keeps just two (mixed, `is_metal=False`); the full reaction-ICOHP descriptor (§6.2) survives only on `bond_type`, not on metallic. This remains a relatively recent construction — its magnitude will need to be reproduced on a genuinely independent dataset before it is taken as a settled result, but surviving full stratification, at $n = 438$ and across five subgroups instead of three, makes it the project’s most robust result to date.

L.5 Correlation with case-1 viability

Every prior correlation of $\Delta(\text{ICOHP antibonding})$ in this appendix is against a *continuous* thermodynamic target (`formation_energy_per_atom/energy_above_hull`), never against the binary `ViabilityLabel` (§6.2) that §6 tests $\Delta(\text{ICOHP})$ (mission 5) against — this had not been tested anywhere in the project before now (`analysis/compute_viability_antibonding_correlation.py`). Same convention as §6.2: `viable = STABLE_ON_HULL  $\cup$  METASTABLE_VIABLE`, not `viable = UNSTABLE_NONEXISTENT` (`insufficient_data` excluded); Mann–Whitney U on the continuous $\Delta(\text{ICOHP antibonding})$, and Fisher’s exact test on $\Delta(\text{ICOHP antibonding}) > 0$ (binary) against `viable` (binary), the same pair of tests already run for mission 5’s sign/continuous split.

Table S18: $\Delta(\text{ICOHP antibonding})$ vs. case-1 viability, by subgroup ($n_{\text{total}} = 434$).

Subgroup	n viable	n not viable	Mann–Whitney p	Fisher OR	Fisher p
all	329	105	$6.7 \times 10^{-12}***$	3.49	$2.4 \times 10^{-7}***$
<code>bond_type=metallic</code>	125	65	0.0028**	2.20	0.021*
<code>is_metal=True</code>	187	99	$6.7 \times 10^{-5}***$	2.38	0.00099***
<code>is_metal=False</code>	142	6 [†]	0.012*	7.88	0.028*

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. [†]only 6 non-viable compounds in this subgroup — flagged, not suppressed, per this project’s $n < 15$ convention. `bond_type=covalent` (3 non-viable) and `bond_type=ionic` (1 non-viable) are too small to test and are omitted. Median $\Delta(\text{ICOHP antibonding})$: 0.0067 eV (viable) vs. 3.2×10^{-6} eV (not viable), all-subgroup.

A compound whose formation *reduces* antibonding character relative to its constituent elements ($\Delta > 0$) is significantly more likely to be `classify_viability()-viable` than one whose formation *increases* it — true overall and within every subgroup large enough to test, including `is_metal=False` despite its thin non-viable class. This is the first time mission 4b’s descriptor has been tested against viability rather than a continuous stability target, and it survives the test cleanly: a second, independent line of evidence (alongside mission 5’s sign-based endobondic/exobondic split, §6.2) that reduced near- E_F antibonding character upon formation is associated with which compounds actually exist or persist, not just with how thermodynamically favorable their formation is.

L.6 Reaction $\Delta(\text{ICOHP})$ restricted to a window near E_F

Extension proposed in the conclusion (§6.2): rather than the fully-integrated ICOHP over the whole occupied range (§6.2) or only the antibonding part near E_F (previous subsections), a third construction — the *signed* ICOHP (bonding *and* antibonding, not clipped) but integrated *only* over the one-sided window ($E_F - \Delta E$, E_F], $\Delta E = 1.0$ eV, summed over *every* bond label (extensive, like the full ICOHP sum) rather than the “average” trace — then combined with the same reaction stoichiometry as the full $\Delta(\text{ICOHP})$ (`reaction_icohp.py`, reactant – predicted products convention). Implementation: `cohp_extraction.icohp_windowed_per_atom()` reads directly the difference between two points on the *cumulative* ICOHP curve LOBSTER already computes in `COHPCAR.lobster` (`ICOHP_cumulative(E_F) - ICOHP_cumulative(E_F - ΔE)`, per bond label), with no new trapezoidal integration — validated: at very large ΔE (covering the whole computed range), this quantity recovers the full ICOHP sum to within $\sim 0.1\%$. Computed for 439 case-1 reactions.

At this expanded scale, this construction also survives on every subgroup tested ($p < 0.01$ throughout) — unlike the full reaction-ICOHP descriptor (§6.2), which loses all `bond_type` stratification, and unlike its own behavior at the earlier, smaller scale (where only ionic and `is_metal` survived). Weaker in magnitude than $\Delta(\text{ICOHP antibonding})$ (previous

Table S19: Spearman correlations, $\Delta(\text{windowed ICOHP near } E_F)$ against `formation_energy_per_atom` (case 1, $n = 286\text{--}434$).

Subgroup	n	ρ
all	434	+0.3566***
covalent	29	+0.6185***
ionic	87	+0.6184***
metallic	286	+0.1801**
mixed	32	+0.4578**
is_metal=False	148	+0.5643***
is_metal=True	286	+0.1801**

** $p < 0.01$; *** $p < 0.001$; no star = not significant. Against `energy_above_hull`: $\rho = +0.073$, $p = 0.13$ overall, only **mixed** reaches significance ($\rho = +0.497$, $p = 0.004$, $n = 32$) — the same pattern as every other ICOHP/ICOBI descriptor in this report against this target.

subsection) on most strata, but just as general in surviving stratification. Cross-correlation with $\Delta(\text{ICOHP antibonding})$: $\rho = -0.567$ ($p < 0.001$, $n = 439$) — strongly *anticorrelated* despite their close construction (same window, same E_F), a sign that truncating to only the antibonding part (previous subsection) captures information qualitatively different from the full signed ICOHP in the same window. Cross-correlation with the full reaction-ICOHP descriptor: $\rho = +0.241$ ($p < 0.001$) — related but clearly distinct, not a simple restriction of the same quantity.

M Reitz & Dronskowski reaction structures

The §6.2 validation (Table 3) only compares the arithmetic of the manuscript’s own published ICOHP values — independent of any local crystal structure. The table below documents, for each of the 16 species involved in these 7 reactions, the structure as described by the manuscript (where it specifies one), the Materials Project/COD entry used in this dataset, and whether an independent verification was possible. Three cases revealed a real bug or ambiguity, since fixed: N_2 (a theoretical polymeric phase substituted for an isolated gaseous dimer, §6.2), $\text{Pb}(\text{N}_3)_2$ (the wrong Pnma polymorph substituted for the Pna2_1 the manuscript cites), and ZnSn ’s competing tin phase (β -tin substituted for α -tin).

Table S20: The 16 species across Reitz & Dronskowski’s 7 reactions (ic-2026-04181q, Table 3): structure as described by the manuscript, the MP/COD entry used in this dataset, and verification status.

Species (role)	Manuscript structure	Our source	Verification
$\text{Pb}(\text{N}_3)_2$ (target)	lead azide, ICSD 28336/16887 (Glen 1963)	mp-620058, Pna2_1, $E_{\text{hull}} = 0.110$	Yes — exact polymorph identified 2026-08-17, substituted for mp-667338 (Pnma) already present in the dataset
Pb (element)	Table S1 (2026-08-18): $Fm\bar{3}m$, $a = 4.85044 \text{ \AA}$	mp-20483, Fm-3m, $E_{\text{hull}} = 0.000$, on-hull; relaxed under the manuscript’s methodology in <code>manuscript_Pb</code> (2026-08-18)	Yes — our relaxation converges to $a = 4.85424 \text{ \AA}$ (conventional cubic cell), 0.08% off

Species (role)	Manuscript structure	Our source	Verification
N ₂ (element, 4 reactions)	gaseous N≡N, $d \approx 1.10$ Å; Table S1 (2026-08-18) fixes the isolated-dimer box at $8 \times 8 \times 8$ Å	<code>gasref_N2_dimerbox</code> , hand-built isolated dimer ($10 \times 10 \times 10$ Å box), $d(\text{N-N}) = 1.113$ Å, ICOHP = -22.998 eV; also relaxed in the manuscript's $8 \times 8 \times 8$ Å box under its own methodology in <code>manuscript_N2</code> (2026-08-18), $d(\text{N-N}) = 1.111$ Å	Yes — within 0.7% of the manuscript's value (-23.161 eV; ICOHP not recomputed for the new box, out of scope for this update). Box size matches Table S1 by construction (a specified input, not an independent check). Before 2026-08-17: mp-1059834 (theoretical, polymeric phase, N-N = 1.296 Å) — a real bug, fixed
S ₄ N ₂ (target)	not specified beyond composition/ICOHP	COD 4031496, <code>P4_2nm</code> ($Z = 4$), no MP entry exists for this composition	no — COD experimental structure used as-is, no independent confirmation it is the manuscript's exact polymorph
S ₈ (element)	not specified	mp-77, <code>Fddd</code> (α -S ₈), $E_{\text{hull}} = 0.000$, on-hull	not required — unambiguous standard reference
S ₄ N ₄ (target)	not specified	COD 7017102, <code>P2_1/c</code> ($Z = 4$)	no
ZnSn (target)	Table S1 (2026-08-18): <i>P6₃/mmc</i> , $a = b = 4.33111$ Å, $c = 5.05386$ Å — not identified before this date, only the (6 Zn-Sn + 1 Zn-Zn per formula unit) coordination was known	<code>gasref_ZnSn_NiAs</code> , hand-built, relaxed under the manuscript's methodology in <code>manuscript_ZnSn</code> (PBEsol+D3(BJ), 2026-08-18)	Yes — our independent relaxation converges to $a = 4.3418/4.3411$ Å, $c = 5.0379$ Å, i.e. $\leq 0.25\%$ off on every parameter
Zn (element)	Table S1 (2026-08-18): <i>P6₃/mmc</i> , $a = 2.60598$ Å, $c = 4.56331$ Å	mp-79, <code>P6_3/mmc</code> , $E_{\text{hull}} = 0.000$, on-hull; relaxed under the manuscript's methodology in <code>manuscript_Zn</code> (2026-08-18)	Yes — our relaxation converges to $a = 2.60786$ Å, $c = 4.55836$ Å ($\leq 0.11\%$ off), correcting a -6.5% c/a anomaly relative to the raw, unrelaxed MP structure
Sn (element, ZnSn's competing phase)	Table S1 (2026-08-18) explicitly confirms <i>Fd$\bar{3}m$</i> , $a = 6.46874$ Å — settles the α/β ambiguity below	mp-117, <i>Fd-3m</i> (α -tin, diamond-cubic), substituted for mp-623511 (β -tin, <i>I4/mmm</i>); relaxed under the manuscript's methodology in <code>manuscript_Sn</code> (2026-08-18)	Yes — our relaxation converges to $a = 6.46694$ Å (conventional cubic cell), i.e. 0.03% off, confirming mp-117 (α -tin) is indeed the manuscript's phase, not mp-623511 (β)

Species (role)	Manuscript structure	Our source	Verification
CaO[sphalerite] (target)	Table S1 (2026-08-18): $F\bar{4}3m$, $a = 5.14699$ Å — confirms the space group already used	<code>gasref_CaO_sphalerite</code> , hand-built (no MP entry for this metastable CaO polymorph), $F\bar{4}3m$ confirmed by symmetry; relaxed under the manuscript's methodology in <code>manuscript_CaO_sphalerite</code> (2026-08-18)	Yes — space group confirmed as before, and our relaxation now also converges to $a = 5.14751$ Å, 0.01% off
CaO[rocksalt] (product)	Table S1 (2026-08-18): $Fm\bar{3}m$, $a = 4.77501$ Å	mp-2605, $Fm\bar{3}m$, $E_{hull} = 0.000$, on-hull; relaxed under the manuscript's methodology in <code>manuscript_CaO_rocksalt</code> (2026-08-18)	Yes — space group confirmed as before; our relaxation converges to $a = 4.73046$ Å, 0.93% off, the largest lattice-parameter deviation of the 2026-08-18 batch (still $\leq 1\%$)
CaN (target)	Table S1 (2026-08-18): $Fm\bar{3}m$, $a = 4.87139$ Å — not identified before this date	mp-1058549, $Fm\bar{3}m$ (rocksalt-type), $E_{hull} = 0.398$; relaxed under the manuscript's methodology in <code>manuscript_CaN</code> (2026-08-18)	Yes — our relaxation converges to $a = 4.87285$ Å, 0.03% off, confirming both space group and lattice parameter
Ca ₃ N ₂ (product)	not specified	mp-844, $Ia\bar{3}$ (anti-bixbyite), $E_{hull} = 0.000$, on-hull	no — the only on-hull entry, the most defensible default choice
Mn ₂ O ₇ (target)	not specified	mp-28338, $P2_1/c$, $E_{hull} = 0.335$	no
MnO ₂ (product)	Table S1 (2026-08-18): $P4_2/mnm$, $a = b = 4.41216$ Å, $c = 2.94150$ Å (AFM ordering per Noda et al., per the manuscript's own methodology)	mp-510408, $P4_2/mnm$ (rutile/pyrolusite); relaxed under the manuscript's own methodology (PBEsol+U(7.09 eV on Mn 3d)→r2SCAN single-point, AFM) in <code>manuscript_MnO2</code> (2026-08-18)	Yes — our relaxation converges to $a = b = 4.41427$ Å (0.05% off), $c = 2.94460$ Å (0.11% off), confirming both space group and lattice parameters
O ₂ (element)	gaseous O ₂ , triplet ground state; Table S1 (2026-08-18) fixes the isolated-dimer box at $8 \times 8 \times 8$ Å	mp-1524462, $C2/m$, $E_{hull} = 0.000$, on-hull, dedicated spin-polarized treatment; also relaxed in the manuscript's $8 \times 8 \times 8$ Å box under its own methodology in <code>manuscript_O2</code> (2026-08-18)	structural — isolated dimer $d(O-O) = 1.222$ Å (dataset default, 10 Å box) / 1.227 Å (manuscript's 8 Å box), both within $\sim 1.6\%$ of 1.208 Å literature. Box size matches Table S1 by construction (a specified input, not an independent check)

Table S21: This project’s own DFT+LOBSTER ICOHP values against the manuscript’s transcribed per-species values (`tests/fixtures/reitz_dronskowski_cases.yaml`), for all 16 species. “Old (PBE)” column: this project’s default DFT methodology (PBE, ENCUT=600 eV, EDIFF=1e-6, no D3) on the structures already present in the dataset before the manuscript-methodology campaign (2 species, S₈ and Pb(N₃)₂, have no such prior entry). “New (PBEsol)” column: the `manuscript_*` campaign reproducing the manuscript’s exact methodology (PBEsol+D3(BJ), ENCUT=700 eV, EDIFF=1e-8, tetrahedron method) — lattice parameters validated to better than 1% of the manuscript’s Table S1 for every species with a citable structure (Table S20). Both columns are summed under the strict single-coordination-shell convention introduced 2026-08-18 and revised 2026-08-20 (`analysis/compute_manuscript_validation_icohp.py`, gap clustering — break the shell at the first gap between sorted bond distances exceeding a 0.03 Å threshold, rather than a fixed tolerance from the pair type’s minimum bond distance — with an explicit two-shell sum for Mn₂O₇’s short+long Mn-O bonds per the manuscript’s own stated scope) rather than `reaction_analysis.parse_lobster`’s general-purpose `bond_filter="nearest_neighbor"`; see the discussion below the table. As of this update (2026-08-20), 5 species (Mn₂O₇, Pb(N₃)₂, S₄N₂, S₄N₄, S₈) are still computing on SLURM and have no “New (PBEsol)” entry yet.

Species / reaction	Manuscript (eV)	New (PBEsol)	Diff.	Old (PBE)	Diff.
Pb(N ₃) ₂ (reactant)	−89.104	pending	—	—	—
Pb (product)	−5.688	−5.703 ^c	0.3%	−5.111 ^c	10.1%
N ₂ (element)	−23.161	−23.126	0.2%	−22.998	0.7%
S ₄ N ₂ (reactant)	−49.240	−49.251 ^d	0.0%	−49.729 ^d	1.0%
S ₈ (product)	−46.800	pending	—	—	—
S ₄ N ₄ (reactant)	−73.840	pending	—	−73.821 ^d	0.0%
ZnSn (reactant)	−16.869	−16.908 ^a	0.2%	−15.598	7.5%
Zn (product)	−12.686	−12.649 ^b	0.3%	−13.393	5.6%
Sn (product)	−7.672	−7.706	0.4%	−7.633 (β, wrong polymorph)	0.5%
CaO[sphalerite] (reactant)	−3.460	−3.459	0.0%	−3.332	3.7%
CaO[rocksalt] (product)	−4.278	−4.316	0.9%	−4.217	1.4%
CaN (reactant)	−4.302	−4.368	1.5%	−4.161	3.3%
Ca ₃ N ₂ (product)	−7.692	−7.699 ^f	0.1%	−7.454	3.1%
Mn ₂ O ₇ (reactant)	−62.210	pending	—	−56.272	9.5%
MnO ₂ (product)	−18.540	−16.975 ^e	8.4%	−18.316	1.2%
O ₂ (product)	−18.050	−16.513 ^e	8.5%	−16.236	10.0%
ΔICOHP, ZnSn→Zn+Sn	−3.489 eV (−337 kJ/mol)	−3.448 eV (−333 kJ/mol)	1.2%	−10.027 eV (−968 kJ/mol)	—

^a excludes a real but manuscript-unaccounted third-shell Sn-Sn contact ($d = 3.54\text{--}3.57$ Å, -1.336 eV/formula unit): the manuscript’s own accounting for ZnSn is 6 Zn-Sn bonds + 1 Zn-Zn bond per formula unit, with no Sn-Sn term at all, even though DFT+LOBSTER reports a genuine, well-separated first-shell Sn-Sn contact at that distance. A scope-of-accounting difference, not a shell-detection error.

^b true first shell only (the 6 basal Zn-Zn bonds at $d = 2.608$ Å); see below for why the dataset’s general first-shell detector instead sums 12 bonds here.

^c a $\sim 2\times$ residual was initially present in *both* columns alike (new PBEsol and old PBE); it was **resolved** 2026-08-19 and is not a shell-detection or bond-counting artifact of the kind fixed for Zn — a specific alternative hypothesis along those lines (LOBSTER double-listing self-image Pb-Pb bonds in this 1-atom primitive cell, flagged by `reaction_analysis.parse_lobster.check_no_reverse_duplicates`) had indeed been investigated and ruled out: that check also flags Zn’s already-validated first shell (12/12 records, same pattern), and for a monoatomic cell each flagged pair (e.g. translations (0,0,−1) and (0,0,1)) is in fact two *distinct* neighbor images in opposite directions, not the same bond listed twice. The real root cause: the Pb-Pb first shell is a single homogeneous coordination-12 shell (FCC structure, all 12 bonds at 3.43246 Å with the identical value -0.95047 eV — `manuscript_Pb/ICOHPLIST.lobster`), with no second sub-shell available to restrict to 6 bonds the way Zn’s real c/a anisotropy splits 6 short basal bonds from 6 longer out-of-plane ones. The raw 12-bond sum (-11.406 eV) is a *single-atom local* coordination total, not the lattice energy per formula unit: since each bond is shared with the periodic image of the same site ($Z = 1$ cell), the standard convention (each bond counted once for the whole crystal) requires dividing by 2, which `analysis/compute_manuscript_validation_icohp.py` was not doing. Once fixed (`halve_for_self_pair=True`, added 2026-08-19), the PBEsol residual drops from 100.5% to 0.3% and the PBE residual from 79.7% to 10.1% — this remaining residual, now comparable in size to other species in this table, is simply explained by the default PBE methodology’s lack of D3 dispersion. **Confirmed post hoc against the manuscript text** (Figure 2b: “Pb–Pb 3.43 Å, ICOHP: -0.95 eV”, a *per-bond* value identical to our raw -0.95047 eV): $-5.688 / -0.95 \approx 6$, whereas Pb’s true FCC coordination number is 12 (verified: all 12 bonds are rigorously equidistant at 3.43246 Å, with no possible sub-shell split). This contrasts with Zn, where the manuscript explicitly states (Figure 5 caption) “six Zn–Zn bonds per Zn atom” as a physically distinct sub-shell (real c/a anisotropy separating 6 short basal bonds from 6 longer ones) — not half of a 12-count. The $/2$ correction is therefore justified only for Pb (a rigorously degenerate, site-centrosymmetric first shell with no possible geometric sub-shell), not for Zn or for any heteronuclear species in the dataset.

^d **resolved 2026-08-20** (previously reported as unrecoverable). The manuscript’s own accounting for S_4N_2 sums three distinct S-N/S-S bond types explicitly (2 bonds each, $-10.30/-8.46/-5.86$ eV — `tests/fixtures/reitz_dronskowski_cases.yaml`); grouping `ICOHPLIST.lobster`’s records only by element pair (all S-N together, all S-S together) collapses these three types into one number apiece and was wrongly assumed unrecoverable. In fact each of the three types occupies its own distance, well separated from the others and from the non-bonding background: N-S at 1.571 Å (-10.301 eV/bond, 8 bonds in the $Z = 4$ cell), N-S at 1.652 Å (-8.456 eV/bond), S-S at 2.068 Å (-5.868 eV/bond) — the next-nearest contact of either pair is > 2.7 Å away, no shell-clustering ambiguity. `analysis/compute_manuscript_validation_icohp.py` now sums each pair type below a 2.2 Å cutoff separately, landing at -49.251 eV (PBEsol) / -49.729 eV (PBE), both within 1% of the manuscript.

A second, independent bug was found and fixed in the same pass: the script normalizes each species’ ICOHP sum by a formula-unit factor z taken from `pymatgen`’s `get_reduced_composition_and_factor()`, which reduces to the simplest empirical ratio ($S_4N_2 \rightarrow S_2N$, $z = 8$; $S_4N_4 \rightarrow SN$, $z = 16$) rather than the manuscript’s own molecular formula unit ($z = 4$ for both, 4 formula units per cell) — silently dividing the per-formula-unit ICOHP by an extra factor of 2 (S_4N_2) or 4 (S_4N_4). Fixed by overriding z with the correct atom count per named formula unit (6 for S_4N_2 , 8 for S_4N_4/S_8) for the handful of species where this applies; verified unchanged for all other species in this table. This bug is confined to this 16-species validation script — the main ~ 600 -compound dataset (`reaction_analysis/parse_lobster.py`) uses the same `pymatgen` reduction for *both* its reaction stoichiometry and its ICOHP normalization consistently (e.g. its S_4N_2/S_4N_4 reaction entries are written in terms of “ S_2N ”/“ SN ”, not the molecular formula), so there is no external-reference mismatch there and no fix is needed. This retroactively explains S_4N_4 ’s old-PBE entry, previously reported here at -16.207 eV (78.1% off) and attributed to its unverified COD polymorph (note e) — that attribution was wrong; the true cause was this same z bug (a factor of 4), and the corrected value (-73.821 eV, 0.0%) shows the COD polymorph is in fact a good match. Its crystallographic verification status (Table S20: “no”) is unaffected by this correction and remains genuinely unconfirmed, just no longer implicated in

any numeric residual.

^e MnO_2 and O_2 's $\sim 8\text{--}10\%$ residuals persist under *both* the new PBEsol and old PBE calculations, so are unlikely to be shell-detection artifacts either — most plausibly a genuine DFT-methodology sensitivity (AFM/Hubbard-U treatment for MnO_2 , spin-polarization of O_2 's triplet ground state) not yet resolved. A two-shell Mn-O sum for MnO_2 (mirroring Mn_2O_7 's explicit short+long convention) was tried and only marginally improves the gap ($8.4\% \rightarrow 7.2\%$), too small to explain the residual on its own. For O_2 specifically, the INCAR-parameter space was exhausted (2026-08-20) by isolating each candidate variable one at a time from the default-PBE baseline, first at fixed geometry and then re-relaxed individually (IBRION=1, ISIF=2, fixed cell volume) to rule out geometry as a confound. MAGMOM, k-point density, D3(BJ) dispersion, ENCUT, and ISMEAR each leave the relaxed O-O distance within $0.0001\text{--}0.0001$ Å of the PBE baseline ($1.233 \rightarrow 1.233$ Å) and the ICOHP within 10.0% of the manuscript value (-16.238 to -16.244 eV, essentially unchanged from the unrelaxed baseline's -16.236 eV). Only the GGA=Ps/PBEsol functional produces a real shift, both in geometry ($1.233 \rightarrow 1.227$ Å, matching the full manuscript_02 relaxation to 0.0001 Å) and in ICOHP (-16.487 eV, 8.7% off) — on its own accounting for nearly the entire gap between the PBE baseline (10.0%) and the full manuscript-methodology reproduction reported above (8.5%). This confirms the residual tracks the exchange-correlation functional itself, not any numerical-convergence or geometry-relaxation parameter, consistent with a genuine DFT-methodology sensitivity rather than a numerical artifact. This is independently corroborated by the literature: GGA functionals are known to systematically overbind the O_2 molecule specifically (an error that "only occurs when the oxidant is O_2 "), commonly corrected with an empirical -1.36 eV total-energy shift per O_2 molecule fitted against non-transition-metal oxide formation enthalpies — Wang, Maxisch & Ceder, *Phys. Rev. B* **73**, 195107 (2006), DOI: 10.1103/PhysRevB.73.195107. While that correction targets total energy rather than ICOHP, it points to the same root cause: O_2 's near-degenerate π^* singly-occupied orbitals are a known pathological case for GGA-family functionals (PBE and PBEsol alike), not a pipeline-specific artifact.

^f a second, distinct occurrence of the near-degenerate-shell sensitivity that motivated note b (Zn), found 2026-08-20. Ca_3N_2 's true Ca-N first shell is not one distance but four near-degenerate sub-shells spanning $2.4021\text{--}2.4239$ Å (96 bonds total across the $Z = 8$ cell, matching the expected 12 Ca-N bonds per formula unit), separated from the next shell by a real 1.74 Å gap. The fixed 0.02 Å tolerance from $d_{\min}$ used since 2026-08-18 captured only the first three sub-shells (72/96 bonds), landing 24% off the manuscript's -7.692 eV. Unlike Zn (where the general dataset-wide detector *over*-includes a genuinely distinct second shell), here the validation script's own fixed tolerance *under*-included part of a single true shell — the opposite failure mode, but the same root cause: assuming one fixed distance window fits every compound's coordination geometry. Fixed (2026-08-20) by switching `_strict_first_shell` to gap clustering (break at the first gap exceeding a threshold, rather than a fixed distance from the minimum); the 0.03 Å threshold was hand-tuned to sit strictly between the widest confirmed within-shell spread (Ca_3N_2 and Mn_2O_7 's own Mn-O/O-O sub-clusters, $0.011\text{--}0.026$ Å) and the tightest genuine second-shell gap found elsewhere in this table (Sn's old-PBE reference: a real, distinct, weaker-ICOHP second shell only 0.047 Å past the first) — verified to leave every other species' bond count in this table unchanged before landing. Ca_3N_2 now matches at 0.1% .

Sn and ZnSn were already resolved in the first pass of the 2026-08-18 update: replacing the default β -tin reference with the manuscript's α -tin phase, computed under its exact methodology, drops Sn's gap from 38.4% to 0.4% — almost the entire Sn gap was a polymorph error, not a genuine DFT disagreement — and excluding the manuscript-unaccounted Sn-Sn term (note a) drops ZnSn from 8.1% to 0.2% .

Zn's 36.0% residual, left open at the time, was resolved the same day. Root cause (`analysis/compute_manuscript_validation_icohp.py`): `reaction_analysis.nearest_neighbor.detect_first_shell()` — the general first-shell detector this project uses dataset-wide via `bond_filter="nearest_neighbor"` for its 588-compound bond-type classification (`analysis/compute_icohp_icobi_bondtype.py`, regression-tested in `tests/test_nearest_neighbor.py`) — cuts at the single LARGEST gap anywhere in a pair type's whole distance spectrum, not the first one. Zn's anomalously low hcp c/a ratio puts a second, chemically distinct Zn-Zn shell (interlayer, $2.729\text{--}2.734$ Å) only 0.12 Å past the true first shell (basal, 2.608 Å) — much closer than the ~ 1.04 Å gap to the real third shell — so the detector locks onto that later gap and sums both shells (12 bonds instead of 6) as one, roughly doubling the ICOHP magnitude. Restricting the sum to the true first shell only (note b) gives -12.649 eV, 0.3% from the manuscript's -12.686 eV.

This is a genuine limitation of applying one general-purpose shell-detection rule across a dataset with widely varying coordination geometries, not a DFT/LOBSTER methodology disagreement — and it is

not evidence against the general detector’s own design choice (the single-largest-gap rule is deliberate and correctly handles the counter-example that motivated it, see `nearest_neighbor.py`’s docstring). `nearest_neighbor.py` itself is left unchanged: Zn’s near-degenerate double shell is not necessarily representative of the 588-compound population, and whether any other compound’s bond-type classification is similarly affected has not been fully closed (see §9).

2026-08-19 extension. The comparison above was extended from the original 3 species (ZnSn, Zn, Sn) to all species with a completed `manuscript_*` directory (10/16 at the time of writing, 6 new since the 2026-08-18 fix: Pb, CaO[sphalerite], CaO[rocksalt], CaN, MnO₂, O₂) and, separately, to every species with a pre-existing PBE-functional dataset entry (14/16 — all but S₈ and Pb(N₃)₂, novel to this campaign). Of the 6 newly-added new-PBEsol comparisons, 3 reproduce the manuscript to within 0.0–1.5% under the same strict single-shell convention used for Zn (CaO[sphalerite], CaO[rocksalt], CaN), with no shell-detection ambiguity found in any of their first-shell distance spectra. Three residuals remain open rather than force-fit to zero: Pb (note c) and MnO₂/O₂ (note e) — all three investigated, all three ruled out as instances of the Zn-type shell-detection bug, and left as genuine, unresolved methodology questions. Mn₂O₇, S₄N₂, S₄N₄, S₈, and Pb(N₃)₂ remain to be compared once their `manuscript_*` SLURM jobs land.

2026-08-20 extension. Ca₃N₂’s `manuscript_*` job landed, extending the comparison to 11/16 species. Its initial 24% residual (note f) was not a new open question but a second, distinct occurrence of the near-degenerate-shell sensitivity first found for Zn: a fixed tolerance from the minimum bond distance is not a safe general assumption for this validation script’s first-shell summation, now replaced by gap clustering (note f), verified not to change any other species’ result in this table. Mn₂O₇, Pb(N₃)₂, S₄N₂, S₄N₄, and S₈ remain to be compared once their `manuscript_*` SLURM jobs land.

With ZnSn/Zn/Sn corrected, the recomputed ΔICOHP for ZnSn→Zn+Sn (−3.448 eV, −333 kJ/mol) is within 1.2% of the manuscript’s −3.489 eV (−337 kJ/mol) — down from 1.9× before this correction, and from 2.9× in the original (pre-manuscript- methodology) calculation. This project’s own DFT+LOBSTER ICOHP methodology reproduces the manuscript’s reported values to within 1.2% for this reaction once (i) the correct polymorphs are used, (ii) the manuscript’s own bond-accounting scope is matched, and (iii) first-coordination-shell summation is not confounded by near-degenerate shells — closing the validation question raised in §6.2 and confirming this project’s ICOHP methodology, not just its arithmetic, matches the manuscript’s for the one reaction where every species is now resolved (§6.2’s own 7-case validation, using only the manuscript’s published values, was never in question and continues to reproduce them almost exactly).

Reading key. “Verification: Yes” means the structure was confirmed to match the manuscript’s explicit description (a cited space group, or a recomputed bond distance compared directly). “Partial” means the geometry/chemistry is consistent with what the manuscript describes but no independent crystallographic citation allowed a strict confirmation. “No” means only the composition was verified: the manuscript does not specify a space group for that species, so the most reasonable MP/COD entry was used with no guarantee it is *the* exact structure the authors used. This does not invalidate the §6.2 validation (which only compares the arithmetic reconstruction of the manuscript’s own published ICOHP values, independent of any local structure) but it does bound the scope of any future comparison between our own DFT+LOBSTER calculations and the manuscript’s numbers for these species.